

Randomized Controlled Trial of Budesonide/Surfactant versus Surfactant alone in Extremely
Preterm Infants (“The Budesonide in Babies (BiB) Trial”)

Manual of Procedures

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ACRONYMS

%	Percent
~	Approximately
<	Less Than
=	Equals
>	Greater Than
±	Plus or Minus
≤	Less Than or Equal To
≥	Greater Than or Equal To
α	Alpha
ADL	Activities of Daily Living
AE	Adverse Event
BiB	Budesonide in Babies
BP	Blood Pressure
BPD	Bronchopulmonary Dysplasia
BSID IV	Bayley's Scales of Infant Development, 4th Edition
BW	Birthweight
CA	Corrected <i>Gestational Age</i>
CFR	Code of Federal Regulations
CLD	Chronic Lung Disease
CMV	Cytomegalovirus
CON MEDS	Concomitant Medications
CP	Cerebral Palsy
CSS	Clinical Safety Specialist
DCC	Data Coordinating Center
DOL	Day of Life
DSMC	Data Safety Monitoring Committee
EDC	Electronic Data Capture
EDD	Estimated Delivery Date
ELBW	Extremely Low Birthweight
EOS	Early Onset Sepsis
FDA	Food and Drug Administration
FU	Follow-Up
g	Grams
GA	Gestational Age
GCP	Good Clinical Practice
GDB	Generic Database of the NICHD NRN
GMFCS	Gross Motor Function Classification System
ICH	International Conference on Harmonization
IND	Investigational New Drug
IRB	Institutional Review Board
ITT	Intention to Treat
IVH	Intraventricular Hemorrhage
kg	Kilograms
LOS	Late Onset Sepsis

LTF	Lost to Follow-Up
MAP	Mean Airway Pressure
ml	Milliliter
MOP	Manual of Procedures
N/A	Not Applicable
NDI	Neurodevelopmental Impairment
NEC	Necrotizing Enterocolitis
NICHHD	National Institute of Child Health and Human Development
NIH	National Institute Of Health
NRN	Neonatal Research Network
pct.	Percentile
PDA	Patent Ductus Arteriosus
PI	Principal Investigator
PMA	Post-Menstrual Age
PRN	pro re nata (as needed)
PVL	Periventricular Leukomalacia
QOL	Quality of Life
ROP	Retinopathy of Prematurity
RTI	Research Triangle Institute International
SAE	Serious Adverse Event
SAF	Safety Population
SAP	Statistical Analysis Plan
SBP	Systolic Blood Pressure
SD	Standard Deviation
SIP	Spontaneous Intestinal Perforation
TPN	Total Parenteral Nutrition
µg	Micrograms
VLBW	Very Low Birth Weight

SUMMARY

Budesonide in Babies (BiB)

Study Hypothesis/Question

Early intratracheal administration of a combination of budesonide with surfactant (Intervention), as compared to surfactant alone (Active control) will reduce the incidence of physiologic bronchopulmonary dysplasia (BPD) or death by 36 weeks' post-menstrual age in extremely preterm infants.

Study Design Type

Randomized, masked, active-controlled multicenter clinical trial with 1:1 allocation to two treatment arms.

Eligibility Criteria

Inclusion criteria: Liveborn infants 22 ⁰/₇ – 28 ⁶/₇ weeks gestation OR 401 – 1000 grams (inclusive) birth weight with clinical decision to give surfactant. Infants must be enrolled into the trial early so that the first dose of study drug can be administered at ≤ 50h postnatal age and may be consented antenatally or postnatally.

Exclusion criteria: Use of surfactant before enrollment (first dose of surfactant must be study drug); infant received systemic steroids prior to enrollment; use of indomethacin, either received by the mother within 24 hours prior to delivery, or received by the infant prior to enrollment, or intent to administer to the infant for IVH prophylaxis or PDA management from enrollment up to within 7 days of final dose of study drug; known congenital infection; serious chromosomal abnormalities or major malformations; a permanent neuromuscular condition that affects respiration; terminal illness (heart rate < 100 beats per minute, unresponsiveness to resuscitation) or unlikely to survive as judged by the clinician; or decision to redirect or limit support; enrollment in a conflicting clinical trial.

Study Intervention/Methods

After informed consent and assessment of eligibility, infants will be randomized 1:1 to either the budesonide + surfactant arm or the surfactant alone arm. Randomization will be stratified by study site and by gestational age (< 26 weeks; ≥ 26 weeks). The research pharmacy will receive a computer-generated code number (via drug kit) that will be linked to each enrolled infant upon randomization. The corresponding budesonide + surfactant or surfactant alone will be administered by an unmasked respiratory therapist or other qualified person in participating neonatal ICUs, while other caregivers will remain masked to treatment group allocation. Multiple gestations (if more than one is eligible) will be randomized independently. The intervention arm will receive surfactant (poractant alfa; Curosurf) 2.5 ml/kg mixed with 0.25 mg/kg budesonide (Pulmicort nebulizing suspension) while the active control arm will receive Curosurf 2.5 ml/kg. If the infant is to receive the second dose of study drug, the dosage of Curosurf will be 1.25 ml/kg for the second dose (noting that the first dose of Curosurf is 2.5 ml/kg, regardless of whether a second dose is anticipated or not).

Study Drug Initiation

Infants must be enrolled into the trial early so that the first dose of study drug can be administered at ≤ 50h postnatal age preferably consented antenatally (so first study drug dose can be given at delivery if needed). If antenatal consent is not available, early postnatal consent can be obtained so that

enrollment and randomization can occur at ≤ 48 h postnatal age and intervention can be started in the window of ≤ 50 h postnatal age

Duration of Study Intervention

A maximum of two doses of study drug can be administered, with the specific criteria for the second dose by ≤ 50 h postnatal age. If subsequent doses of surfactant are required, usual clinical practice at the neonatal ICU will be followed

Primary Outcome

Physiologic BPD or death by 36 weeks post-menstrual age (PMA)

Secondary Outcomes

Death by 36w PMA; Physiologic BPD at 36w PMA; BPD severity by the Jensen et al. (2019) definition at 36w PMA (No BPD/Grade 1/Grade 2/Grade 3); Grade 3 BPD at 36w PMA; use of additional postnatal steroids for chronic lung disease by 36w PMA; severe neurodevelopmental impairment (NDI) assessed at 22-26 months corrected age; death by 22-26 months corrected age assessment; severe NDI/death by 22-26 months corrected age

Exploratory Outcomes

BPD severity (None/Mild/Moderate/Severe) at 36w PMA by the NICHD definition (Jobe and Bancalari 2001); number of days on invasive mechanical ventilation by postnatal day 28; number of days on invasive mechanical ventilation by 36w PMA; intubation after the treatment window and by postnatal day 28; intubation after the treatment window and by 36w PMA; administration of repeat surfactant doses after the treatment window; respiratory outcomes (recurrent wheezing and chronic coughing) by 22-26 months corrected age

Safety Outcomes

Within the first week following last study drug administration (hypertension, hyperglycemia, prolonged hypoxemia with bradycardia, endotracheal tube blockage, hypotension, pulmonary air leak, sepsis, intracranial hemorrhage); within 30 days following last study drug administration (spontaneous intestinal perforation, periventricular leukomalacia); impairment of growth parameters assessed at 36w PMA (weight, length, head circumference)

CHAPTER 1. ADMINISTRATION

1.1 Study Organization

The NICHD Neonatal Research Network is conducting this study. The Network is funded by the NICHD under cooperative agreements with fifteen institutions, comprising Clinical Centers and a Data Coordinating Center (DCC).

1.2 Participating Institutions

1.2.1 BiB Study Subcommittee

The BiB Study subcommittee is responsible for the preparation and maintenance of the Protocol, Manual of Operations and data forms. Members of this subcommittee are:

Subcommittee Members (in alphabetical order):

Michele Walsh MD, NIH
 Luc Brion MD, UTSW
 Waldemar A. Carlo MD, UAB at Birmingham
 Michael Cotten MD, Duke University
 Carl D'Angio MD, University of Rochester
 Abhik Das PhD, RTI International
 Martin Keszler MD, Brown University
 Matthew Laughon MD, University of North Carolina
 Namasivayam Ambalavanan MD, UAB at Birmingham (Chair)
 Kayla Nowak BS, RTI International
 Brenda Poindexter MD, Emory (V. Chair)
 Kristi Watterberg MD, UNM
 Beth Wiener DrPH, RTI International

Research Coordinators:

Shirley Cosby BSN RN, UAB at Birmingham
 Carrie Rau BSN RN University of Utah

Collaborators:

Roberta Ballard MD, UCSF
 Phil Ballard MD, UCSF
 Alan Jobe MD, CCHMC
 Cindy McEvoy MD, OHSU

1.2.2 Participating NICHD Neonatal Research Network Centers

Centers from the NICHD Neonatal Research Network participating in the trial are listed below. The NICHD center number is indicated in parentheses next to the name of each center and the principal investigator is in the second column with clinical research coordinators in the third.

Case Western Reserve Univ. (3)	Anna Maria Hibbs, MD	Nancy Newman, RN
University of Texas- Dallas (4)	Myra Wyckoff, MD	Joanne Duran, RNC-NIC
Emory University (9)	Ravi Patel, MD	Yvonne Loggins, BSN
Cincinnati Children's Hospital (11)	Stephanie Merhar, MD	Cathy Grisby, BSN, CCRC
*Brown University (14)	Abbot Laptook, MD	Angelita Hensman, PhD RNC-NIC
Stanford University (15)	Krisa Van Meurs, MD	M. Bethany Ball, BS, CCRC
University of Alabama (16)	Waldemar A. Carlo, MD	Shirley Cosby, RN
University of Texas- Houston (18)	Jon Tyson, MD	Emily Stephens, BSN, RNC-NIC
Duke University (19)	C. Michael Cotten, MD	Joanne Finkle, RN, JD
University of Iowa (24)	Edward Bell, MD	Karen Johnson, BSN
University of Utah (25)	Robin Ohls, MD	Carrie Rau, BSN, CCRC
University of New Mexico (26)	Kristi Watterberg, MD	Sandra Beauman, MSN
University of Pennsylvania (27)	Eric Eichenwald, MD	Toni Mancini, RN, BSN, CCRC
*University of Rochester (28)	Carl D'Angio, MD	Rachel Jones RN
*Nationwide Children's Hospital (30)	Pablo Sanchez, MD	Patty Luzader, RN
Northwestern University	Aaron Hamvas, MD	Erin Lonergan, RN

University of Mississippi
Sharp Mary Birch Hospital

Abhay Bhatt, MD
Anup Katheria, MD

Chelsea Giachelli, BS
Felix Ines, RCP-RRT

* The recruitment period was 2021-2023

1.3 Responsibilities

The roles and responsibilities of the NRN Clinical Centers, the Data Coordinating Center, and NICHD program staff are outlined in the Policy and Procedures Manual. This information is posted on the private gateway of the NRN website.

CHAPTER 2. STUDY OVERVIEW

2.1 Study Population

2.1.1 Inclusion Criteria

- Liveborn infants 22 0/7 – 28 6/7 weeks gestation **OR** 401 – 1000 grams (inclusive) birth weight
- Clinical decision to give surfactant, and
- Less than or equal to 48 hours postnatal age (infants must be enrolled into the trial early enough (via either antenatal consent or early postnatal consent) so that randomization can be started in the window of ≤ 48 h postnatal age (study drug initiation can occur at ≤ 50 hours postnatal age)).

2.1.2 Exclusion Criteria

- Terminal illness (heart rate < 100 beats per minute, unresponsiveness to resuscitation) or unlikely to survive as judged by the clinician,
- Decision to redirect or limit support,
- Use of surfactant before enrollment (first dose of surfactant must be study drug),
- Infant received systemic steroids prior to enrollment,
- Use of indomethacin, either received by the mother within 24 hours prior to delivery, received by the infant prior to enrollment, or intent to administer to the infant for IVH prophylaxis or PDA management from enrollment up to 7 days of final dose of study drug,
- Serious chromosomal abnormalities or major malformations,
- Known congenital infections including, but not limited to, confirmed sepsis, congenital CMV, etc.,
- Infants with a permanent neuromuscular condition that affects respiration, or
- Enrollment in a conflicting clinical trial.

2.2 Detailed Study Procedures

2.2.1 Screening

Clinical research coordinators will screen pregnant women admitted at 22-28 weeks of gestation who are considered at risk of preterm delivery and will screen extremely preterm infants admitted to the neonatal intensive care unit for meeting eligibility criteria.

2.2.2 Consent

Consent can be obtained either before birth of the infant (antenatal consent) or after birth of the infant (postnatal consent), as long as the consent is obtained early enough to assess eligibility and randomize the infant in the trial by 48h postnatal age. The consensus of the subcommittee is to try to obtain antenatal consent, but if not possible, to try to obtain postnatal consent to maximize enrollment. A +2h grace period will be allotted for attaining and administering the dose (study drug may be administered up to 50h after birth)

2.2.3 Randomization

The BiB trial will aim to enroll a total of 1160 infants, equally allocated to the two treatment arms.

Randomization must occur less than or equal to 48 hours postnatal age. This approach takes

motivation from the step-forward randomization idea of “Use Next” drug kits. Due to the need for physical queues of randomized study drug for rapid access, 1:1 randomization to the budesonide + surfactant arm or the surfactant alone arm will be stratified by NRN site and gestational age (<26 weeks vs. ≥26 weeks), drug kits of budesonide respules or sham will be randomized, assembled, masked, and labeled by the Investigational Drug/Research Pharmacy, then stored in a locked location for rapid access by the respiratory therapist (or other qualified designated person). The research pharmacy will receive a computer-generated code number from dispensed (opened) and returned drug kits that will be linked to each enrolled infant upon randomization.

Drug kits will be added to the queue such that there are at least three sets of kits available in each gestational age stratum at a given time, and the Investigational Drug/Research Pharmacy will be responsible for ordering, stocking, and supplying the respules of budesonide and sham in study drug kits to the NICU (or secure location). Two drug kits will be prepared for each randomization number in case of the need for a second dose of study drug within the treatment window, thus a set of drug kits includes two drug kits, with two respules each, for a total of four respules in a set of drug kits (i.e., four respules per randomization ID).

If multiple births are expected, then the corresponding number of kits will be taken from the queue in the randomization number ordering, as multiple gestation births will be randomized independently once eligibility criteria are met and consent is obtained.

Infants will be considered randomized and assigned to the randomization number when the masked drug kit is opened, which will occur after assessment of eligibility and directly before treatment administration. If the drug kit is not opened, then it may be returned to the queue in the locked location and the randomization number is not considered spent.

Refer to the Investigational Drug Manual of Procedures (V1.0, 15 May 2020) for process details.

2.2.4 Study Intervention and Comparison

Study Drug: Infants randomized to the active control arm will receive a dose of surfactant (poractant alfa; Curosurf 2.5 ml/kg) while infants randomized to the intervention arm will receive a dose of surfactant mixed with budesonide (Pulmicort nebulizing suspension). The dose of budesonide will be 0.25 mg/kg in a volume of 1 ml/kg, for a total volume of 2.5 ml/kg of Curosurf + 1 ml/kg of budesonide. If the infant is to receive the second dose of study drug, the dosage of Curosurf will be 1.25 ml/kg for the second dose (noting that the first dose of Curosurf is 2.5 ml/kg, regardless of whether a second dose is anticipated or not).

The study drug will be administered for at least one dose to infants who meet criteria for surfactant administration, and for a maximum of two doses. After 50h of life, subsequent uses and type of surfactant will not be directed by study protocol but will be by clinician preference and center practice. The exact process of surfactant administration (size of aliquot, infant positioning, etc.) will be as per clinician preference and center practice.

2.2.5 Control or Monitoring of Co-Interventions

Early (< 7 days after the last dose of study drug) postnatal systemic steroids will be discouraged and reserved only for infants (at clinician discretion) who have **fluid-resistant hypotension OR vasopressor-resistant hypotension, OR evidence of hyponatremia and low serum cortisol concentrations** (as per local NICU definitions, with suggested definitions including:

Hypotension Definitions

Fluid Resistant Hypotension	Receiving 20 ml/kg or more of fluid infusion for hypotension in < 2 hours time
Vasopressor Resistant Hypotension	Receipt of treatment with at least one vasopressor at a higher dose for example dopamine at 10 mcg/kg/min or more

The use of postnatal steroids at later time points will be considered a secondary outcome and it is suggested that use of postnatal systemic steroids not exceed the dosing used in DART or the Hydrocortisone trial (if hydrocortisone is used), except in patients with demonstrated adrenal insufficiency.

Other co-interventions (e.g., use of ibuprofen or other agents for PDA management) are permissible as per clinician preference or unit practice. However, prophylactic or therapeutic use of indomethacin during the seven days following the last dose of study drug is not permissible in this study due to the possibility of increased risk for SIP. All reportable, concomitant medications will be collected and summarized.

2.3 Research Design

2.3.1 Blinding/Masking

Drug kits of budesonide respules or sham will be randomized, assembled, masked, and labeled by the Investigational Drug/Research Pharmacy, then stored in pharmacy or a locked location for rapid access by the respiratory therapist (or other qualified designated person). In sites where it is necessary that the Investigational Drug/Research Pharmacy prepares the study drug, the unmasked Investigational Drug/Research Pharmacist will prepare and dispense the prepared study drug for administration by the masked respiratory therapist (or other qualified designated person).

Curosurf will be available as clinical supply and housed separately from the drug kits following institutional /unit practice. It is not masked.

In addition to labeling the drug kits with the required cautionary statement that the drug product is only for investigational purposes, the sequential randomization number will also be labeled on each kit, such that the designated person (e.g., respiratory therapist) will select the next sequential randomization number in the queue.

The unmasked respiratory therapist (or other qualified designated person) will mix the budesonide with Curosurf (or mimic with active control/sham) and administer it to the infant. The used respules will then be placed back into the masked drug kits labelled with the infant randomization number and EMR

sticker to identify the kit back to the infant. The Investigational Drug/Research Pharmacy will track usage of the Curosurf, budesonide, and sham by evaluating used respules. Other clinical caregivers will therefore remain masked to treatment group allocation.

Separate research personnel will be responsible for data collection (other than those data collected by the research pharmacy) and will remain masked to the treatment arm (this includes the Network GDB). The parents, medical providers, and neurodevelopmental follow up personnel will be masked to the treatment arm.

2.3.2 Primary Outcome

The primary outcome is physiologic BPD or death by 36 weeks' post-menstrual age in all randomized extremely preterm infants and will be recorded by the Research Coordinator at participating centers, using existing GDB criteria.

2.3.3 Secondary Outcomes

The following secondary outcomes will be assessed for all randomized subjects, with many being standard in the NRN:

- Death by 36w PMA
- Physiologic BPD at 36w PMA
- BPD severity as defined by Jensen et al. (2019) (35) at 36w PMA according to treatment with the following support:
 - No BPD, room air;
 - Grade 1, nasal cannula at flow rates ≤ 2 L/min;
 - Grade 2, nasal cannula at flow rates > 2 L/min or noninvasive positive airway pressure;
 - Grade 3, invasive mechanical ventilation.
- Grade 3 BPD at 36w PMA according to the Jensen et al. (2019) definition
- Use of postnatal steroids for treatment of evolving chronic lung disease from 7 days post final dose of study drug through 36w PMA
- Severe NDI assessed at 2-year follow-up (22-26 months corrected age). In light of the upcoming transition to BSID IV, severe NDI assessment will be based on the agreed-upon definition at the time of the first participant's follow-up assessment. An example of such a definition could be presence of any of the following:
 - Cognitive composite score on the BSID IV < 70 ,
 - GMFCS level 3-5,
 - Severe hearing impairment (no functional hearing despite amplification), or
 - Bilateral severe visual impairment (bilateral blindness despite correction).
- Death by 2-year follow-up (22-26 months corrected age)
- Severe NDI or death as assessed at 2-year follow-up (22-26 months corrected age)

2.3.3.1 Exploratory Outcomes

The following exploratory outcomes will be collected for all randomized subjects:

- BPD severity (none/mild/moderate/severe) by the NIH (NICHD/NHLBI June 2000 Workshop) definition at 36w PMA (37)
- Number of days on invasive mechanical ventilation by postnatal day 28
- Number of days on invasive mechanical ventilation by 36w PMA
- Intubation after the treatment window (50h postnatal age) and within the first 28 postnatal days

- Intubation after the treatment window (50h postnatal age) and within 36w PMA
- Administration of repeat surfactant doses after the treatment window (50h postnatal age) and within 7 days after last dose of study drug
- Reported respiratory outcomes as assessed at 2-year follow-up (22-26 months corrected age), specifically:
 - Recurrent wheezing
 - Chronic coughing

Possible ancillary studies will evaluate development/analysis of a biorepository (genomic/proteomic analysis, similar to the Cytokine study) in this cohort.

2.3.3.2 Safety Outcomes

Adverse events (AEs) are monitored during the study (from treatment initiation through 7 days (168 hours) post final study drug administration) for all AEs, and then continuous monitoring within 30 days of final study drug administration for specified AEs (e.g., spontaneous intestinal perforation). The following are pre-specified events to be monitored (see *Appendix A Toxicity Table Severity/Intensity Grading Scale* for specific diagnostic and grading criteria):

- Within 7 days of final study drug administration:
 - Hyperglycemia
 - Hypertension
 - Prolonged hypoxemia with bradycardia
 - Endotracheal tube blockage
 - Any other adverse event deemed moderate or worse in severity. Mild events are not recorded unless the event is also considered unexpected.
- Within 30 days of final study drug administration:
 - Spontaneous intestinal perforation (SIP)

All serious adverse events (SAE) and those events requiring IND safety reporting will be followed until satisfactory resolution or until the site investigator deems the event to be chronic or the adherence to be stable or by study status of 36 weeks' post-menstrual age.

Neonatal morbidities and outcomes occurring during the study-defined AE monitoring period are captured as an AE and recorded in lesser detail in the NRN Generic Database (GDB). The following list denotes additional pre-specified events to be monitored, however, these outcomes are deemed to be risks of prematurity, and therefore possible, but unlikely to be related to study intervention. Specific diagnostic and grading criteria for these events can also be found in Appendix A .

- Within 7 days of final study drug administration:
 - Hypotension
 - Pulmonary air leak
 - Culture positive sepsis (early-onset sepsis or late-onset sepsis)
 - Intracranial hemorrhage (ICH in preterm infants usually happens in the first 72hours (90% by 72h) and is usually determined by a routine head ultrasound on days 5-7. As some infants may not get their head ultrasounds on a weekend, we are providing 3 days additional window.)
- Within 30 days of final study drug administration

- Periventricular leukomalacia (PVL is usually diagnosed closer to one month of age (usually on a day 28 head ultrasound))

Neonatal morbidities and outcomes occurring after the study-defined AE monitoring period are also recorded in the GDB in lesser details. Impairment of growth parameters (weight, length, head circumference) at 36 weeks PMA will also be collected and summarized from the GDB.

2.3.4 Compliance Monitoring

Other than study intervention and restricted use of postnatal systemic corticosteroids, other clinical practices are per usual clinical protocol and clinician preference at site. Strategies to monitor adherence to the study protocol will include the following:

- Monthly recruitment reports of infants screened and enrolled (accrual figures)
- Monthly reports detailing data received at the Data Coordinating Center, data consistency, missing data, performance measures and adherence to the study protocol (with appropriate measures taken to preserve the blinding of study personnel and investigators)
- Periodic reports for the DSMC detailing data quality, protocol adherence, participant withdrawals, adverse events, losses to follow-up and study futility/efficacy
- Supplementary blinded reports requested by the study investigators or sub-committee that do not disclose allocation group specific outcomes (primary, secondary or any safety outcomes).

2.3.5 Study Specimens

None specifically for study, although ancillary studies are possible for evaluating tracheal aspirates, blood, and urine for genomic, transcriptomic, microbiomic, and other biomarkers (e.g. inflammatory markers) associated with BPD and other outcomes (NDI, BPD/death).

2.3.6 Post-hospital Procedures

None

2.3.7 Follow-up at 24 Months

Neurodevelopmental evaluation will be performed by certified follow up examiners at 22-26 months corrected age per the NRN Follow up protocol. Evaluations will consist of the following: BSID IV, neurologic evaluation, anthropometric measurements, and assessment of severity of neurodevelopmental impairments (NDI). For the purposes of this study, severe, moderate or mild NDI will be defined as follows.

Definition of neurodevelopmental impairment (NDI)

	Normal	Mild	Moderate	Severe
Cognitive	≥85	70-84 or ≥85 and any below	70-84	<70
(GMFCS)		1	2	3-5
Vision	No apparent problem	No apparent problem	Limited with correction/ unilateral blind	Bilateral blind (cannot correct)
Hearing	No apparent problem	Hearing loss not requiring amplification	Moderate, can hear and follow directions with amplification	No functional hearing with amplification

2.3.8 Additional Follow-up Assessments

Long-term respiratory outcomes will be collected at 22-26 months corrected age in order to evaluate early childhood pulmonary outcomes, specifically: recurrent wheezing and chronic coughing (CRF BB10).

2.4 Study Documentation

This study will follow good clinical practices (GCPs) according to the International Conference on Harmonization Guideline E6 (ICH E6) in conducting research (<http://ich.org>). Essential documents will be maintained at the clinical site. The ICH guidelines provide an overview of essential documents, which individually and collectively, permit the evaluation of the conduct of the trial and the quality of the data produced. The site is responsible for maintaining a Regulatory Binder or files for monitoring and auditing purposes and should follow the ICH guidelines and local site policy to establish an essential document system.

The Data Coordinating Center (RTI) will compile and maintain a listing of qualified (by training and certification) research personnel, their defined roles, contact information (email) and specific access rights (or restrictions) prior to study start. Centers are responsible for notifying RTI for any modifications to the initial listing. Additions /deletions will be granted as needed during the conduct of the study by contacting RTI with the appropriate information (NRN Technical Support (NRNTechSupport@rti.org)).

2.4.1 Regulatory and Administrative Study Binder

The BiB clinical research centers will follow investigator/institution responsibility for ICH E6(R1) 8.0 ESSENTIAL DOCUMENTS FOR THE CONDUCT OF A CLINICAL TRIAL

The minimum list of essential documents can be found in the following location <https://www.fda.gov/media/93884/download> and has been provided to the clinical centers via the Regulatory and Administrative Study Binder. The various documents are grouped in three sections per the stage of the trial during which they will normally be generated: 1) before the clinical phase of the trial commences, 2) during the clinical conduct of the trial, and 3) after completion or termination of the trial. A description is given of the purpose of each document, and whether it should be filed in either the investigator/institution or sponsor files, or both. It is acceptable to combine some of the documents, provided the individual elements are readily identifiable.

Any or all of the documents addressed in this guideline may be subject to, and should be available for, audit by the sponsor's auditor and inspection by the regulatory authority(ies).

NOTE: External to the Regulatory or Administrative Study Binder, all completed study forms must be kept in separate and secure Subject Study File per infant case.

2.4.2 Unmasking

If requested to unmask by the physicians caring for an infant, the pharmacist would be required to contact the center study PI and the overall study PI, Namasivayam Ambalavanan MD, UAB at Birmingham (cell 205-790-4817). Either the center study PI or the overall study PI will then contact NICHD Program Official Michele Walsh, MD (cell 216-570-1786 office 301-402-7886) before releasing this information. If unmasking does occur, the pharmacist will notify the DCC (NRNProtocolSupport@rti.org), the study coordinator and PI at their own center that unmasking has been requested and/or done. In such an event, a [Protocol Violations/Deviation Form \(BB06\)](#) should be

completed (CHAPTER 8). Parents requesting unmasking should be informed they can be told the treatment arm as soon as the study is over. However, sites should consult their local IRB and institutional policy (the same process above applies).

CHAPTER 3. STUDY SCREENING

3.1 General Form Completion

Liveborn infants (inborn/outborn) 1) between 401 grams and 1000 grams inclusive birth weight or 2) between 22 0/7 to 28 6/7 weeks gestational age and 3) less than or equal to 48 hours postnatal age or 4) antenatally consented mothers who did not deliver an infant in this window (as Infant birth order #1) should be entered on this CRF. Cases recorded on this [BB01 form](#) should have a corresponding [BB02 form](#) completed.

Appendix G: “Scenarios on How to Complete Forms” will help guide you on which forms should be completed given a scenario that others have encountered.

3.1.1 Dates and Times

For this study, all dates and times will be recorded in the following format:

Dates

All dates must be recorded in the day/month/year format (15/JUL/2020 as example)

Time

All times must be recorded in the 24-hour format (hr:min), 00:00 to 23:59.

3.1.2 Base Record and Header Information

Selected infant identifying information will be entered into the Medidata RAVE™ EDC. A Study Number will be obtained from the Medidata RAVE™ EDC for each infant screened. The following information is used to identify each infant and is placed as the heading of each study form:

Center Number

The assigned Network center number

Site Number

Centers with more than one participating hospital also have a site code

Study Number will serve as the unique infant identifier used for randomized and non-randomized but screened infants (inborn and outborn) and antenatally consented mothers who did not deliver an infant in this window to link different data forms across an individual case. A research staff member will log into the web-based EDC system (RAVE™) and select the ‘add a new patient’ button. RAVE™ will then generate **a five-digit (alphanumeric) Study Number (B + 4-digit family number + birth order)**. Antenatally consented mothers will default to the birth order number of 1. Multiple gestation infants will be randomized independently.

Network Number

The Network Number will be obtained in the web-based EDC system (Beacon) for the Generic Database (GDB) to link infant cases with registry data. (See [GDB MOP Section 3.1.4](#) for details)

Mother’s Initials

This is used at the convenience of the coordinators and will **not** be transmitted to RTI. Mothers should carry the same initials throughout the study even if her name changes.

3.2 Instructions for Completing Screening Log ([BB01](#))

The purpose of the Screening Log is to list each infant who is liveborn (inborn or outborn) between 401 grams and 1000 grams inclusive birth weight OR between 22 0/7 to 28 6/7 weeks gestational age

and less than or equal to 48 hours postnatal age. Additionally, this form tracks antenatally consented mothers who did not deliver an infant in this window.

The Network Coordinator, or her/his designee, enters daily onto the log the following information:

- 1. Infant's Name (Last, First)**
Record the infant's name using the format of last name first followed by the infant's first name. This **will not** be entered into the EDC.
- 2. Infant's Hospital Number**
Record the infant's medical record number or other identifying hospital number. This **will not** be entered into the EDC.
- 3. Last Date Eligible**
Record the final date of eligibility for the infant. Record as DD/MMM/YYYY e.g. 01/JAN/2020. This last date/time to enroll refers to the 48-hour enrollment window. Infants must be enrolled/randomized within 48 hours of birth but may be treated up to 50 hours. This **will not** be entered into the EDC.
- 4. Last Time Eligible**
Record the final time of eligibility for the infant. Record using 24-hour clock time HH:MM e.g. 18:00. This **will not** be entered into the EDC.
- 5. Consent (Y/N)**
Record Y if consent is signed otherwise record N. Antenatal consent is preferable so first study drug dose can be given at delivery if needed. If antenatal consent is not available, early postnatal consent can be obtained so that enrollment and randomization can occur ≤ 48 h postnatal age and intervention can be started in the window of ≤ 50 h postnatal age. This **will not** be entered into the EDC.
- 6. Enrolled/Randomized (Y/N)**
Enrolled = Randomized = Treatment kit opened. Infants will be considered enrolled and assigned to the randomization number when the masked drug kit is opened, which will occur after assessment of eligibility and directly before treatment administration. Record Y if a signed consent was obtained AND the infant was enrolled. If the infant was not enrolled record N.
- 7. Study Number**
Study Number is generated by the web-based EDC system (RAVE™) when study staff select the 'add a new patient' button. RAVE™ will then generate **a five-digit (alphanumeric) Study Number (B + 4-digit family number + birth order)**. Multiple gestation infants will be independently randomized and enrolled where eligible.
- 8. Comments**
This section may be used for any comments related to the infant.

A **BB02 Enrollment and Study Drug Initiation Form** must be completed for each infant who is listed on the **BB01 Screening Log**.

3.3 Instructions for Completing Antenatal Tracking Log (BB01Ante)

This log is used as a tracking tool for pregnant women with anticipated delivery between 22 0/7 – 28 6/7 weeks gestational age. This is optional and will not be entered into the EDC.

Aggregated numbers of antenatally screened mothers who decline consent will be tracked using a survey tool in the NRN Beacon EDC: <https://nrn.rti.org> by selecting the “BiB Antenatal Summary Count” that appears in the list after logging in and changing to survey mode.

Antenatally consented mothers who do not deliver an infant in the eligibility window and liveborn infants will be entered on form BB01 and corresponding BB02.

CHAPTER 4. ELIGIBILITY, CONSENT AND RANDOMIZATION

4.1 Eligibility

This is a randomized, masked, active controlled clinical study in which enrolled infants will receive budesonide + surfactant (intervention group) or surfactant alone (active control group) as soon as the clinical decision to give surfactant is made in the window of $\leq 48\text{h}$ postnatal age (study drug initiation can occur at ≤ 50 hours postnatal age). **Inclusion/exclusion criteria are further defined in section 4.6.2 Section B. Eligibility Criteria.**

4.2 Consent

Parents who are likely to deliver infants with GA 22 0/7 – 28 6/7 weeks **or** between 401 - 1000 grams (inclusive) birth weight who are not known to have any exclusion criteria may be approached for consent before delivery if allowed by the local IRB. Parents/guardians of infants who meet inclusion criteria and who do not have any exclusion criteria may also be approached for consent after delivery but must be enrolled and randomized up to and including 48h postnatal age.

Consenting should include informing families that the clinician may choose to not enroll the infant in the study, and that if enrolled, study drug may be administered in the delivery room by a respiratory therapist (or other qualified staff).

4.3 Randomization Procedures

Infants are stratified by study site and by gestational age (< 26 weeks; ≥ 26 weeks). The research pharmacy will receive a computer-generated code number (via drug kit) that will be linked to each enrolled infant upon randomization. Infants are considered **randomized** and assigned to the randomization number **when the masked drug kit is opened**, which will occur after assessment of eligibility, consent, and directly before treatment administration. Multiple gestations (if more than one is eligible) will be randomized independently. See **Chapter 3 Investigational Drug Supply/Preparation/Administration** of the [Investigational Drug Manual of Procedures](#) for specific details of randomization.

The infant study ID will be obtained from the research coordinator once parental permission is obtained and the infant is assessed for eligibility (or post randomization as needed for deliveries occurring quickly overnight or during the weekend).

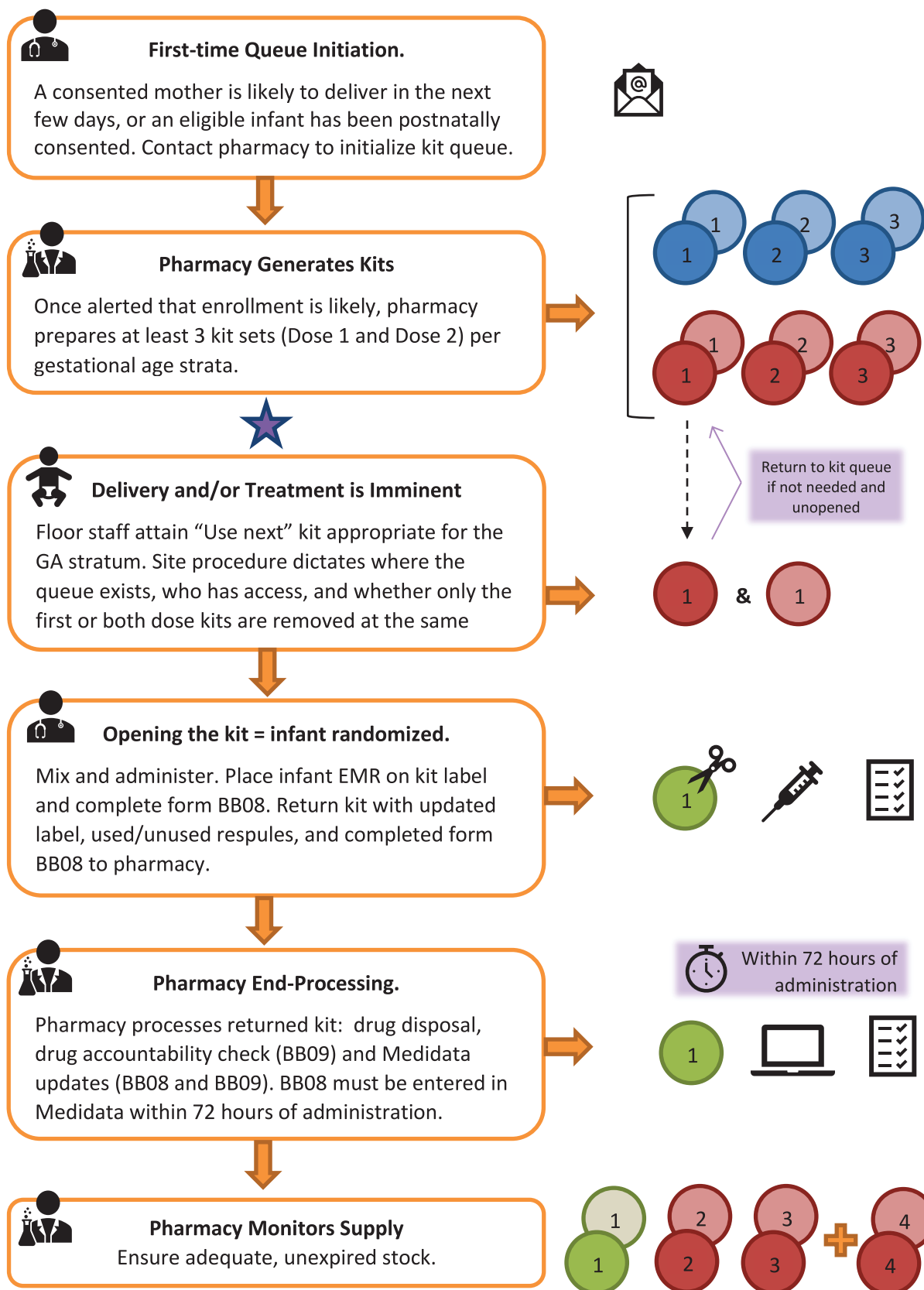
Pharmacy will perform end processing and record study drug disposal and accountability checks on the [Pharmacy Drug Accountability Form BB09](#) and enter these data in the Medidata EDC <https://www.imedidata.com>.

4.4 Study Intervention/Methods

After informed consent and assessment of eligibility, infants are randomized 1:1 to either the intervention arm: surfactant (poractant alfa; Curosurf) 2.5 ml/kg mixed with 0.25 mg/kg budesonide (Pulmicort nebulizing suspension) or the active control arm: Curosurf 2.5 ml/kg. If the infant is to receive the second dose of study drug, the dosage of Curosurf will be 1.25 ml/kg for the second dose (noting that the first dose of Curosurf is 2.5 ml/kg, regardless of whether a second dose is anticipated or not). A maximum of two doses of study drug can be administered, with specific criteria for the second dose at $\leq 50\text{h}$ postnatal age. The corresponding budesonide + surfactant or surfactant alone will be administered by an unmasked respiratory therapist or other qualified person in participating neonatal

ICUs, while other caregivers will remain masked to treatment group allocation. See **Chapter 3 Investigational Drug Supply/Preparation/Administration** of the *Investigational Drug Manual of Procedures* for specific details of masked study drug administration.

4.5 Study Initiation Flow



4.6 Instructions for Completing Enrollment and Study Drug Initiation Form (BB02)

This form should be completed for all cases entered on the [BB01 Screening Form](#). If the infant is randomized, with or without receiving study drug, complete all GDB and BiB forms as indicated in the study MOP (refer to Appendix C).

Complete each section before moving down to the next section. The form is designed to be completed in alphabetical order of sections.

4.6.1 Section A. Consent

Complete this section for all cases entered on BB01 study. [Note, this will always be marked YES for antenatally consented mothers]

1. Consent Status

Record 'Y' if a signed consent was obtained.

- a. If "Y" recorded select appropriate code for when consent was granted.
1 = Antenatal
2 = Postnatal
- b. Date (required) and Time of consent (if available):
 1. Record the date of consent: Using the following format dd/mmm/yyyy format. Note that the day begins at 00:00 and ends at 23:59.
 2. Time of consent if available: Record the time using the 24-hour clock format (00:00-23:59).
- c. If "N" is recorded. Select the appropriate Consent Status for this subject from the boxed options.
0 = Not eligible
1 = Parent/Guardian refused consent
2 = No parental/guardian decision. This option is for the parents who defer decision making repeatedly until the infant is too old to participate. This is often perceived by staff as a passive form of refusal to participate in the study.
3 = Parent/Guardian unavailable
4 = Parent/Guardian not approached
5 = Physician refused study participation
6 = Other (specify
7 = COVID-19 Institutional policy
 - a. If Code '2', '3', '4', '5' or '6' specify reason.

4.6.2 Section B. Eligibility Criteria

Complete this section for all cases entered on the BB01. All inclusion criteria must be marked "YES" and all exclusion criteria must be marked "NO" for the infant to be eligible. If antenatally consented mother's deliver outside the eligibility window, select "NO" to access and select the appropriate failed inclusion criteria.

1. Did the subject meet all eligibility criteria?

Record 'Y' if the infant satisfies all of the inclusion criteria AND does not meet any of the exclusion criteria. All inclusion criteria must be marked 'Y' for the infant to be eligible. All exclusion criteria must be marked 'N' for the infant to be eligible. Record 'N' otherwise.

If Yes to Q.A.1 and Q.B.1, proceed to Section C. Initiation of Study Drug

If No, select criteria for ineligibility and proceed to Section D. Form Completion

a. What criterion code(s) did they fail? (Enter all that apply)

Select from the drop-down menu the code for all inclusion criteria that were **NOT** met.

[Note: This includes antenatally consented mother's delivering outside of eligibility].

Select the code for all exclusion criteria that WERE met. Select ALL codes that apply.

NOTE: The form [BB02](#) includes space to mark 'Y' or 'N' to indicate whether the inclusion or exclusion criterion was met. However, the space to mark individual responses for inclusion and exclusion criteria are only for worksheet purposes.

NOTE: If a baby is consented antenatally but then lost to follow-up before delivery please mark 'N' for Live Born (IN001)

INCLUSION CRITERIA CODES

IN001 = Liveborn

IN002 = Infant's gestational age is between 22 0/7 - 28 6/7 weeks at birth or between 401 - 1000 grams (inclusive) birth weight

Record 'Y' if infant is between 22 0/7 - 28 6/7 weeks gestational age at birth or between 401 - 1000 grams (inclusive) birth weight

IN003 = Clinical decision to give surfactant

Record 'Y' if clinical decision to administer surfactant.

IN004 = Infant is $\leq$ 48 hours postnatal age at the time of randomization

Record 'Y' if infant postnatal age $\leq$ 48 hours at the time of randomization.

Record 'N' if otherwise.

EXCLUSION CRITERIA CODES

EX001 = Infant is terminally ill (heart rate $<$ 100 beats per minute, unresponsiveness to resuscitation) or unlikely to survive as judged by the clinician

Record 'Y' if in the clinician's opinion, the infant is terminally ill or unlikely to survive.

Record 'N' otherwise.

EX002 = Decision to redirect or limit support

Record 'Y' if the decision was made to provide comfort/palliative care and not full treatment. Record 'N' otherwise.

EX003 = Use of surfactant before enrollment

Record 'Y' if the infant received at least 1 dose of surfactant prior to enrollment (defined as randomization = study drug kit opened). First dose of surfactant must be study drug. Record 'N' otherwise.

EX004 = Previously received systemic steroids or indomethacin

Record 'Y' if the infant received systemic steroids or indomethacin prior to randomization. Record 'N' otherwise.

EX005 = Use of indomethacin

Record yes if indomethacin use by either the mother < 24 hours prior to delivery, the infant prior to enrollment, or intent to administer to the infant < 168h post study drug. Record 'N' otherwise.

EX006 = Serious chromosomal abnormalities or major malformations

Record 'Y' if the infant has known complex congenital heart disease, CNS anomalies, chromosomal and non-chromosomal syndromes known to affect neurodevelopmental outcomes. Refer to Appendix D for a list of major congenital/chromosomal anomalies. Record 'N' if no major congenital or chromosomal anomalies are identified prior to randomization.

EX007 = Known congenital infections included but not limited to confirmed sepsis, congenital CMV

Record 'Y' if the infant has confirmed congenital infection. Record 'N' if no confirmed congenital infection.

EX008 = Permanent neuromuscular condition that affects respiration

Record 'Y' if the infant has known neuromuscular condition that affects respiration. Permanent neuromuscular condition that affects respiration. This is not always related to a central nervous system defect and could be related to myopathy or congenital myasthenia, etc. Refer to Appendix D for a list of major congenital/chromosomal anomalies. Record 'N' if no neuromuscular conditions that affects respiration are identified prior to randomization.

EX009 = Infant enrolled in a conflicting trial

Record 'Y' if the infant is enrolled in a conflicting trial at the time of determining eligibility. Record 'N' otherwise.

4.6.3 Section C. Initiation of Study Drug

The Medidata RAVE™ EDC will prompt users with the following question:

1. Date and time of birth

- a. **Record the date of birth:** Using the following format dd/mmm/yyyy format. Note that the day begins at 00:00 and ends at 23:59.
- b. **Time of birth:** Record the time using the 24-hour clock format (00:00-23:59).

2. Was the infant enrolled (randomized)?

Record 'Y' if the study kit selected was opened, regardless of whether or not study drug was administered. Record 'N' only if a study drug kit remained unopened.

Note: If a randomized infant is unable to receive the per-protocol dose due to damaged/unavailable respules, the administration (or randomization and lack of administration) will be reported as a protocol deviation.

a. Record Gestational Age (GA) to confirm strata for randomization

Record the gestational age at birth as indicated in the eMR. This is used to confirm the correct study kit was obtained and/or study drug prepared and administered as stratified by gestational age.

Note: Infants outside the gestational age strata but eligible based on birth weight will still be stratified by gestational age i.e., < 26 weeks; ≥ 26 weeks.

Note: If a kit is removed from its secure location antenatally in anticipation of a delivery and the GA stratum of the infant changes before actual delivery (e.g. labor on 25 6/7 weeks GA that extends past midnight to 26 0/7 weeks GA) this should be documented as a protocol deviation.

b. Record Randomization Date

Record the randomization date in DD/MMM/YYYY format as documented by the unmasked personnel who randomized the infant (opened the study drug kit).

If IDS pharmacy prepares and dispenses the study drug in a masked syringe, the date of randomization would equate to the date the study drug was dispensed to the floor.

c. Record Randomization Number

Record the six-digit randomization number as identified on the masked study drug kit label.

d. If No, record reason infant not randomized

Code the single best reason the infant was not randomized from the codes listed on BB02 under question C.3.a.

Codes for Question C.2.d and C.3.a:

- 1 = Infant became too medically unstable to participate
- 2 = Unmasked study personnel not available
- 3 = Study drug not available
- 4 = Parent(s) decision not to proceed with initiation of study drug (withdrawal of consent)
- 5 = Clinician decision not to proceed with initiation of study drug
- 6 = Other (specify): _____

3. Was study drug administered?

Record 'Y' if study drug was given. Record 'N' if otherwise. Study must be initiated ≤ 50 hours postnatal age. ***If study drug initiation cannot be started at ≤ 50 hours postnatal age, complete either a protocol deviation/violation form (BB06) or withdrawal form (BB07)***

a. If No, record reason study drug not given

Code the single best reason the infant was not randomized from the codes listed on [BB02](#) under question C.3.a.

If YES,

Codes for Question C.2.d and C.3.a:

- 1 = Infant became too medically unstable to participate
- 2 = Unmasked study personnel not available
- 3 = Study drug not available
- 4 = Parent(s) decision not to proceed with initiation of study drug
(withdrawal of consent)
- 5 = Clinician decision not to proceed with initiation of study drug
- 6 = Other (specify): _____

b. Was study drug/placebo started less than or equal to 50 hours of life?

Record 'Y' if study drug was given less than or equal to 50 hours postnatal age. Record "N" if otherwise and code the reason study drug was not given within the study specified timeframe. Complete applicable [BB06](#)

- 1 = Drug delayed for medical indication
- 2 = Problem with kit contents
- 3 = Unmasked study personnel not available
- 4 = Study drug administration delay at bedside
- 5 = Other

If Other, specify reason study drug was not started $\leq$ 50 hours postnatal age

Note: Study drug should NOT be given > 50 hours postnatal age even if randomized

4.6.4 Section F. Form Completion

1. Initials of person completing this form.

CHAPTER 5. RESPIRATORY STATUS/LABORATORY MEASURES

5.1 Overview

This chapter records the type and magnitude of respiratory support needed by the infant in the first week after birth along with associated blood gas values and blood glucose values during the same time period.

5.1.1 Instructions for Completing Respiratory Status/Laboratory Measures Form (BB03)

This form will be completed for all enrolled infants.

Section A. Respiratory Support/Vital Sign/Blood Gas/Glucose Measures

Record respiratory support and vital sign measures **q12hr x 7 days (168 hours of life)**. Record all respiratory support from birth (baseline) regardless of when study drug is initiated. Baseline measures are from the initial NICU admission for infants who receive study drug in the delivery room. For outborn infants, data collection may begin on NRN NICU admission unless transfer charts contain these data.

1. Date

Record the date the respiratory measures were obtained. Daily measures to start with unit policy whether routine monitoring begins at 0700 or 0800 (or other timeframe) daily.

2. Time

Record at 12-hour interval time points (+/- 1hr) through DOL 7 following unit practice e.g. 07:00, 19:00.

3. Mode of Respiratory Support

Code the mode of respiratory support the infant is receiving at the interval time point during the study day.

0= None

1= HFV/HFOV

2= CV/IMV/SIMV

3= CPAP/SiPAP (includes nasal CPAP, nasal positive pressure ventilation)

4= Hood oxygen >21%

5= Nasal cannula oxygen >21% and/or ≤ 2.0 lpm (FiO₂ is not required for this code)

6 = High flow nasal cannula > 21% and > 2.0 lpm. Note: lower flow (e.g. 1.5 L), should be documented under Code 5.

For infants on HFOV record Hertz and ΔP in lieu of rate and PIP, respectively, and no PEEP. The table below indicates which values are expected for a specific respiratory support code. If an expected respiratory code is not available, enter permanently missing in that field. If a value is not expected, you may leave that field blank or enter a value if available.

	Expected for	NOT Expected for
4. FiO₂	1, 2, 3, 4, 6	0, 5
5. Rate or Hz	1, 2	0, 3, 4, 5, 6
6. PIP or ΔP	1, 2	0, 3, 4, 5, 6

7. PEEP	2, 3	0,1, 4, 5, 6
8. MAP	1	0, 2, 3, 4, 5, 6

9. SpO2

Record the infant's oxygen saturation at the interval time point during the study day. Disregard any temporary desaturation episodes, apnea, bradycardia or procedures where the infant returns to his/her previous saturation in a reasonable amount of time (≤ 30 minutes).

10. Heart Rate (HR):

Record the heart rate at the interval time point during the study day. Disregard any temporary desaturation episodes, apnea, bradycardia or procedures where the infant returns to his/her previous heart rate in a reasonable amount of time (≤ 30 minutes).

11. Respiratory Rate (RR)

Record the true respiratory rate as seen visually or on the monitor at the interval time point during the study day. If the infant is intubated, then the RR will be ventilated + spontaneous respiration. Disregard any temporary desaturation episodes, apnea, bradycardia or procedures where the infant returns to his/her previous respiratory rate in a reasonable amount of time (≤ 30 minutes).

12. Mean Blood Pressure (BP)

Record the mean BP at the interval time point during the study day. Preference is to record arterial measures over cuff. When collecting MAPs, choose the reading closest to the 12-hour timepoint. i.e., if the BP is measured at 10 am then that value would be used for the 12 pm entry if your unit policy uses that 12-hour period.

Clinical Blood Gas: Record clinically obtained blood gas and glucose results +/- 4 hours closest to Respiratory measures at the stated timepoint.

13. Blood Gas Performed?

Record "Yes" if a blood gas was obtained within the window and complete measures as indicated. Record "No" otherwise and this column/section will be complete.

If Yes, complete the following questions 13a-13i:

13a. Date

Record the date the clinical blood gas was obtained.

13b. Time Measured

Record the time the blood gas was obtained (as opposed to reported). Record the measure closest to (+/- 4 hours) the Respiratory and Vital sign measures for DOL 1 – 7.

13c. Source

Record the source from which the blood gas was obtained:

- 1 = Arterial
- 2 = Venous
- 3 = Capillary
- 4 = Unknown

13d. Mode of Respiratory Support at the time of blood gas collection

Record the mode of support at the time the clinical blood gas measure was obtained (drawn)

0= None

1= HFV/HFOV2= CV/IMV/SIMV

3= CPAP/SiPAP (includes nasal CPAP, nasal positive pressure ventilation)

4= Hood oxygen >21%

5 = Nasal cannula oxygen >21% and/or ≤ 2.0 lpm (FiO₂ is not required for this code)

6 = High flow nasal cannula > 21% and > 2.0 lpm. Note: lower flow (e.g. 1.5L), should be documented under Code 5.

13e-i Blood Gas Values. Record the following measures

d. pH

e. pCO₂

f. pO₂

g. HCO₃

h. FiO₂

Clinical Blood Glucose: Record clinically obtained glucose results +/- 4 hours closest to Respiratory measures at the stated timepoint.

14. Clinical Glucose Performed?

Record "Yes" if a clinical glucose was obtained within the window and complete measures as indicated. Record "No" otherwise and this column/section will be complete.

14a. Date

Record the date the clinical glucose measure was obtained using the dd/MMM/yyyy format.

14b. Time Measured

Record the time the clinical glucose labs were obtained (drawn)

14c. Glucose (mg/dL)

Record the results to the clinically obtained glucose.

14d. Method

Record the method used to obtain the clinical glucose. If 2 or more measures using different methods are collected during this time point, document the measure obtained centrally. If 2 measures using the same method are collected during this timepoint, record the one closest (+/-) to the stated 12hr mark.

Record the following collection method

a. Point-of-care (POC)

b. Central (Cnt)

c. Unknown (Unk)

Section B Intubation/Extubation

Complete **Table B** for all intubations/extubations from baseline (birth) through 3 Days of Life (72 hours). Subsequent data will be collected per the GDB NG07. Note: Less Invasive Surfactant Administration (LISA) does NOT count as intubation and is not currently permitted for surfactant administration in BiB.

1. Date of intubation/extubation

Enter the date of the event using the dd/MMM/yyyy format.

2. Time of intubation/extubation

Enter the time of the event using the 00:00 format.

3. Action

Enter the action, either intubation or extubation. Record each action in a separate column regardless of date (i.e., record all actions on any given date).

1= Intubation

2= Extubation

4. Mode of Respiratory Support

Record the respiratory settings just after intubation, and settings just after extubation.

0= None

1= HFV/HFOV

2= CV/IMV/SIMV

3= CPAP/SiPAP (includes nasal CPAP, nasal positive pressure ventilation)

4= Hood oxygen >21%

5= Nasal cannula oxygen >21% and/or $\leq$ 2.0 lpm (FiO₂ is not required for this code)

6 = High flow nasal cannula > 21% and > 2.0 lpm.

5. Reasons for Intubation:

Select all reasons for intubation that apply per event.

0 = Planned Extubation

1 = Surfactant administration

2 = Apnea / hypoventilation/hypercapnia

3 = Increased respiratory effort

4 = Sepsis/Possible sepsis

5 = Atelectasis

6 = Required for procedures

7 = Upper airway abnormality

8 = Self/Unplanned

9 = Plugged tube

10 = Other reasons (Specify)

Section C Adrenocortical Insufficiency**1. Was the infant diagnosed with adrenocortical insufficiency at any time through 15 days (360 hours) after birth?**

Record "Yes", if in the medical records there is a documented diagnosis to adrenal insufficiency.

Record "No" otherwise and this section is complete.

2. If yes, was a clinical cortisol value recorded?

Record "Yes", if in the medical records there is a documented cortisol value. Record "No" otherwise and this section is complete.

a. If yes, date and time of lowest cortisol level

Enter the date of the event using the dd/MMM/yyyy format. Enter the time of the event using the 00:00 format.

b. Lowest cortisol level (mcg/dL)

Record the lowest cortisol level in mcg/dL. If result is reported to 2 decimal digits (hundredths), round to tenths (one decimal)

Section D. Postnatal Steroids

1. Was infant given postnatal steroids for treatment of evolving Bronchopulmonary Dysplasia/Chronic Lung Disease from 7 days post final dose of study drug through 36 completed weeks PMA?

Record "Yes or No"

CHAPTER 6. CONCOMITANT MEDICATION

6.1 Overview

Based on a medication order (first and last dose delivered within that order), record all medications given intravenously, intra-arterially, intramuscularly, subcutaneously, intratracheally, inhaled (example: inhaled steroids, bronchodilators or iNO), as eye drops, or enterally. Non-systemic medications (topical ointments) will be excluded.

Medication Course: For each course of medication (example: An antibiotic is given on DOL 3 x 72 hours), record the date of the first and last dose for this episode. If the same medication is started again later (e.g., antibiotic is started again in 5 days), record the medication again and record the date of the first and last dose for this episode.

Bolus/Maintenance Medications: Treat bolus/maintenance medications as a course if given consecutively (at least daily). If there are separate orders and/or a break in delivery of the same medication, record separate entries.

PRN: Treat prn medications as a course if given consecutively (at least daily). Once there is a break in delivery, record that as the stop date. If the prn order still stands and more medication is given 2 days later, treat that as a new course.

TPN: Medications added to parenteral nutrition (examples would be iron, erythropoietin, etc.) will be considered medications (refer to Appendix F : Nutrition vs Medications).

Single Dose: Single dose medications will be documented by recording the same date in the 'Start Date' and 'Stop Date' column.

6.1.1 Instructions for Completing Concomitant Medication Form (BB04)

Complete this form for each infant who is randomized and receives study drug. Document all concomitant medications, supplements, and vitamins, other than study drug ([BB04 Section A](#)). The start and stop times for each exposure are recorded from birth through 7 days (168 hours) post final dose of study drug or discharge or transfer if sooner. Postnatal steroids for treatment of evolving Bronchopulmonary Dysplasia (BPD)/Chronic Lung Disease (CLD) will be collected from 7 days post final dose of study drug through 36 completed weeks PMA ([BB04 Section B](#)).

1. Medication Name

Record the name of the medication given. The iMedidata RAVE EDC will provide a dropdown list for medication selection. Generic names (rather than trade names) are to be used whenever possible. Refer to Appendix F .

2. Start Date

Record the date of the first dose of each medication administered as documented in the medical records. If part or all of start or stop date is unknown then report as much of the date that is known and report Unknown for the remaining portions.

3. Stop Date

Record the date of the last dose of each medication administered as documented in the medical records. If part or all of start or stop date is unknown, then report as much of date that is known and report Unknown for the remaining portions.

4. Continuing

For medication continued > 7 days post last study drug dose or continued at transfer or discharge to home and this information is documented in the medical record check “continuing”.

5. Unknown

For medication continued > 7 days of life or continued at transfer or discharge to home and this information is not documented in the medical record check “unknown” (UNK).

6. Comments

Record comment as necessary.

CHAPTER 7. ADVERSE EVENTS

7.1 Adverse Event (AE) / Serious Adverse Events (SAE)

Definition of Events

7.1.1 Adverse Events (AE)

An **adverse event (AE)** is defined as any untoward medical occurrence in a research subject temporally associated with the use of a drug in humans, whether or not considered drug related. Adverse Events occurring during the safety monitoring period (from treatment initiation through 7 days (168 hours) post final study drug discontinuation) are recoded on the [Adverse Events Form BB05](#). Any worsening of a pre-existing condition is considered an adverse event. Common neonatal morbidities and outcomes occurring after the study-defined AE monitoring period are only recorded in lesser detail on the NRN Generic Database (GDB). Laboratory abnormalities and changes in vital signs are considered to be adverse events only if they meet protocol-specific criteria. Refer to Appendix A Toxicity Table for guidance on adverse event severity.

7.1.2 Serious Adverse Event (SAE)

A **serious adverse event (SAE)** is defined as any adverse event that results in any of the following outcomes:

Outcome	Additional Information/Description
Death	
Life-Threatening AE	An event that, in the opinion of the Investigator, would have resulted in immediate fatality if medical intervention had not been taken. This does not include an event that would have been fatal if it had occurred in a more severe form.
Inpatient hospitalization or prolongation of an existing hospitalization	
Results in persistent or significant disability/incapacity	An event that results in a substantial disruption or impairment in ability to conduct activities of daily living (ADL) and impacts quality of life (QOL).
Other Important Medical Event	Any other AE that a study investigator judges to be serious, as it may jeopardize the participant or require medical or surgical intervention to prevent one of the above listed SAE outcomes above.

Note: Based on the premature infant population studied, the serious adverse event categories that result in congenital anomaly/birth defect or require intervention to prevent permanent impairment or damage (devices), or require hospitalization, do not apply when defining events for these infants.

7.2 Collection of AEs and SAEs

7.2.1 Adverse Event Collection

Events will be collected via medical assessment and medical record review. Events will be documented on the adverse event CRF (BB05) and entered into iMedidata RAVE EDC system. The study AE Log is to be completed for every participant who initiates study drug. Adverse event (AE) monitoring is from treatment initiation through 7 days (168 hours) post final study drug discontinuation and then continuous monitoring within 30 days of final study drug administration for adverse events of special interest (e.g. spontaneous intestinal perforation (SIP) or Periventricular Leukomalacia (PVL)) to ensure timely detection of events that may affect the safety or continued participation of research subjects. The following are pre-specified events to be monitored (see Appendix A Toxicity Table Severity/Intensity Grading Scale). The adverse event CRF requires recording of the date and time of onset and resolution for each adverse event, (or documentation that it was not resolved) from initial study drug administration through 7 days (168 hours) after final discontinuation of study drug.

Sites should be aware of acute adverse events monitored for this study – Please refer to Toxicity Table Appendix A).

Do not report any mild categories unless they are also unexpected.

7-Day Post Study Drug Safety Monitoring

Endotracheal tube blockage:

- Moderate category: Endotracheal tube occlusion requiring tube replacement.
- Severe category: Endotracheal tube occlusion requiring tube replacement, associated with prolonged hypoxemia and bradycardia (HR<80 for >30s, with SpO2 <80 for >30s).
- Life-threatening category: Endotracheal tube occlusion requiring tube replacement and resuscitation with chest compressions.

Prolonged hypoxemia and bradycardia associated with study drug administration:

- Moderate category: Not applicable.
- Severe category: Heart rate less than 80 beats per minute for at least 30 seconds, in association with a pulse oximetry reading of less than or equal to 80% for at least 30 seconds.
- Life-threatening category: Same as grade III (severe) but with slow response or no response to treatment.

Pulmonary Air leak:

- Moderate category: Pneumothorax with increased observation or needle aspiration OR PIE on x-ray.
- Severe category: Pneumothorax with chest tube placement or PIE with use of high frequency ventilation.
- Life-threatening category: Chest tube placed; on invasive ventilation with O2 >80%.

Hypotension:

- Moderate category: Use of a single vasoconstrictor agent or a single fluid bolus (20 ml/kg or less in 4-hour period).
- Severe category: Use of two vasoconstrictor agents or fluid boluses >40 ml/kg in 4 hours or single vasoconstrictor PLUS a fluid bolus 20 ml/kg or more in a 4-hour period.
- Life-threatening category: Sustained decrease in BP requiring at least two vasoconstrictor agents for 12 hours.

Hypertension:

- Moderate category: A mean blood pressure (2-3 determinations over at least 12 hours) which is >95thile but without use of antihypertensive medication.
- Severe category: A mean blood pressure (at least 4 determinations over at least 12 hours) which is >95thile or any use of antihypertensive medication.
- Life-threatening category: Sustained increase in BP >95thile for at least 12 hours requiring multiple treatment types or prolonging hospitalization.

Hyperglycemia:

- Moderate category: 120-180 mg/dL on 2 consecutive determinations at least 6 hours apart, or more than 2 determinations within a 24-hour period with no in between normal range values.
- Severe category: > 180 mg/dL on 2 consecutive determinations at least 6 hours apart, or more than 2 determinations within a 24-hour period with no in between normal range values. Or initiation of insulin.
- Life-threatening category: > 600 mg/dL on 2 consecutive determinations at least 4 hours apart, or more than 2 determinations within a 24-hour period with no in between normal range values.

Intracranial Hemorrhage (+ 3-day window to receive report)

- Moderate category: Unilateral or bilateral blood/echodensity in the ventricle without distention of the ventricle and without a midline shift; if subarchnoid or subdural blood without a midline shift is detected, it will be included in this category.
- Severe category: Unilateral or bilateral blood/ echodensity filling the ventricle(s) resulting in distension without parenchymal bleeding and without a midline shift.
- Life-threatening category: Parenchymal hemorrhage/ echodensity (including cerebellar hemorrhage) and / or any bleed/ echodensity associated with a midline shift.

Sepsis (culture positive): ☐ EOS ($\leq$ 72 hours postnatal age) or ☐ LOS (>72 hours postnatal age)

- Moderate category: Treated > 5 days of consecutive antibiotics/virals/fungals, no cardiovascular instability.
- Severe category: Treated for systemic symptoms and with >5 days with anti-infective agents, and additional symptomatic support needed, responded to treatment.
- Life-threatening category: Same as grade III (severe) but with cardiovascular instability and slow response or no response to treatment.

Other events to include:

- Moderate Category: The adverse event causes the subject discomfort or interrupts the subject's usual function.
- Severe Category: The adverse event causes considerable interference with the subject's usual function.
- Life Threatening Category: The adverse event put the subject at substantial risk of dying at the time of the adverse event.

30-Day Post Study Drug Safety Monitoring

Spontaneous intestinal perforation, without NEC: Moderate category: Not applicable

- Severe category: Intestinal perforation/free air without pneumatosis or bowel necrosis; no shock or respiratory deterioration or hypotension and in the absence of a known gastrointestinal malformation or therapeutic misadventure.
- Life-threatening category: Intestinal perforation/ free air; systemic symptoms including, but not limited to tachycardia, hypotension, shock, respiratory deterioration and/or placement of drain or lap.

Periventricular Leukomalacia (PVL):

- Moderate category: Per radiology report indication of moderate PVL.
- Severe category: Per radiology report indication of extensive PVL.
- Life-threatening category: Not applicable

7.2.2 Serious Adverse Event Collection

If the event meets serious criterion the site must also provide a narrative describing the event, treatment and diagnostic details ([BB05 Section B](#)). The following reporting timelines should be used:

- All deaths and immediately life-threatening events, whether related or unrelated, will be recorded on the SAE Form and submitted to the NICHD and the DCC within 24 hours of site awareness (see Section 7.8.1 Section A. Adverse Events (AE) for reporting procedures to NICHD). Reporting to local IRB is per institutional policy. Such events must also be entered into the iMedidata RAVE EDC within 24 hours of site awareness. Enter the event with as much detail as you have available at the time of awareness. It can be updated.
- Other SAEs, regardless of relationship, will be submitted to the DCC and NICHD project scientist within 72 hours of site awareness. This event must also be entered into the iMedidata RAVE EDC within 72 hours of site awareness. Enter the event with as much detail as you have available at the time of awareness. It can be updated.

All SAEs will be followed until satisfactory resolution or until the site investigator deems the event to be chronic, observed to be stable or until 36 completed weeks PMA (study status) whichever comes first. Other supporting documentation of the event may be requested by the DCC or NICHD and should be provided as soon as possible.

7.2.3 Expedited Reporting

The study Sponsor (or designee per the Transfer or Regulatory Responsibility (TORO)) is responsible for notifying the FDA of any unexpected and fatal or life-threatening suspected adverse event as soon as possible but in no case later than 7 calendar days after the information is submitted by the clinical site. NICHD, in collaboration with the study PI, will determine if individual serious adverse events a) meet the criteria for expedited reporting in accordance with 21CFR 312.32 IND safety reporting and b) if so, are promptly reported to the FDA and the local IRB according to their policies.

All adverse events that meet mandatory FDA reporting criteria are those that are:

- Serious; and
- Unexpected (unanticipated); and
- Definitely, probably, or possibly related to the study drug.

The DCC safety desk will provide regulatory support to facilitate electronic reporting to SAEs to the FDA.

7.3 Attribution/Causality Assessment (AE/SAE Relationship to Study Intervention)

The investigator will assess the relationship of an AE/SAE to the study intervention(s). Relationships will be recorded as:

Attribution/Causality	Additional Information/Description
Not related	The AE is completely independent of study intervention and/or evidence exists that the event is definitely related to another etiology. There must be an alternative, definitive etiology documented by the clinician
Unlikely	A clinical event, including an abnormal laboratory test result, whose temporal relationship to study intervention(s) makes a causal relationship improbable (e.g., the event did not occur within a reasonable time after administration of the trial medication) and in which other drugs or chemicals or underlying disease provides plausible explanations (e.g., the participant's clinical condition, other concomitant treatments).
Possibly related	There is some evidence to suggest a causal relationship [e.g., the event occurred within a reasonable time after study intervention(s)]. However, other factors may have contributed to the event (e.g., the participant's clinical condition, other concomitant events). Although an AE may rate only as "possibly related" soon after discovery, it can be flagged as requiring more information and later be upgraded to "probably related" or "definitely related," as appropriate.
Probably related	The study agent and/or placebo is more likely the cause of the adverse event than other factors.
Definitely related	The adverse event can be fully explained by administration of the study intervention and/or placebo (where applicable).

The causality assessment from both the NICHD and the study site investigator will be reported in the SAE narrative. SAEs that are at least possibly related to study will be submitted to the Data Safety Monitoring Committee (DSMC), regardless of expectation, under expedited criteria, by the DCC.

7.4 Determining Expectedness

The PI or study investigator will assess expectation for all adverse events and should use the following guidance. NICHD in collaboration with the NRN DSMC will make independent expectation assessments as needed for SAEs only.

An **unexpected** adverse event is any adverse event occurring in one or more subjects participating in a research protocol, the nature, severity, or frequency of which is **not** consistent with either:

- The known or foreseeable risk of adverse events associated with the procedures involved in the research that are described in (a) the protocol-related documents, such as the IRB-approved research protocol, any applicable investigator brochure, and the current IRB-approved informed

consent document, and (b) other relevant sources of information, such as product labeling and package inserts; or

b. The expected natural progression of any underlying disease, disorder, or condition of the subject(s) experiencing the adverse event and the subject's predisposing risk factor profile for the adverse event.

7.5 AE Severity/Grade

The clinical research team should use the grading scale outline in the Toxicity Table (Appendix A):

The term "severe" is a measure of intensity and not all severe AEs are serious. However, a severity rating of life-threatening (4) or death (5) will always be considered serious and should be reported as an SAE.

7.6 Recording Procedures for AEs

The site will be responsible for entering all AEs into iMedidata RAVE. In order to warrant reporting, an event must be new [i.e. must have started after the initiation of study intervention(s)] or, if present prior to entry into the study, must be worsening in severity (or frequency) or deemed serious. The site Principal Investigator will review each AE to assess its severity, possible relationship (attribution) to study intervention, expectedness and actions required at a minimum.

If symptom(s) are reported initially, the AE should be updated to reflect the final diagnosis. Only one verbatim term/event/condition should be reported/recorded per CFR/EDC field. Report the full duration of the AE and ultimately record the highest grade of severity/seriousness that occurred during the entire event (i.e. the start and end date would include the entire period of AE, regardless of grade changes).

If an AE occurs, returns to normal (resolves), and then reoccurs later, report as separate events.

If existing multiple single AEs are already entered into Medidata to reflect one chronic AE as determined by the clinical research team; then research staff would delete all rows except one and edit the start and/or end date accordingly. If the grades were not the same, use the highest grade that occurred during the entire AE period.

7.7 Recording Procedures for SAEs

Serious Adverse Event (SAE)

If an SAE occurs or an AE becomes a SAE, the same convention described above applies. Report the full duration of the AE and record only the highest severity/intensity grade that occurred during the time period (i.e. the start and end date would include the entire period of AE, regardless of grade changes, with the designation of only the highest grade during that time.)

The clinical study team will be responsible for evaluating all information pertaining to a serious adverse event (SAE) (particularly hospital records and discharge summaries, as applicable) and entering into the Medidata RAVE EDC, an appropriate summary of any relevant information specific to the reported SAE. Unless otherwise requested, the DCC does not expect receipt of source documentation from the investigator sites as a standard process. If however, source documentation is required by reviewing monitors, the DCC will contact the investigator site directly for required information.

The DCC will be notified of Serious Adverse Events (SAEs) via receipt of an electronic data capture (EDC) trigger from iMedidata RAVE. If clarification and/or additional information is required, the Clinical Safety Specialist (CSS) or designee will enter safety related queries within Medidata RAVE for resolution by the investigator site. All query responses should be entered into Medidata RAVE. Once the query is resolved, the CSS or designee will close the safety query within Medidata RAVE.

In the event that the EDC is not available due to system failure for more than 24-hours, the Investigator will submit a completed paper SAE Report Form (BB05) to the DCC Central Safety Desk inbox: RTI-Safety-Desk@rti.org. Once Medidata RAVE is functional, the site is required to enter the SAE data as appropriate, within 24-hours.

The DCC CSS, or designee, will send an Acknowledgement of Receipt (AOR) of the back-up paper SAE Report Form to the reporting site. If no AOR is received within 24 hours, please send an alert email to RTI-Safety-Desk@rti.org and copy the DCC protocol coordinator NRNProtocolSupport@rti.org.

7.7.1 Reporting Criteria for Initial AE/SAE Reporting

To define the official receipt date of a SAE report and complete an initial report, several pieces of information for the event are required. However, the site should report as much information as they have available at the time the site becomes aware of an adverse event. The data can be updated in iMedidata RAVE. Those requirements are:

1. Adverse Event Awareness Date
2. Adverse Event Term
3. Date of Event Onset
4. Date of Event End (leave blank if ongoing)
5. Severity (Toxicity Grade)
6. Relationship to Study Intervention(s) (causality, per site investigator)
7. Action (intervention) Taken
8. Final Outcome
9. Expectedness
10. Continuing/Ongoing Status
11. Seriousness
12. Comments

7.8 Instructions for Completing Adverse Event (AE) and Serious Adverse Event (SAE) CRF (BB05):

7.8.1 Section A. Adverse Events (AE)

AE#:

The AE# will be autogenerated by the iMedidata RAVE EDC system when you add a new log line.

Awareness Date and Time:

Enter the date and time clinical research staff became aware of the AE event. Document the 'awareness' date and time in your research records for source verification as needed. Use the following convention DD/MMM/YYYY and HH:MM (24 hour).

Adverse Event Term:

A prepopulated drop-down menu will be available for you to select an immediate adverse event or report an 'other' adverse event term. Avoid reporting symptoms for 'other event' and record the diagnosis. If a diagnosis is not available, the best clinical descriptor of the event should be used. Entry of an adverse event into the adverse event log should not be delayed due to lack of a diagnosis. Use the event term to describe the event, one event per line. Abbreviations should be avoided.

Event Start/Onset Date and Time:

Record the onset date and time per medical records or clinical assessment. For events present prior to study drug administration but have worsened since, the start date for the sake of the study is the date the event worsened since study drug initiation and thus qualified as a reportable AE. Use the following convention for date DD/MMM/YYYY and use the following convention for time HH:MM (24-hour clock). If the actual time of AE onset is not available, please provide an estimated onset time and select the box "Estimate," otherwise, leave onset time blank and select the box "Unknown".

Event Stop/End Date and Time:

Record the date and time the adverse event resolved per medical records or clinical assessment. If at the time of reporting the event is ongoing leave the date blank. Update it when the event has resolved. If an AE has ended but the actual end time is not available, please provide an estimate end time and select the box "Estimate," otherwise, leave end time blank and select the box "Unknown".

All adverse events (AE) will be followed until satisfactory resolution or checked as 'Continuing/Ongoing' if the event has not resolved by 7 days (168 hours) post last study drug administration. The Final Outcome would be coded as either Resolving/Recovering or Not resolved/Not recovered with the exception of continuous monitoring within 30 days of final study drug administration for adverse events of special interest (e.g. spontaneous intestinal perforation (SIP), which can occur after postnatal day 7).

All 'serious' adverse events (SAE) and those events requiring IND safety reporting will be followed until satisfactory resolution or until the site investigator deems the event to be chronic or the adherence to be stable or by study status of 36 weeks' post-menstrual age, whichever comes first.

Severity/Grade:

Use the criteria specified in the Toxicity Table to make the determination. Adverse events that are mild in severity, transient, and expected in the study population will not be recorded unless it is considered unexpected.

- 1 = Mild
- 2 = Moderate
- 3 = Severe
- 4 = Life Threatening
- 5 = Fatal

Relation to Study Intervention(s):

Refer to criteria specified in Section 7.3 Attribution/Causality Assessment (AE/SAE Relationship to Study Intervention) to make the determination.

- 1 = Not Related
- 2 = Unlikely Related
- 3 = Possibly Related

- 4 = Probably Related
- 5 = Definitely Related

Action Taken with Study Treatment:

The numerical order of the options is applicable to BiB study only. Select the appropriate action taken.

- 1 – Dose not changed/No change/None
- 4 – Dose Interrupted
- 5 – Drug Withdrawn

Outcome:

- 1 – Resolved/Recovered
- 2 – Resolved/Recovered with Sequelae
- 3 – Resolving/Recovering
- 4 – Not resolved/Not recovered
- 5 – Fatal

Expected:

Use the criteria specified in Section 7.4 Determining Expectedness to make the determination.

- 1 – Yes
- 2 – No

Continuing/Ongoing:

Indicate if the AE is ongoing at the time of submission. If an (S)AE was ongoing at the time of submission, but ends before the end of the safety monitoring period, change the field to 2 (No); otherwise, leave as 1 (Yes).

- 1 – Yes
- 2 – No

Serious AE:

Use the criteria specified in Section 7.1.2 Serious Adverse Event (SAE) to make the determination

- 1 – Yes
- 2 – No

If the event qualifies as serious select “yes”. Complete an instance of CRF [BB05 Section B](#). See the following MOP section (7.8.2 Section B. Serious Adverse Event (SAE)) for additional details on SAE reporting.

7.8.2 Section B. Serious Adverse Event (SAE)

Serious adverse events (SAE), regardless of causality/attribution and expectation assessment require additional detail to create a summary report for review by the NICHD project scientist and/or other authority.

Once an event has been identified as serious (Serious = Yes) a SAE eCRF will populate in the list of available forms for that infant case in the EDC and the site should complete it immediately. If the EDC system is not available for 24 hours, the paper SAE form should be completed then entered once the

EDC system is functional. **If the paper form is used, the complete SAE form should be sent to the DCC via RTI-Safety-Desk@rti.org.**

Once the SAE is reported to the DCC an alert notifications via email from RTI-Safety-Desk@rti.org will be generated by the DCC Clinical Safety Specialist (CSS) and sent to the NICHD Program Scientist for review. Should that initial review warrant further DSMC review, the DCC will formulate that notification, track responses, query the clinical center as needed, and inform NICHD and the site PI of determination. A manual notification process via email will be used should the EDC be unavailable for 24 hours or longer.

All AE fields with the exception of those listed below will be auto-populated with the responses from the AE log form. The following fields will require data entry:

Serious Criterion:

Select at least one of the criteria as the reason the event is serious. A check mark will appear next to the selected criteria. Use criteria specified in Section 7.1.2 Serious Adverse Event (SAE) to make the determination.

- Other (Important Medical Event)
- Persistent or significant disability/ incapacity
- Life threatening
- Congenital anomaly/birth defect
- Inpatient hospitalization or prolongation of hospitalization
- Death

If outcome 'Fatal': Include death date (in DD/MMM/YYYY format), Confirm if an autopsy was performed (Yes/No), and if 'Yes' list primary cause of death found on autopsy report.

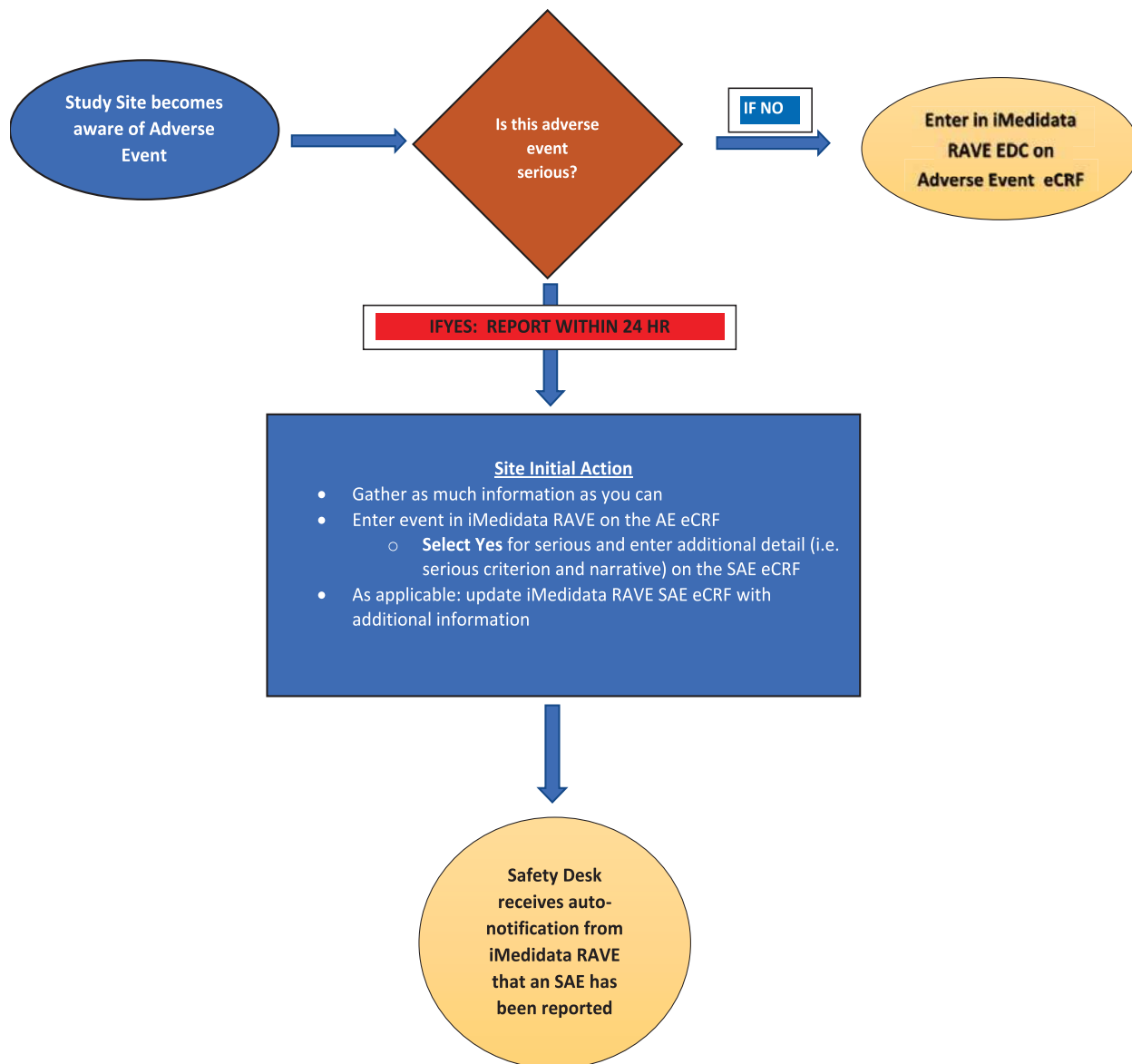
SAE Narrative: Write a summary of the SAE. Summary may include the following but do not provide identifiable information.

1. Presenting signs and symptoms
2. Physical assessment findings
3. Vital signs (at event onset and resolution)
4. Diagnosis (if signs and symptoms were initially given)
5. Consultations (e.g., infectious disease), include date of the consultation and outcome
6. Treatments given as result of the event (include dose, route, and dates given)
7. Clinical course
8. If discharged, provide status of the participant at time of discharge (e.g., stable).
9. Relevant and possibly contributing medical history
10. If SAE is considered related to something other than study drug, provide the cause (e.g., other procedures, concurrent illness, other therapies, etc.)
11. Hospitalization and discharge dates, if applicable

NOTE: There is a character size limit of < 2000. Try to avoid spacing between sentences, creating separate paragraphs or indenting.

7.8.3 Section C. Form Completion

Record the initials of the person completing the SAE form and date the event data were reviewed by the site (or study specific) PI.

Process Workflow for Site Reporting of Serious Adverse Event/Adverse Event

CHAPTER 8. PROTOCOL VIOLATIONS AND DEVIATIONS

8.1 Explanation of Protocol Violations and Deviation

The purpose of the protocol violation and/or deviation report is to facilitate the accurate documentation of the facts and circumstances surrounding a deviation from or violation of the protocol.

Protocol Deviation-A protocol deviation is any change, divergence, or departure from the study design or procedures of a research protocol that is under the investigator's control and is not part of the IRB approved protocol. Upon discovery, the Principal Investigator is responsible for reporting protocol deviations to the IRB using the standard reporting form.

Any change, divergence, or departure from the study design or procedures of a research protocol that affects the subject's rights, safety, or well-being and/or the completeness, accuracy and reliability of the study data constitutes a **protocol violation**. Changes or alterations in the conduct of the trial which do not have a major impact on the subject's rights, safety or well-being, or the completeness, accuracy and reliability of the study data are considered protocol deviations.

Protocol Violation-A protocol violation is a deviation from the IRB approved protocol that may affect the subject's rights, safety, or well-being and/or the completeness, accuracy and reliability of the study data.

If the deviation meets any of the following criteria, it is considered a **protocol violation**. Example list is not exhaustive.

I. The deviation has harmed or posed a significant or substantive risk of harm to the research subject.

Examples:

- A research subject received the wrong treatment or incorrect dose.
- A research subject met withdrawal criteria during the study but was not withdrawn.
- A research subject received an excluded concomitant medication.

II. The deviation compromises the scientific integrity of the data collected for the study.

Examples:

- A research subject was enrolled but does not meet the protocol's eligibility criteria.
- Failure to treat research subjects per protocol procedures that specifically relate to primary efficacy outcomes. (if it involves patient safety it meets the first category above)
- Changing the protocol without prior IRB approval.
- Failure to obtain or inadvertent loss of required lab samples or data.
- Unmasking of the treatment arm.

III. The deviation is a willful or knowing breach of human subject protection regulations, policies, or procedures on the part of the investigator(s).

Examples:

- Failure to obtain informed consent prior to initiation of study-related procedures
- Falsifying research or medical records.
- Performing tests or procedures beyond the individual's professional scope or privilege status (credentialing)

IV. The deviation involves a serious or continuing noncompliance with federal, state, local or institutional human subject protection regulations, policies, or procedures.

Examples:

- Working under an expired professional license or certification
- Failure to follow federal and/or local regulations, and intramural research or CC policies
- Repeated minor deviations.

V. The deviation is inconsistent with the NIH Human Research Protection Program's research, medical, and ethical principles.

Examples:

- A breach of confidentiality.
- Inadequate or improper informed consent procedure.

Minor Protocol Deviation- A minor protocol deviation is any change, divergence, or departure from the study design or procedures of an IRB approved research protocol which DOES NOT have a major impact on the subject's rights, safety or well-being, or the completeness, accuracy and reliability of the study data.

If the deviation meets any of the following criteria, it is considered a protocol violation. Example list is not exhaustive.

Example:

- Infant randomization GA strata is different from GA at birth.
- Open label surfactant replaced intended second dose of study drug.
- BPD Assessment not administered per protocol.
- Other Deviation (specify under circumstances)

8.2 Instructions for Completing Protocol Violation and Deviation Form (BB06)

This form should be completed for all randomized patients whenever a protocol violation/deviation is encountered by the study personnel. A new form should be completed for each protocol deviation or violation. Since more than one deviation or violation per patient may occur, record the Report number in the header. The Report Number is sequential starting with the number 1 for the first violation/deviation per Study Number.

At end of study, if no deviations occurred, designated masked and unmasked personnel should select "no deviations" in Medidata EDC per infant case. Masked Events are documented by those personnel given restricted access (*see MOP Section 2.4 Study Documentation*).

1. Date of Event:

Record the date on which the protocol violation/deviation occurred using the DD/MMM/YYYY format.

2. Submitted to IRB?

Code 'Y' if this protocol violation/deviation was submitted to your local IRB as required per institutional policy (if yes, complete Q2a and 2b below). Code 'N' if otherwise and continue to Q.3.

2a. If this protocol deviation/violation was submitted to your local IRB, record the date the event was submitted to the IRB.

2b. If this protocol violation/deviation reported to your IRB considered a serious or continued non-compliance event; code 'Y' if this event is considered a serious or continued non-compliance event and code 'N' if otherwise.

3. Event (choose one, and identify associated dose if applicable):

Code the most appropriate protocol violations/deviation. Only one can be reported per form.

Protocol Violations (report to IRB following Institutional Policy)

1. Lack of consent

Record 'Y' if no consent was obtained.

Record 'N' if this is not the violation being reported.

2. Ineligible infant randomized

Record 'Y' if an inclusion or exclusion criteria was known/present prior to randomization.

Record 'N' if this is not the violation being reported.

3. Randomized infant never initiated study drug

Record 'Y' if study drug was never initiated. This would include if the randomized kit contents were compromised, and the infant did not receive the randomized treatment assignment.

Record 'N' if this is not the violation being reported.

4. Infant randomized to wrong GA strata (*see #13 for GA time-out)

Record 'Y' if the incorrect gestational age-stratified/color-coded study drug kit was opened (randomized) to an infant of different GA at birth.

Record 'N' if this is not the violation being reported.

5. Infant was enrolled/randomized > 48 hours postnatal age

Record 'Y' if randomization occurred > 48 hours postnatal age.

Record 'N' if this is not the violation being reported.

6. *MASKED EVENT* Infant received wrong study drug (treatment arm)

Record 'Y' if the randomized kit contents administered to the infant was not the randomized treatment assigned. Record the "Associated Dose"

Record 'N' if this is not the violation being reported.

7. *MASKED EVENT* Infant received incorrect dose (volume and/or concentration)

Record 'Y' if the infant received incorrect dose (volume and/or concentration) of either surfactant (Curosurf) or Budesonide (Pulmicort) per actual infant weight. Record the "Associated Dose"

Record 'N' if this is not the violation being reported.

8. Infant received study drug > 50 hours postnatal age

Record "Y" if the infant received study drug > 50 hours postnatal age even if ordered prior.

Record the "Associated Dose"

Record 'N' if this is not the violation being reported.

9. *MASKED EVENT* Preparation and/or physical administration of study drug was not per-protocol

Record 'Y' if study drug was not prepared following methods stated in CHAPTER 5 (i.e., not using an aseptic field or aseptic technique to prepare study drug or using clinical supply of budesonide should study supply (kits) not be available as example). Record the "Associated Dose".

Record 'N' if this is not the violation being reported.

10. Unintentional Unmasking

Record 'Y' if study drug treatment assignment was inadvertently disclosed any masked parties i.e., caregivers or parent(s)/guardian(s), masked study or clinical staff.

Record 'N' if this is not the violation being reported.

11. Indomethacin used during the 7 days following the last study drug dose (i.e., up to 9 days postnatal).

Record 'Y' if indomethacin is used during the 7 days following the last study drug dose (i.e., up to 9 days postnatal). Systemic steroids or indomethacin use prior to randomization is a study exclusion. Early postnatal systemic steroids will be discouraged and reserved only for infants (at clinician discretion) who have fluid-resistant hypotension OR vasopressor-resistant hypotension, OR evidence of hyponatremia and low serum cortisol concentrations (See [Protocol Section 4.2.6](#) and MOP Section 2.2.5 Control or Monitoring of Co-Interventions). If indomethacin is used for these reasons, no protocol violation is reported. As such, a comment should be added to that medication row on the concomitant medications form [BB04](#).

Record 'N' if this is not the violation being reported.

12. Other violation (specify under circumstances)

Record 'Y' if violations occurred that are not otherwise listed above. Record the "Associated Dose".

Record 'N' if this is not the violation being reported.

Protocol Deviations

13. Infant randomization GA strata is different from GA at birth

Record 'Y' if the consented and eligible infant crossed GA strata (timed out) from birth (< 26 weeks; ≥ 26 weeks) and was randomized to the incorrect GA stratification.

Record 'N' if this is not the violation being reported.

14. Open label surfactant replaced intended second dose of study drug

Record 'Y' if an infant was randomized and received a study intervention (from a kit) on initial dose and was administered open-label surfactant clinically (from the clinical supply, not a kit) for the second dose, not per protocol. This deviation is only applicable if an infant required a second dose of surfactant within 50h PNA, but a BiB drug kit was not used for that second dose. This may happen if kit #2 cannot be accessed (e.g., kit #2 was damaged, IDS pharmacy was unavailable), or when a clinical care decision* is made to stop study intervention for the infant (e.g., in response to an SAE, in response to a protocol violation).

*If the second dose was clinical supply due to a clinical decision, the participant may qualify for withdrawal from the study. Open label surfactant administered after withdrawal from the study is not a protocol deviation.

Record 'N' if this is not the violation being reported **15. BPD Assessment not administered per protocol.**

Record 'Y' if the *Physiologic Evaluation for BPD* was not conducted per the NRN GDB guidelines ([NRN GDB MOP Section 5.1.2](#)) in addition to determining BPD by the Jensen et al. (2019) criteria at 36w PMA.

Record 'N' if this is not the violation being reported.

16. Other Deviation (specify under circumstances)

Record 'Y' if deviations occurred that are not otherwise listed above. Record the "Associated Dose".

Record 'N' if this is not the violation being reported.

4. Circumstances of the Protocol Violation/Deviation:

Briefly describe the circumstances that led to the protocol violation/deviation. In particular, if there were medical reasons for deviating the protocol, describe them.

5. Additional Comments:

Record any additional comments or information here such as steps taken to prevent recurrence of the violation/deviation.

6. Name of the person reporting the violation/deviation from protocol:

The name of the individual making the report should be recorded here.

A report form ([BB06](#)) should be completed each time a deviation or violation occurs and reported via the RAVE™ web-based EDC in as close to real-time as possible. All protocol violations will periodically be reviewed by the DSMC.

CHAPTER 9. STUDY WITHDRAWAL

9.1 Study Withdrawal CRF

Parents or legal guardians can elect to withdraw their infant(s) from the study at any time without compromise to the infants' care. Infants will be withdrawn from the study immediately if the Investigator believes it to be in the best interest of the subject. Infants who are withdrawn from the study are not to be replaced, and their data will be included in the final analyses up to the point when parents withdraw permission to collect further data.

Decisions to discontinue study therapy do not implicitly constitute a withdrawal from study. Specifically, parents may discontinue an infant from the study treatment but allow for study follow-up participation or allow for continued data collection from clinical visits (i.e., safety and outcome measures).

9.2 Instructions for completing the Study Withdrawal Form (BB07)

Complete this form for all **randomized** infants. Sections B and C should be completed when a randomized infant has been withdrawn from study participation. This form will cover the study time from randomization to study status (36 completed weeks PMA, death, D/C or transfer whichever is sooner).

It is critical to collect data about and explore the reasons why an infant has been withdrawn from the study. The use of the comments or specific fields on this form can more fully describe the circumstances for why parent(s) or physician(s) withdraw infants from study participation or intervention or declare that no further data can be obtained. The field size for these questions will allow up to 800 characters.

9.2.1 Section A. Occurrence of Withdrawal

1. Was the infant withdrawn from study participation at any time during the study period?

Record 'Y' if the infant was withdrawn from study intervention or participation at any time during the study period. Record 'N' otherwise.

If Yes, complete Sections B and C.

If No, form is complete.

9.2.2 Section B. Withdrawal

1. Date infant withdrawn

Record the documented date the infant has been withdrawn from study. Withdrawal is a process and this date should reflect the date study drug or study-specific assessments have been halted or the date no further data can be obtained.

2. Select the withdraw status appropriate for this infant

Code the appropriate status for withdrawal

1= Withdrawn from study with permissions to continue data collection per protocol

(i.e. no additional study-specific assessments will be completed but data collected as part of standard clinical practice may be used for study purposes). Administration of Study Drug is halted.

2= Withdrawn from study with discontinuation of further data collection (i.e. no further/future data to be reported about this infant for this study). Note that data that has already been collected, and those in the source documents prior to the date of

withdrawal are part of the research record. These data, up to the time permission to collect data was withdrawn, will be expected to be reported by the FDA.

3 = Discontinuation of study intervention with permission to continue study -specific assessment and data collection (i.e. study intervention will be halted but other study-specific assessments can be conducted/data obtained.

3. Infant withdrawn by:

Use the following codes to identify who withdrew consent:

1 = Parent(s) or Legal Guardian(s)

2 = Neonatologist

3 = Other (specify in the space provided)

This may include court appointed guardian

4. Select Primary Reason(s) infant was withdrawn

Use the following codes to identify the reason the infant was withdrawn:

1 = Infant too medically unstable to continue study participation

*If the withdrawal is a direct result of an adverse event occurring **after at least one dose of study drug**, ensure that the adverse events form (BB05) section A and B are completed where appropriate.*

2 = Reason not requested (specify below)

Explain why a reason was not requested

3 = Reason not stated (specify below)

Explain briefly, as feasible, why a reason was not provided.

4 = Other (specify below)

Use this code to explain the reason given for withdrawing the infant from the study if you are able to learn about this.

9.2.3 Section C. Additional Comments

1. Comments

Use this space for additional comments.

2. Name of Person Obtaining Withdrawal

Record the name of the person who obtained the withdrawal of consent from Parent(s) or Legal Guardian(s), a Physician or other persons as specified above.

3. Date form completed

Enter the date study staff complete data captured on this form.

4. Initials of person completing this form.

Record the initials of the person completing the form.

CHAPTER 10. FOLLOW UP RESPIRATORY OUTCOMES**10.1 Overview**

The purpose of this questionnaire is to obtain an interval respiratory history. Questions are designed to ascertain the frequency and severity of wheezing and coughing episodes in enrolled infants. The person being interviewed does not need to be the primary caretaker (see relationship codes below). This interview should be administered by a trained study interviewer at the 22-26 month corrected age follow-up visit.

10.1.1 Completing the Follow-Up Respiratory Support CRF (BB10)**A. Interview Conducted****2. Was the interview conducted?**

Record "Y" if the interview was conducted at the 22-26 month follow up visit.

- a. If "N" and indication the reason why, then skip to Section E

- 1 = Lost to Follow-Up
- 2 = Interview Refused
- 3 = Child died
- 4 = Other, SPECIFY _____

B. Interview Overview

Parents or guardians expressing concerns regarding their child's breathing should be advised to discuss them with the family's primary care physician.

1. Today's Date

Record the date of the parent(s) or guardian completed the questionnaire in DD/MMM/YYYY format

3. Name and Relationship code of the person being interviewed

- a. Specify the name of the person being interviewed.
- b. Specify relationship to the infant using the relationship codes used in the Network Follow up Program specified below:

001 - Mother of Child	081 - Maternal female cousin
002 - Father of Child	082 - Paternal (SO) female cousin
011 - Husband, Significant Other (SO) (if different from 002)	091 - Maternal male cousin
012 - Wife, Girlfriend (if different from 001)	092 - Paternal (SO) male cousin
021 - Maternal grandmother	101 - Other maternal relative
022 - Paternal (SO) grandmother	102 - Other paternal (SO) relative
031 - Maternal grandfather	201 - Foster mother
032 - Paternal (SO) grandfather	202 - Foster father
041 - Maternal aunt	301 - Adoptive mother
042 - Paternal (SO) aunt	302 - Adoptive father
051 - Maternal uncle	401 - Other non-relative
052 - Paternal (SO) uncle	402 - Social worker/case worker

061 - Brother
062 - Stepbrother
071 - Sister
072 - Stepsister

501 - Staff in congregate care
502 - Still hospitalized
504 - Unknown

C. **Wheezing Outcomes**

Interviewers should inform the parents/guardian about these questions as follows:

“As part of today’s follow-up visit, I will be asking you some questions about your child’s breathing over the past 6 months.

The first questions are about wheezing. By wheezing we mean an expiratory sound (a sound that is made when breathing out, not in) that comes from the chest, sometimes described as whistling or musical. The next questions are about coughing”

1. **In the past 6 months, has your child’s chest sounded like this? (wheezy or whistling?)**

Record “Y” if the parent(s) or guardian reports that the baby’s chest has sounded wheezy or whistling. The interviewer may repeat the verbal descriptions of wheezing to the respondent. If respondent answers, “I don’t know”, interviewer should ask respondent to think back over the time period, repeating the description. If parent is still not sure, record “Unknown (Unk)”.

If “N”, skip to Section D.

2. **If “Y”, in the past 6 months, during the worst 2-week period, how often as your child’s chest sounded wheezy or whistling like this?**

Ask the respondent, during the worst 2-week period (of the past 6 months), how often the child’s chest sounded wheezy or whistling? Interviewer may repeat the choices or help respondent settle upon the choice as follows:

- 1 = Unsure/Unknown
- 2 = Twice a week or less
- 3 = More than two times a week, but not every day
- 4 = Everyday, but not all the time
- 5 = Everyday, all the time

D. **Coughing Outcomes**

1. **In the past 6 months, has your child had a cough for more than 3 days when he/she did not have a cold?**

Record “Y” if the respondent reports the baby coughed for more than 3 days when otherwise well. Do not include coughing associated with eating, drinking or choking.

A “cold” is an upper respiratory infection; other phrases for a “cold” include, “head cold”, “rhinitis”, “runny or watery nose” or “sniffles”. A cold may be complicated by an otitis media (ear infection).

A “cold” does not include “chest cold”, “bronchitis”, “pneumonia”, or “bronchiolitis”.

If the respondent answers “don’t know”, repeat the question and if he/she still does not know, then code the answer “Unknown (Unk)”.

If “N”, skip to Section E.

2. If “Y, in the past 6 months, during the worst 2-week period, how often has your child had coughing when he/she did not have a cold?”

Ask the respondent, during the worst 2-week period (of the past 6 months), how often the child experienced coughing without having a cold? Interviewer may repeat the choices or help respondent settle upon the choice as follows:

- 1 = Unsure/Unknown
- 2 = Twice a week or less
- 3 = More than two times a week, but not every day
- 4 = Everyday, but not all the time
- 5 = Everyday, all the time

E. Forms Completion

1. Record the Initials of person completing this form

Appendix A Toxicity Table

Guidance Toxicity Table for Premature Neonates in Budesonide in Babies Study Version March 1, 2020 Eunice Kennedy Shriver NICHD Neonatal Research Network

Evaluation of event severity (intensity) and seriousness is the responsibility and at the discretion of the investigator. However, this Toxicity Table is provided as guidance to assist with evaluation of intensity and seriousness of adverse events that will be collected per protocol. Adverse events that are mild in severity, transient, AND expected in the study population will not be recorded unless it is considered unexpected. An event that is present at baseline is not an AE UNLESS there is an increase in grade (i.e. worsening in severity or frequency) OR the event resolves and then returns.

Event	Grade I Mild (Report only if unexpected)	Grade II Moderate	Grade III Severe	Grade IV Life Threatening	Grade V Death
Endotracheal tube blockage	Endotracheal tube occlusion requiring suctioning	Endotracheal tube occlusion requiring tube replacement	Endotracheal tube occlusion requiring tube replacement, associated with prolonged hypoxemia and bradycardia (HR<80 for ≥30s, with SpO2 <80 for ≥30s)	Endotracheal tube occlusion requiring tube replacement and resuscitation with chest compressions	Demise associated with endotracheal tube blockage
Prolonged hypoxemia and bradycardia associated with study drug administration	Not applicable	Not applicable	Heart rate less than 80 beats per minute for at least 30 seconds, in association with a pulse oximetry reading of less than or equal to 80% for at least 30 seconds.	Same as grade III (severe) but with slow response or no response to treatment	Demise associated with hypoxemic and bradycardic episode
Pulmonary Air leak	Not applicable	Pneumothorax with increased observation or needle aspiration OR PIE on x-ray	Pneumothorax with chest tube placement or PIE with use of high frequency ventilation	Chest tube placed; on invasive ventilation with O2 >80%	Demise associated with hypertensive event

Event	Grade I Mild (Report only if unexpected)	Grade II Moderate	Grade III Severe	Grade IV Life Threatening	Grade V Death
Hypotension	Hypotension lasting less than 24 hours and no treatment	Use of a single vasoconstrictor agent or a single fluid bolus (20 ml/kg or less in 4-hour period)	Use of two vasoconstrictor agents or fluid boluses ≥ 40 ml/kg in 4 hours or single vasoconstrictor PLUS a fluid bolus 20 ml/kg or more in a 4-hour period	Sustained decrease in BP requiring at least two vasoconstrictor agents for 12 hours	Demise associated with hypotensive episode
Hypertension	A mean blood pressure (1 determination over a 12 hour period) which is $>95^{\text{th}}$ ile but without use of antihypertensive medication	A mean blood pressure (2-3 determinations over at least 12 hours) which is $>95^{\text{th}}$ ile but without use of antihypertensive medication	A mean blood pressure (at least 4 determinations over at least 12 hours) which is $>95^{\text{th}}$ ile or any use of antihypertensive medication	Sustained increase in BP $>95^{\text{th}}$ ile for at least 12 h requiring multiple treatment types or prolonging hospitalization	Demise associated with hypertensive event
Hyperglycemia	120-180 mg/dL on one determination but not on determination 6 hours apart	120-180 mg/dL on 2 consecutive determinations at least 6 hours apart	> 180 mg/dL on 2 consecutive determinations at least 6 hours apart or initiation of insulin	> 600 mg/dL on 2 determinations at least 4 hours apart	Demise associated with hyperglycemia >600 mg/dL and no other evident cause
Intracranial Hem (these findings must be detected during the monitoring period, i.e., onset of study drug to 7 days (168 hours) after final study drug)	Unilateral or bilateral blood or echodensity in the sub-ependymal region	Unilateral or bilateral blood/echodensity in the ventricle without distention of the ventricle and without a midline shift; if subarchnoid or subdural blood without a midline shift is detected, it will be included in this category.	unilateral or bilateral blood/echodensity filling the ventricle(s) resulting in distension without parenchymal bleeding and without a midline shift.	Parenchymal hemorrhage/echodensity (including cerebellar hemorrhage) and / or any bleed/echodensity associated with a midline shift.	Death secondary to Parenchymal hemorrhage/echodensity
Sepsis (culture positive)	Not applicable	Treated ≥ 5 days of consecutive antibiotics/virals/fungals, no cardiovascular instability	Treated for systemic symptoms and with ≥ 5 days with anti-infective agents, and additional symptomatic support needed, responded to treatment	Same as grade III (severe) but with cardiovascular instability and slow response or no response to treatment.	Demise associated with sepsis

Event	Grade I Mild (Report only if unexpected)	Grade II Moderate	Grade III Severe	Grade IV Life Threatening	Grade V Death
Other events not listed above	The adverse event is transient, is very common in this population and easily tolerated by the subject.	The adverse event causes the subject discomfort or interrupts the subject's usual function.	The adverse event causes considerable interference with the subject's usual function.	The adverse event put the subject at substantial risk of dying at the time of the adverse event.	Demise associated with event
PVL (30-day Safety Window)	Per radiology report (minimal) If not quantified, report minimal	Per radiology report (Moderate)	Per radiology report (Extensive)	N/A	N/A
Spontaneous intestinal perforation, without NEC (30-day Safety Window)	Not applicable	Not applicable	Intestinal perforation/free air without pneumatosis or bowel necrosis; no shock or respiratory deterioration or hypotension and in the absence of a known gastrointestinal malformation or therapeutic misadventure.	Intestinal perforation/free air; systemic symptoms including, but not limited to tachycardia, hypotension, shock, respiratory deterioration and/or placement of drain or lap	Demise associated with Spontaneous intestinal perforation, without NEC

Appendix B Hypertension Tables and Graphs

First Two Weeks After Birth

Table 1 The 95th percentile for Systolic, Mean, and Diastolic Blood Pressures during the first 2 weeks following birth in infants of 23-25 weeks gestation. Based on the Table 1 data. Reference: Batton

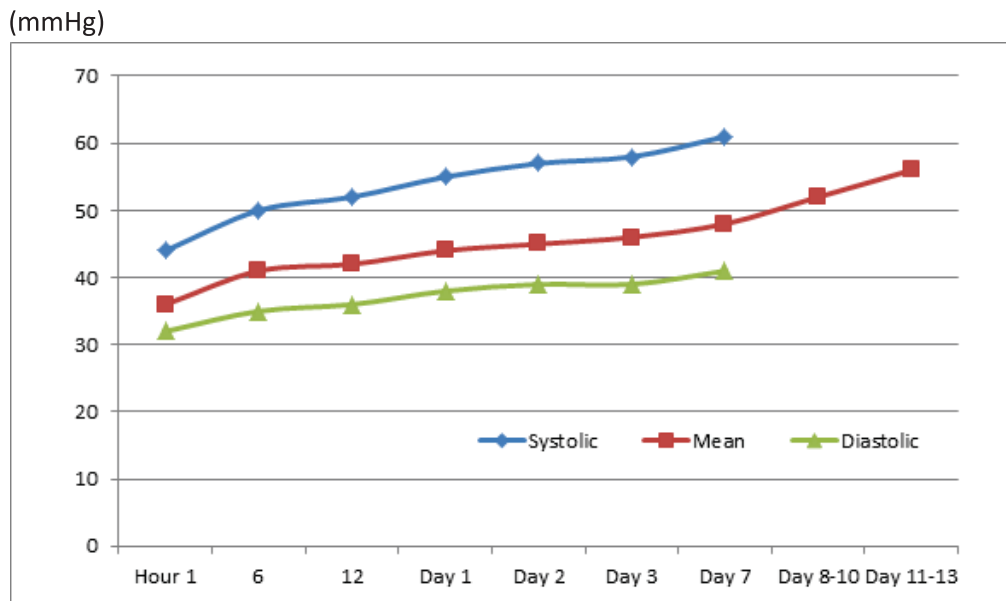


Table 2 COMPRESSED Table. Mean Blood Pressures at the 95th percentile in Stable Preterm infants of 23-25 weeks Gestational age during the first week following birth (Batton) and extrapolated in the second week

Age After Birth	95 th percentile MEAN BP
1 hour	36
6 hours	41
12 hours	42
24 hours	44
Day 2 48 hours	45
Day 3 72 hours	46
Day 7	48
Ages 8, 9, 10 days	52
Ages 11, 12, 13 days	56

Numbers provided in italics and Times Roman Font are extrapolated/interpolated, and not in the original table. Example: 52

Appendix C Electronic Data Capture System and Data Forms

The following contains a listing of the data forms for Randomized Controlled Trial of Budesonide/Surfactant versus Surfactant alone in Extremely Preterm Infants.

BB01	Screening Log CRF
BB01Ante	Antenatal Screening Log
BB02	Enrollment and Study Drug Initiation CRF
BB03	Respiratory Status/Laboratory Measures CRF
BB04	Concomitant Medicine CRF
BB05	Adverse Event (AE)/Serious Adverse Event (SAE) CRF
BB06	Protocol Violation and Deviation CRF
BB07	Study Withdrawal CRF
BB08a	Unmasked Study Drug Administration (Dose #1) CRF (<i>see Investigational Drug MOP</i>)
BB08b	Unmasked Study Drug Administration (Dose #2_ CRF (<i>see Investigational Drug MOP</i>)
BB09	Pharmacy Drug Accountability CRF (<i>see Investigational Drug MOP</i>)
BB010	Follow-Up Respiratory Outcomes CRF

Data from this trial will be entered in the Medidata RAVE™ Electronic Data Capture System (EDC): (<https://www.imedidata.com>)

All staff that enter data, reviews data or downloads reports must undergo remote EDC training. Those seeking certification should contact NRN Technical Support (NRNTechSupport@rti.org) with a valid email address. A new account will be set up and the individual must review training modules, known as “e-Learning”, through the RAVE™ EDC. These are self-guided and can be done on any computer with an internet connection. Staff previously certified with active accounts in RAVE™ do not need to re-certify.

Data from the Neonatal Research Network (NRN) Generic Database (GDB) and Follow Up study for enrolled trial infants will be entered in the NRN web-based EDC (Beacon): <https://nrn.rti.org/MyAccount/LogOn?ReturnUrl=%2f>

****NOTE:** The primary outcome is physiologic BPD or death by 36 weeks’ post-menstrual age in all randomized extremely preterm infants and will be recorded by study personnel at participating centers, using existing GDB criteria (5.1.2 Section B - PHYSIOLOGIC EVALUATION) on the RESPIRATORY SUPPORT FORM (NG07).

Appendix D List of Major Congenital and Chromosomal Anomalies

Central Nervous System Defects

Anencephaly
Congenital Hydrocephalus
Encephalocele
Holoprosencephaly
Hydranencephaly
Meningomyelocele
Neuronal Migration Defects

Chromosomal Abnormalities

Trisomy 13
Trisomy 18
Trisomy 21

Congenital Heart Defects

Complete Atrio-Ventricular Canal
Double Outlet Right Ventricle
Hypoplastic Left Heart Syndrome
Interrupted Aortic Arch
Pulmonary Atresia

Single Ventricle
Tetralogy of Fallot
Total Anomalous Pulmonary Venous Return
Transposition of the Great Vessels
Tricuspid Atresia
Truncus Arteriosus

Gastrointestinal Defects

Atresia of large bowel or rectum
Cleft Palate
Duodenal Atresia
Esophageal Atresia
Gastroschisis
Ileal Atresia
Imperforate anus
Jejunal Atresia
Omphalocele
Tracheo-Esophageal Fistula

Genitourinary Defects

Bilateral polycystic, multicystic, or dysplastic kidneys
Bilateral Renal Agenesis
Exstrophy of the Urinary Bladder
Obstructive Uropathy with Congenital Hydronephrosis

Other Birth Defects

Permanent neuromuscular condition that affects respiration
Congenital Diaphragmatic Hernia
Inborn Error of Metabolism (DESCRIBE w/ Comment)
Other Serious and/or Life-Threatening Birth Defect
Skeletal Dysplasia (DESCRIBE w/ Comment)

Pulmonary

Cystic Adenomatoid Malformation (CAM)

Appendix E Concomitant Medications Code List

Newborn ICU Pharmacy List

Generic Name	Generic Name
ACETAMINOPHEN	ANTIHEMOPHILIC FACTOR/VON WILLEBRAND
ACETAMINOPHEN/CODEINE	ANTIHEMOPHILIC PORCINE FACTOR VIII
ACETAZOLAMIDE	ANTI-INHIBITOR COAGULANT COMPLEX
ACETIC ACID	ANTIPYRINE/BENZOCAINE
ACETYLCYSTEINE	ANTITHROMBIN III
ACYCLOVIR	ANTITHYMOCYTE GLOBULIN
ADENOSINE	APRACLOPIDINE
ALBUMIN	APREPITANT
ALBUTEROL	ARGININE HYDROCHLORIDE
ALEMTUZUMAB	ARTIFICIAL TEARS
ALGLUCOSIDASE ALFA	ASCORBIC ACID
ALLOPURINOL	ASPARAGINASE
ALPHA-LIPOIC ACID	ASPIRIN
ALPRAZOLAM	ATENOLOL
ALPROSTADIL (PROSTAGLANDIN E1)	ATORVASTATIN
ALTEPLASE	ATROPINE
ALTEPLASE, RECOMBINANT	AZATHIOPRINE
ALUMINUM CHLORIDE HEXAHYDRATE	AZITHROMYCIN
ALUMINUM SULFATE AND CALCIUM ACETATE	AZTREONAM
AMIKACIN	BACITRACIN
AMINOCAPROIC ACID	BACITRACIN/NEOMYCIN/POLYMYXIN B
AMINOPHYLLINE	BACITRACIN/NEOMYCIN/POLYMYXIN B/ HYDROCORTISONE
AMIODARONE	BACLOFEN
AMITRIPTYLINE HYDROCHLORIDE	BECLOMETHASONE DIPROPIONATE
AMLODIPINE	BELLADONNA/OPIUM
AMMONIA SPIRIT, AROMATIC	BENZOCAINE
AMOXICILLIN	BENZOCAINE/BUTAMBEN/TERACAINE
AMOXICILLIN/CLAVULANIC ACID	BENZON TINCTURE
AMPHOTERICIN B	BENZOYL PEROXIDE
AMPHOTERICIN B LIPID COMPLEX (ABELCET)	BENZTROPINE MESYLATE
AMPHOTERICIN B LIPOSOME (AMBISOME)	BENZYL PENICILLIN (PENICILLIN G)
AMPICILLIN	BERACTANT
AMPICILLIN/SULBACTAM	BETAMETHASONE
AMYL NITRITE	BETAXOLOL
ANIDULAFUNGIN	BETHANECHOL CHLORIDE
ANTIHEMOPHILIC FACTOR IX	BIOTIN
ANTIHEMOPHILIC FACTOR VIIA	BISACODYL
ANTIHEMOPHILIC FACTOR VIII	

BISMUTH SUBSALICYLATE	CETIRIZINE
BLEOMYCIN SULFATE	Generic Name
Generic Name	CETIRIZINE
BOTULINUM TOXIN TYPE A	CHARCOAL
BROMPHENIRAMINE/PSEUDOEPHEDRINE	CHLORAL HYDRATE
BUDESONIDE	CHLORAMBUCIL
BUMETANIDE	CHLORAMPHENICOL
BUPIVACAINE HYDROCHLORIDE	CHLORHEXIDINE GLUCONATE
BUSULFAN	CHLOROPROCAINE
CAFFEINE CITRATE	CHLOROQUINE PHOSPHATE
CAFFEINE/SODIUM BENZOATE	CHLOROTHIAZIDE
CALAMINE LOTION	CHLORPHENIRAMINE MALEATE
CALCITONIN	CHLORPROMAZINE HYDROCHLORIDE
CALCITRIOL	CHOLESTYRAMINE
CALCIUM CARBONATE	CHORIONIC GONADOTROPIN
CALCIUM CHLORIDE	CIMETIDINE
CALCIUM GLUBIONATE	CIPROFLOXACIN
CALCIUM GLUCONATE 10%	CIPROFLOXACIN/DEXAMETHASONE
CALFACTANT	CISAPRIDE
CANTHARIDIN	CISPLATIN
CAPTOPRIL	CITALOPRAM
CAPTOPRIL	CITRATE/CITRIC ACID
CARBAMAZEPINE	CITRULLINE
CARBAMIDE PEROXIDE	CLADRIBINE
CARBENICILLIN	CLARITHROMYCIN
CARBOPROST TROMETHAMINE	CLINDAMYCIN
CARMUSTINE	CLOFARABINE
CARVEDILOL	CLOMIPRAMINE HYDROCHLORIDE
CASPOFUNGIN	CLONAZEPAM
CASTOR OIL	CLONIDINE HYDROCHLORIDE
CEFACLOL	CLOPIDOGREL
CEFAZOLIN	CLORAZEPATE DIPOTASSIUM
CEFEPIME	CLOTIMAZOLE
CEFIXIME	CODEINE
CEFOTAXIME	COENZYME Q10
CEFOTETAN	COLFOSCERIL PALMITATE
CEFOXITIN	COLISTIMETHATE
CEFOXITIN SODIUM	CORTICOTROPIN
CEFTAZIDIME	COSYNTROPIN
CEFTRIAXONE SODIUM	CROMOLYN SODIUM
CEFTRIZOXIME	CROTAMITON
CEFUROXIME	CYANOCOBALAMIN
CEPHALEXIN MONOHYDRATE	CYCLOBENZAPRINE HYDROCHLORIDE
CEPHALOTHIN	CYCLOPENTOLATE (OPHTHALMIC)

Generic Name	
CYCLOPENTOLATE HYDROCHLORIDE	DOCUSATE SODIUM
CYCLOPENTOLATE/PHENYLEPHRINE	DOCUSATE/SENNA
CYCLOPHOSPHAMIDE	DOPAMINE
CYCLOSPORINE	DORNASE ALFA
CYPROHEPTADINE HYDROCHLORIDE	DOXORUBICIN HYDROCHLORIDE
CYTARABINE	DOXYCYCLINE
CYTARABINE HYDROCHLORIDE	DRONABINOL
CYTOMEGALOVIRUS IMMUNE GLOBULIN	DROPERIDOL
DACARBAZINE	DTaP VACCINE
DACLIZUMAB	DTP, Hepatitis B (Recombinant), Poliovirus Vaccine
DACTINOMYCIN	DULOXETINE
DANTROLENE SODIUM	ENALAPRIL
DAPSONE	ENALAPRIL MALEATE
DARBEPOETIN ALFA	ENALAPRILAT
DAUNORUBICIN HYDROCHLORIDE	ENDROPHONIUM CHLORIDE
DEFEROXAMINE MESYLATE	ENOXAPARIN
DESIPRAMINE HYDROCHLORIDE	EPHEDRINE
DESMOPRESSIN ACETATE	EPINEPHRINE
DESONIDE	EPOPROSTENOL
DEXAMETHASONE	ERGOCALCIFEROL
DEXMEDETOMIDINE	ERGOTAMINE
DEXRAZOXANE	ERYTHROMYCIN
DEXTRAN	ERYTHROMYCIN ETHYLSUCCINATE (PO)
DEXTROAMPHETAMINE SULFATE	ERYTHROMYCIN LACTOBIANATE
DEXTROMETHORPHAN	ERYTHROMYCIN OINTMENT (Topical)
DEXTROSE	ERYTHROMYCIN/SULFISOXAZOLE
DIATRIZOATE MEGLUMINE/DIATRIZAOTE SODIUM	ERYTHROPOIETIN
DIAZEPAM	ESMOLOL
DIAZOXIDE	ESTRADIOL
DICLOXACILLIN	ESTROGEN
DIGOXIN	ETANERCEPT
DIGOXIN IMMUNE FAB	ETHAMBUTOL HYDROCHLORIDE
DILTIAZEM	ETHANOLAMINE OLEATE
DIMENHYDRINATE	ETHINYL ESTRADIOL
DIMERCAPROL	ETHOSUXIMIDE
DIMETHYL SULFOXIDE	ETHYL ALCOHOL
DINOPROSTONE	ETHYL CHLORIDE
DIPHENHYDRAMINE HYDROCHLORIDE	ETOMIDATE
DIPHENOXYLATE/ATROPINE	ETONOGESTREL
DIPYRIDAMOLE	ETOPOSIDE
DOBUTAMINE	ETOPOSIDE PHOSPHATE
Generic Name	Generic Name
	FAMOTIDINE

FAT EMULSION	GUAIFENESIN/CODEINE
FELBAMATE	GUAIGENESIN/DEXTROMETHORPHAN
FENOLDOPAM	GUAIGENESIN/PSEUDOEPHEDRINE
FENTANYL	Haemophilus Influenzae, type B (Hib) Vaccine
FENTANYL CITRATE	HALOPERIDOL
FERRIC GLUCONATE COMPLEX	HEMORROIDAL PREPARATIONS
FERROUS GLUCONATE	HEMORROIDAL
FERROUS SULFATE	PREPARATIONS/HYDROCORTISONE
FILGRASTIM	HEPARIN
FLECAINIDE	HEPATITIS B Vaccine
FLUCONAZOLE	HEPATITIS B IMMUNE GLOBULIN
FLUCYTOSINE	HETASTARCH
FLUDARABINE PHOSPHATE	HEXACHLOROPHENE
FLUDROCORTISONE ACETATE	HOMATROPINE HYDROBROMIDE
FLUMAZENIL	HYALURONIDASE
FLUNISOLIDE	HYDRALAZINE
FLUOCINONIDE	HYDROCHLOROTHIAZIDE (HCTZ)
FLUORESCEIN SODIUM	HYDROCHLOROTHIAZIDE
FLUORESCEIN/BENOXINATE	(HCTZ)/SPRIONOLACTONE
FLUOROMETHOLONE	HYDROCODONE/ACETAMINOPHEN
FLUOROURACIL	HYDROCORTISONE
FLUOXETINE HYDROCHLORIDE	HYDROGEN PEROXIDE
FLUTICASONE	HYDROMORPHONE HYDROCHLORIDE
FLUTICASONE/SALMETEROL	HYDROXOCOBALAMIN
FOLIC ACID	HYDROXYLCHLOROQUINE SULFATE
FOSPHENYTOIN	HYDROXYPROPYL METHYLCELLULOSE
FUROSEMIDE	HYDROXYUREA
GABAPENTIN	HYDROXYZINE
GADOPENTETATE DIMEGLUMINE	HYOSCYAMINE SULFATE
GANCICLOVIR	HYOSCYAMINE/ATROPINE/SCOPOLAMINE/ PHENOBARBITAL
GEMCITABINE	IBUPROFEN
GENTAMICIN	IBUPROFEN LYSINE
GENTIAN VIOLET	IDARUBICIN HYDROCHLORID
GLIPIZIDE	IFOSFAMIDE
GLUCAGON	IMATINIB
GLUTAMINE	IMIPENEM
GLYBURIDE	IMIPRAMINE HYDROCHLORIDE
GLYCERIN	INDOMETHACIN
GLYCOPYRROLATE	INFLIXIMAB
GRAMICIDIN/NEOMYCIN/POLYMYXIN B	INSULIN
GRANISETRON	Generic Name
Generic Name	INTERFERON ALFA 2A
GRISEOFULVIN	INTERFERON ALFA 2B
GUAIFENESIN	

INTRAMUSCULAR IMMUNE GLOBULIN	LIOTHYRONINE
INTRAVENOUS IMMUNE GLOBULIN	LISINOPRIL
IODIXANOL	LITHIUM CARBONATE
IDOQUINOL	LOMUSTINE
IOHEXOL	LOPERAMIDE HYDROCHLORIDE
IPRATROPIUM	LORATADINE
IPRATROPIUM/ALBUTEROL	LORAZEPAM
IRINOTECAN	LUCINACTANT (SURFAXIN)
IRON	LYMPHOCYTE IMMUNE GLOBULIN
ISOLUECINE	MAGNESIUM CITRATE
ISONIAZID	MAGNESIUM GLUCONATE
ISOPROTERENOL	MAGNESIUM HYDROXIDE
ISOTRETINOIN	MAGNESIUM OXIDE
ISRADIPINE	MAGNESIUM SULFATE
ITRACONAZOLE	Magnesium/Aluminum Hydroxides
KANAMYCIN	MAGNESIUM-PROTEIN COMPLEX
KETAMINE HYDROCHLORIDE	MALT SOUP EXTRACT
KETOCONAZOL	MANNITOL
KETOROLAC TROMETHAMINE	MEBENDAZOLE
LABETALOL	MECHLORETHAMINE HYDROCHLORIDE
LACTOBACILLUS ACIDOPHILUS	MECLIZINE HYDROCHLORIDE
LACTULOSE	MEDIUM CHAIN TRIGLYCERIDES
LAMIVUDINE (3TC)	MEDROXYPROGESTERONE ACETATE
LAMOTRIGINE	MEGESTROL
LANSOPRAZOLE	MELPHALAN
LATANOPROST	MEPERIDINE
LEUCOVORIN CALCIUM	MERCAPTOPYRINE
LEUPROLIDE	MEROPENEM
LEVALBUTEROL	MESALAMINE
LEVETIRACETAM	MESNA
LEVOBUPIVACAINE	MESTRANOL/NORETHINDRONE
LEVOCARNITINE	METFORMIN
LEVOFLOXACIN	METHADONE
LEVONORGESTREL	METHICILLIN
LEVOTHYROXINE (T4)	METHIONINE
LIDOCAINE	METHOCARBAMOL
LIDOCAINE HYDROCHLORIDE	METHOTREXATE
LIDOCAINE/EPINEPHRINE	METHYLDOPA
LIDOCAINE/PRILOCAINE	METHYLENE BLUE
LIDOCAINE/TERACAINE	Generic Name
Generic Name	METHYLERGONOVINE
LIGNOCAINE	METHYLPHENIDATE HYDROCHLORIDE
LINDANE	METHYLPREDNISOLONE
LINEZOLID	METOCLOPRAMIDE

Appendix E: Concomitant Medications Code List

METOLAZONE
METOPROLOL
METRONIDAZOLE
MEZLOCILLIN
MICA FUNGIN
MICONAZOLE NITRATE
MICROFIBRILLAR COLLAGEN HEMOSTAT
MIDAZOLAM
MILRINONE
MINERAL OIL
MINERAL OIL/ROSEMARY OIL/PETROLATUM
MINOXIDIL
MISOPROSTOL
MITOXANTRONE HYDROCHLORIDE
MIVACURIUM
MMR
MONTELUKAST
MORPHINE SULFATE
MOXALACTAM
MOXIFLOXACIN
MUPIROCIN (PSEUDOMONIC ACID A)
MUROMONAB-CD3
MVI INTRAVENOUS VITAMINS
MYCOPHENOLATE MOFETIL
NAFCILLIN
NALBUPHINE HYDROCHLORIDE
NALOXONE
NAPHAZOLINE HYDROCHLORIDE
NAPHAZOLINE/ANTAZOLINE
NAPROXEN
NEOMYCIN SULFATE
NEOMYCIN/POLYMYXIN B
NEOMYCIN/POLYMYXIN B/DEXAMETHASONE
NEOMYCIN/POLYMYXIN B/HYDROCORTISONE
NEOMYCIN/POLYMYXIN B/PREDNISOLONE
NEOSTIGMINE
NESIRITIDE
NETILMICIN
Generic Name
NEVIRAPINE
NIACIN
NIACINAMIDE
NICARDIPINE
NIFEDIPINE

NIMODIPINE
NITRIC OXIDE
NITROFURANTOIN
NITROGLYCERIN
NITROGLYCERIN OINTMENT, 2%
NIZATIDINE
NOREPINEPHRINE
NORTRIPTYLINE HYDROCHLORIDE
NYSTATIN
NYSTATIN/TRIAMCINOLONE
OATMEAL, COLLOIDAL
OCTREOTIDE
OCULAR LUBRICANT
OFLOXACIN
OMALIZUMAB
OMEPRazole
ONDANSETRON
OSELTAMIVIR
OTHER
OXACILLIN
OXCARBAZEPINE
OXYBUTYNNIN CHLORIDE
OXYCODONE
OXYCODONE/ACETAMINOPHEN
OXYMETAZOLINE HYDROCHLORIDE
OXYTOCIN
PACLITAXEL
PALIFERMIN
PALIVIZUMAB
PALONOSETRON
PAMIDRONATE
PANCRELIPASE
PANCURONIUM BROMIDE
PANTOPRAZOLE
PANTOTHENIC ACID
PAPAVERINE
PAPILLOMAVIRUS VACCINE
Generic Name
PARALDEHYDE
PAREGORIC
PEGASPARGASE
PEGFILGRASTIM
PEMOLINE
PENICILLAMINE

PENICILLIN G	PROCARBAZINE HYDROCHLORIDE
PENICILLIN G PROCAINE	PROCHLORPERAZINE
PENICILLIN G BENZATHINE	PROMETHAZINE HYDROCHLORIDE
PENICILLIN V	PROMETHAZINE/CODEINE
PENTACEL VACCINE (combination CTAp/IPV/Hib)	PROPAFENONE HYDROCHLORIDE
PENTAMIDINE ISETHIONATE	PROPANTHELINE BROMIDE
PENTOBARBITAL	PROPARACAINE
PERMETHRIN	PROPOFOL
PHENAZOPYRIDINE HYDROCHLORIDE	PROPOXYPHENE
PHENOBARBITAL	PROPOXYPHENE/ACTEAMINOPHEN
PHENOL	PROPRANOLOL HYDROCHLORIDE
PHENOXYBENZAMINE HYDROCHLORIDE	PROPYLTHIOURACIL
PENTOLAMINE	PROSTAGLANDIN E1 (PGE-1)
PHENYLEPHRINE	PROTAMINE SULFATE
PHENYTOIN	PSEUDOEPHEDRINE
PHYSOSTIGMINE	PSYLLIUM
PILOCARPINE HYDROCHLORIDE	PYRANTEL PAMOATE
PIPERACILLIN	PYRAZINAMIDE
PIPERACILLIN/TAZOBACTAM (ZOSYN)	PYRIDOXINE HYDROCHLORIDE (Vitamin B6)
PNEUMOCOCCAL 7-VALENT CONJUGATE VACCINE	PYRIMETHAMINE
Polio Vaccine Inactivated	QUINIDINE
POLYETHYLENE GLYCOL-ELECTROLYTE SOLUTION	QUINUPRISTIN/DALFOPRISTIN
POLYMYXIN B SULFATE	RACEMIC EPINEPHRINE
PORACTANT ALFA	RANITIDINE
POSACONAZOLE	RASBURICASE
POSOPHYLLIN RESIN	REMIFENTANIL
POTASSIUM ACETATE	RHO(D) IMMUNE GLOBULIN
POTASSIUM CHLORIDE	RIBAVIRIN
POTASSIUM IODIDE	RIBOFLAVIN
POTASSIUM PHOSPHATE	RIFAMPIN
POTASSIUM PHOSPHATE/SODIUM PHOSPHATE	RITUXIMAB
POVIDONE-IODINE	ROCURONIUM BROMIDE
PRAMOXINE HYDROCHLORIDE	ROTAVIRUS VACCINE
PREDNISOLONE	SALICYLIC ACID
PREDNISONE	SALMETEROL XINAFOATE
Generic Name	Generic Name
PREGABALIN	SCOPOLAMINE
PREVNAR 13	SELENIUM
PRIMAQUINE PHOSPHATE	SELENIUM SUFLIDE
PRIMIDONE	SENNA
PROBENECID	SERTRALINE
PROBIOTICS	SEVELAMER
PROCAINAMIDE	SILDENAFIL
	SILVER NITRATE

SILVER SULFADIAZINE
 SIMETHICONE
 SINCALIDE (Synthetic CCK)
 SIROLIMUS
 SODIUM ACETATE
 SODIUM BENZOATE
 SODIUM BICARBONATE
 SODIUM CHLORIDE (NaCl)
 SODIUM FLUORIDE
 SODIUM HYPOCHLORITE SOLUTION
 SODIUM NITROPRUSSIDE
 SODIUM PHENYLACETATE/SODIUM BENZOATE
 SODIUM PHENYLBUTYRATE
 SODIUM PHOSPHATE
 SODIUM POLYSTYRENE SULFONATE
 SODIUM SULFACETAMIDE/PREDNISOLONE
 ACETATE
 SODIUM THIOSULFATE
 SORBITOL
 SOTALOL
 SPIRONOLACTONE
 SQUARIC ACID DIBUTLYESTER
 STREPTOKINASE
 SUCCINYLCHOLINE CHLORIDE
 SUCRALFATE
 SUCROSE
 SUFENTANIL CITRATE
 SULFACETAMIDE SODIUM
 SULFAMETHOXAZOLE/TRIMETHOPRIM
 SULFASALAZINE
 SULINDAC
 SUMATRIPTAN SUCCINATE
 SURFACTANT
 TACROLIMUS
Generic Name

 TEMOZOLOMIDE
 TENIPOSIDE
 TERBUTALINE SULFATE
 TESTOSTERONE
 TETANUS IMMUNE GLOBULIN
 TETRACAINE HYDROCHLORIDE
 TETRACYCLINE
 TETRAHYDROZOLINE HYDROCHLORIDE
 THEOPHYLLINE

THIABENDAZOLE
 THIAMINE HYDROCHLORIDE
 THIOGUANINE
 THIOPENTAL SODIUM
 THIORIDAZINE
 THIOTEPA
 THIOTHIXENE HYDROCHLORIDE
 THROMBIN
 TIAGABINE
 TICARCILLIN DISODIUM
 TICARCILLIN/CLAVULANTE
 TIMOLOL MALEATE
 TINCTURE OF OPIUM
 TOBRAMYCIN
 TOBRAMYCIN/DEXAMETHASONE
 TOCOPHEROL ACETATE (Vitamin E)
 TOLAZOLINE
 TOPIRAMATE
 TOPOTECAN HYDROCHLORIDE
 TOTAL PARENTERAL NUTRITION (TPN)
 TRAZODONE
 TREPROSTINIL
 TRETINOIN
 TRIAL DRUG/PLACEBO (HYDROCORTISONE
 FOR BPD)
 TRIAL DRUG/PLACEBO (OTHER)
 TRIAMCINOLONE
 TRIAZOLAM
 TRIFLURIDINE
 TRIHEXYPHENIDYL HYDROCHLORIDE
 TRIMETHOPRIM / SULFAMETHOXAZOLE
 TRIMETHOPRIM/POLYMYXIN B
 TRIPROLIDINE/PSEUDOEPHEDRINE
Generic Name

 TRISHYDROMETHYLAMINOMETHANE
 TROMETHAMINE
 TROPICAMIDE
 UREA
 UROKINASE
 URSODIOL (URSODEOXYCHOLIC ACID)
 VALACYCLOVIR
 VALGANCICLOVIR
 VALINE
 VALPROIC ACID

VANCOMYCIN	VITAMIN D
VARICELLA VACCINE	VITAMIN E
VASOPRESSIN	VITAMIN K ₁ (PHYTONADIONE)
VECURONIUM	VITAMIN, MULTIVITAMIN
VENLAFAXINE	VORICONAZOLE
VERAPAMIL HYDROCHLORIDE	WARFARIN SODIUM
VIDARABINE	WHEAT DEXTRIN
VINBLASTINE SULFATE	ZIDOVUDINE
Generic Name	ZINC
VINCRIStINE SULFATE	ZINC OXIDE/COD LIVER OIL/TALC
VINORELBINE	ZINC SULFATE
VITAMIN A	ZOLPIDEM
VITAMIN B	ZONISAMIDE
VITAMIN C	

Appendix F (Supplement): Nutrition vs. Medications Table of TPN components and Hydration Fluids

Items that are considered a nutrient when in TPN or when given with TPN (e.g. Lipids) (<i>not</i> considered a Concomitant Medication)	Medications Related to TPN, but are considered a Concomitant medication	Comments
	This list is limited to meds related to TPN, of having been added to TPN for convenience: See the Medication Table for a full listing of all meds	
Intravenous Nutrition		
Amino Acid solutions	Insulin	
Glucose (aka Dextrose) solutions		
Sodium chloride		
Sodium acetate		
Potassium chloride		
Calcium gluconate		
Calcium chloride		
Calcium citrate		
Phosphorous		
Magnesium	Vitamin D	
Pediatric MVI	Vitamin E , E-ferol	
intravenous lipid formulation eg. intralipid or SMOF	Investigational IV lipids if on an IND (investigational)	
Carnitine (levocarnitine)		
Trace metal/elements supplements (Zn, copper, selenium, chromium, manganese, iron) or the variation without copper	Imferon/IV Iron as a 'large' periodic dose (other than in a trace elements package)	
Albumin in small amounts as a protein nutrition supplement	Cimetidine or other H2 blockers or Proton Pump inhibitors	
Other		
*Heparin line maintenance (low does continuous)	Heparin for treatment of clotting disorders/thromboses	
*Heparin line maintenance Small "heparin flush"		

Items that are considered a nutrient when in TPN (<u>not</u> considered a Concomitant Medication)	Medications Related to TPN, but are considered a Concomitant medication	Comments
Intravenous hydration Fluids		This category is unrelated to TPN These volume expanders do not contain medications, and need not be coded as 'Concurrent medications'
Sodium chloride (NaCl)		
Potassium chloride (KCl)		
Saline (0.9% NaCL)		
"half normal saline"		
Ringer's Lactate		
Blood Products	Specific products (e.g. Epo, Darbe, Neupogen etc) that increase blood components	

Appendix G Scenarios on How to Complete Forms

Scenarios	BiB Forms to Complete by Coordinators	BiB Forms to Complete by Unmasked Pharmacist
Infant consented (antenatally or postnatally), randomized/enrolled and dose(s) administered per protocol	BB01Ante ¹ ; BB01; BB02; BB03; BB04; BB05; BB06 (as applicable); BB07; BB10	BB06 (as applicable); BB08a; BB08b (if second dose of surfactant (Curosurf) is ordered clinically to be administered ≤ 50h of age); BB09
Parents approached antenatally and refused antenatal consent	BB01Ante ¹ ; Quarterly entry in NRN Beacon EDC: https://nrn.rti.org by selecting the “BiB Antenatal Summary Count” that appears in the list after logging in and changing to survey mode.	No forms to complete
Parents refused consent postnatally	BB01; BB02	No forms to complete
Mother/Parents consented antenatally, and eligible infants screened postnatally (regardless of consent status).	BB01Ante ¹ ; BB01; BB02	No forms to complete
Mother/Parents consented antenatally, but baby born outside of eligibility window	BB01Ante ¹ ; BB01; BB02	No forms to complete
Mother/Parents consented antenatally, but mom is transferred prior to delivery	BB01Ante ¹ ; BB01; BB02 (IN003 = N)	No forms to complete
Infant consented, randomized/enrolled (≤ 48h postnatal age) without receiving study drug (≤ 50 hours postnatal age)	BB01Ante ¹ ; BB01; BB02; BB03; BB06; BB07; BB10	BB08a (not BB08b) and indicate “No” to Administered Per Protocol; BB09 (for Kit dispensed and end processing)
Infant consented, randomized/enrolled (≤ 48h postnatal age) and received incorrect treatment assignment (administered ≤ 50 hours postnatal age) *The initial treatment administered should be the same treatment for second dose if surfactant (Curosurf) is clinically ordered.	BB01Ante ¹ ; BB01; BB02; BB03; BB04; BB05; BB07; BB10	BB06; BB08a; BB08b (if second dose of surfactant (Curosurf) is ordered clinically to be administered ≤ 50h of age); BB09
Infant consented, randomized/enrolled (≤ 48h postnatal age) and infant withdrawn before study drug administration	BB01Ante ¹ ; BB01; BB02; BB03; BB06 (if deviation or violation occurred); BB07	BB08a and indicate “No” to Administered Per Protocol; BB09

Infant consented, randomized/enrolled (≤ 48 h postnatal age), study drug administered (≤ 50 hours postnatal age and infant withdrawn	BB01Ante ¹ ; BB01; BB02; BB03; BB04; BB05; BB06 (if deviation or violation occurred); BB07; BB10	BB08a, BB08b (if second dose given); BB09
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Notes:

1. BB01Ante is not keyed in the EDC and is a clinical tool that applies to pregnant women at risk of preterm delivery between 22 ⁰/₇ and 28 ⁶/₇ weeks' gestation.
2. Enrolled=Randomization=Study Kit Opened
3. If the infant is to receive the second dose of study drug, the dosage of Curosurf will be 1.25 ml/kg for the second dose (noting that the first dose of Curosurf is 2.5 ml/kg, regardless of whether a second dose is anticipated or not). A maximum of two doses of study drug can be administered, with specific criteria for the second dose at ≤ 50 h postnatal age.

SURVEY OF MORBIDITY AND MORTALITY AMONG HIGH RISK PRETERM INFANTS (GDB)

***Eunice Kennedy Shriver* NICHD
Neonatal Research Network**

Manual of Operations

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Chapter 1

Objectives and Study Designs

1.1 Introduction

This manual gives detailed instructions for the **Survey of Morbidity and Mortality Among High Risk Preterm Infants**. It is meant to serve as a reference guide for study staff, including investigators, coordinators, technicians and data managers. This study is being conducted by the *Eunice Kennedy Shriver* NICHD Neonatal Research Network.

1.2 Survey of Morbidity and Mortality Among High Risk Preterm Infants

The purpose of this study is to provide a registry of baseline and outcome data for high risk preterm infants, based on data collected in a uniform manner from neonatal intensive care units (NICUs) at institutions participating in the *Eunice Kennedy Shriver* NICHD Neonatal Research Network. These data, although not representative of a regional sample, do represent a number of major tertiary care academic centers. Although centers serve varying populations, they exemplify the neonatal morbidity problems of the 1980s through the 2010s and into the next decade. These data will be used to characterize the infants admitted to the units, to examine the relationships between certain entry characteristics and outcome, to measure trends in incidence of various disease entities, and to provide the basis for hypothesis formulation for Network multi-center studies.

Baseline and outcome data will be collected on all liveborn infants who are inborn and between 21 0/7 to 27 6/7 weeks (<28 weeks) inclusive gestational age. Infants enrolled in a Network RCT or prospective observational study (inborn and transferred infants) may be included in the GDB dataset if requested by the study subcommittee and approved by the NRN Steering Committee. These may be otherwise extremely premature infants outside the GDB gestational criteria. In addition, infants who meet the above criteria with a heart rate at birth, who died in the delivery room will be included. These data will be obtained by review of the mothers' and babies' charts. The data forms (listed in **Appendix F**) for the survey have been named '**generic data forms**' in recognition of the fact that the information collected is of universal interest and not specific to a particular disease or treatment. They provide a descriptive summary of the babies' background, perinatal and neonatal experience. Baseline data will be obtained soon after admission to the NICU and the outcome data will be obtained at the time of death or discharge from the hospital, transfer or 120 days.

Chapter 2 Administration

2.1 Organizational Structure

The Survey of Morbidity and Mortality Among High Risk Preterm Infants is being conducted by the NICHD Neonatal Research Network. The NICHD Neonatal Research Network was established by the Center for Research for Mothers and Children in 1986 to conduct multi-center clinical trials in neonatal medicine and management. The Network is funded as a cooperative agreement between the Clinical Centers, the Data Coordinating Center (DCC) and the NICHD. The Steering Committee for the Network is limited to the Principal Investigator from each Clinical Center, the Data Coordinating Center, and the NICHD Neonatal Research Program Official. Non-voting Steering Committee participants include the Director of the Center for Developmental Biology & Perinatal Medicine (CDBPM) and the Steering Committee Chairman, who is appointed by NICHD. The Steering Committee has the responsibility to develop study protocols and monitor their implementation.

2.1.1 Participating Centers

2.1.1.1 Clinical Centers

The Principal Investigators representing the Clinical Centers have agreed to abide by the study protocols and, in addition, to have comparable staff, facilities, and equipment. To ensure that centers meet standards for procedures, equipment and staffing, each center is certified prior to participation in Network studies.

2.1.1.2 Data Coordinating Center

The Data Coordinating Center (DCC) collaborates with the Steering Committee on protocol design, data management, data collection systems (including the final versions of protocols, forms and manual of operations), and analysis. The DCC conducts the interim and final statistical analyses and collaborates with the Steering Committee members in the preparation of publications based on the study results. The Principal Investigator of the DCC reports to the Steering Committee and the Data Monitoring and Safety Committee.

2.1.2 NICHD

In addition to its role as the funding agency, the NICHD participates as a voting member of the Steering Committee (the Program Official). NICHD staff also participates in the development of protocols and in assisting the Steering Committee in the coordination and publication of the studies conducted by the Network.

2.2 Committees

2.2.1 Steering Committee

This committee is comprised of the Principal Investigators from each of the Clinical Centers and the Data Coordinating Center, the NICHD Program Official, and the Chairman of the Steering Committee. The Steering Committee has the responsibility for identifying topics for network studies, designing study protocols, monitoring study implementation, and recruitment. The Steering Committee will also make recommendations for changes to study protocols if it deems

necessary. This committee receives recommendations from the Data Monitoring and Safety Committee via NICHD.

2.2.2 Generic Database Subcommittee

The Generic Database Subcommittee is responsible for the design of the generic data forms and for monitoring the conduct of the study. This subcommittee reports to the Steering Committee.

2.2.3 Publication Committee

The Publication Committee is responsible for developing the publication policy for the NICHD Neonatal Research Network and for writing the policy for the use of the Generic Data Base for publication.

Chapter 3

Survey of Morbidity and Mortality - Enrollment and Baseline Data

3.1 Enrollment

3.1.1 Eligibility

All infants who are inborn and between 21 0/7 to 27 6/7 weeks (<28 weeks) inclusive gestational age are eligible for the study. Infants enrolled in a Network RCT or prospective observational study (inborn and transferred infants) may be included in the GDB dataset if requested by the study subcommittee and approved by the NRN Steering Committee. These may be otherwise extremely premature infants outside the GDB gestational criteria. In addition, all inborn, liveborn infants who meet the above criteria and die prior to admission to the NICU are enrolled posthumously.

3.1.2 Screening Log Entry - Form NG01

The purpose of the Screening Log is to record all eligible infants admitted to all network centers. It serves as a cross check for identification and ensures that no infants are omitted for data collection. This form will not be entered into the center-based computer system. All infants should be entered on this log regardless if consent is required for GDB data collection.

3.1.2.1 NICU Admissions

For every infant admitted to the NICU who meet the above eligibility criteria, the Network Coordinator enters daily onto the log the baby's name, hospital record number, date of birth, birth weight, gestational age, if consent is obtained (if required at center) and whether or not enrolled in a NRN study.

The Network Number is each patient's unique identifier made up of a five digit Family Number (assigned to the pregnancy) plus the Birth Order Number. The Family Number is a consecutive number assigned (computer generated) when the infant is entered in the data management system (DMS). A Birth Order Number is assigned by the coordinator for all infants; for singleton or first born of a multiple birth, the code will be '1', each additional infant of a multiple birth will be assigned '2', '3', etc. If all infants of a multiple birth are not eligible for GDB, assign birth order number(s) starting with '1' for those infants who meet entry criteria.

This log may be modified to meet the particular needs of individual centers.

3.1.2.2 Deaths Prior to NICU Admission

The Network Coordinator will create a system to identify liveborn infants in the weight and gestational age range who expired prior to NICU admission. This is Center dependent and may be from weekly review of delivery room records, monthly reports from Morbidity & Mortality committees, etc. Name, hospital record number, date of birth, birth weight (measurement from the delivery room record), and Network number (as above) will be recorded on the screening log.

3.1.3 Data Forms

If the baby meets GDB eligibility criteria, forms NG02, NG03 and NG07 should be obtained and placed in a file folder designated for that baby. If the infant dies in 12 hours or less then replace the NG03 with the NG03E and do not enter an NG07. Headers on all forms should be

completed including Center number, Site number, infant's Network Number and mother's initials (optional)

Date and time format:

Date is mm\dd\yyyy unless otherwise specified

Time is recorded as 24 hour clock, day begins at midnight, 00:00.

3.1.4 Assignment of Network Number by Computer

The infant's identifying information should be entered into the DMS as soon as possible. This is known as the 'base' screen in the DMS and corresponds to the NG01 Screening Log.

The following information is contained on the NG01 and components of the base record:

Header Information

- **Center**
Two digit number for each NRN center, assigned by the DCC.
- **Site**
The NRN center assigns these to their hospitals that comprise the NRN center. Any site letter or number is acceptable. The site letter/number must be unique within the NRN center, consistent and never change even when a hospital is no longer part of the NRN center.

Data not keyed into the DMS

- **Infant's Name**
Last name followed by first name.
- **Infant's Hospital Number**
Two digit number for a center, assigned by the DCC.
- **GDB Consent**
Whether or not consent was obtained, if required.

Data keyed into the DMS

- **Date of Birth**
- **Birth Weight**
Record in grams.
- **Gestational Age**
Record the best estimate of gestational age, in weeks and days, using the following hierarchy:

Best OB estimate: Obstetrical measures based on last menstrual period, obstetrical parameters, and/or early (first trimester) prenatal ultrasound as recorded in the maternal chart.

Best Neonatologist estimate: Neonatologist's estimate based on physical criteria, neurologic examination, combined physical and gestational age exam (Ballard or Dubowitz), or examination of the lens. In instances when the gestational age in days is not recorded, enter '0' in the days field.

If the OB date based on last menstrual period (not early ultrasound) disagrees with the Neonatologist's estimate by more than 2 weeks, the Neonatologist's estimate should prevail.

Network

In the web-based EDC, the Network ID must be 6 digits long consisting of a 5 digit family ID (used to link siblings together from a multiple birth) and a 1 digit birth order number. The Network ID must begin with a value of 1 through 9, for example values of 100221 and 310002 are acceptable NETWORK IDs. The increased length of the NETWORK ID will ensure there are no duplicate IDs across the history of the GDB data collection and will quickly identify that the patient was keyed into the web-based EDC rather than the MS Access system.

Birth Number Assignment: for a single birth or first born of a multiple birth enter '1'.

For the second born code '2' etc. It does not necessarily have to conform to strict birth order from a multiple birth, but must be kept consistent for each baby. If any infant(s) of a multiple birth is not eligible for GDB, assign birth order number(s) starting with '1' for only those infants who meet entry criteria.

Users may use the EDC to automatically generate the Network ID or they may create it themselves. To automatically generate the Network ID click on the "Auto Generate Family Number" link found immediately below Q. 4 "Family Number" on the NG01 electronic case report form (eCRF). The system will populate a five digit family ID. The user will then key the birth number and the system will combine the two values and populate the Network ID in Q. 6 "Network" on the screen. Please record this ID on your paper case report forms.

To create the Network ID yourself simply key a unique family ID in Q. 4 "Family Number", enter the birth number and the EDC will combine the two values and populate the Network ID in Q. 6.

- **Infant enrolled in an NRN study**

For infants outside of the GDB gestational age and birthweight criteria

Infants enrolled in a Network RCT or prospective observational study (inborn and transferred infants) may be included in the GDB dataset if requested by the study subcommittee and approved by the NRN Steering Committee.

To identify these infants as correctly included on the NG01 "Enrolled in NRN Study" will be marked as "Y."

All GDB infants who meet usual GDB criteria do not require this documentation.

- **Enrolled in PAIR Study?**

Check the box for 'Yes' if the infant is also enrolled in the Parent and Infant Registry (PAIR) Study. Otherwise leave the box blank.

It is important to enter a 'base' screen for an infant as soon as possible, since at the time of entering data on the 'base' screen that infant is assigned a Network Number. This number is the infant's unique identifying number and should be used for all subsequent data forms.

When keying test data into the Data Management System (DMS), the user access mode should be set to “Training” (please review the DMS User’s Manual for specific instructions). The DMS assigns a number starting with a 'T', such as 'T0001' to test records. When entering real data, the system will request the first 4 to 5 digits of the patient ID (dependent on the current size of your patient IDs), referred to as the Family ID. It then requests the patient’s birth order number from the user. Lastly, the DMS creates the patient’s NETWORK ID by combining the two numbers together. For example, Network ID 688751 is made up of the family ID, 68875, and the birth order number, 1. The Network number can be up to 6 digits long.

3.1.5 Adding Network Numbers to NG01: Screening Log after Base Record Completion

After the base record is entered, enter the network number on the screening log and all of the infant’s generic forms.

3.2 Baseline Data Collection

When a baby has been enrolled in this study, the coordinator should complete form NG02 with baseline information. Coding instructions are presented below. Most of the information required is standard in nature, and is to be obtained from the baby's chart and from the mother's medical record.

3.2.1 For Infants Participating in Clinical Trials

Masked Clinical Trials

For any question which asks if the infant received a medication with a ‘Y/N’ response, a code ‘T’ will be used when an infant has received the medication under a **masked** study protocol in which the infant may or may not have received the drug (for example, a placebo-controlled trial).

If at another time during the hospitalization or at discharge the infant receives the known medication as part of clinical care, the response would be changed to ‘Y’.

When a **date of exposure/first dose** is required for any medication question coded as “T” for trial, record the date the infant first received the study medication (active or placebo) under a masked protocol. If at another time during the hospitalization or at discharge the infant receives the known medication as part of clinical care, the date entry would be changed to the date of the known medication administration. A comment will be added to state the infant previously received the medication under a masked study protocol.

When the **number of courses received** is required for any medication question coded as “T” for Trial the total number of courses will be left blank. If at another time during the hospitalization or at discharge the infant receives the known medication as part of clinical care, the number of courses would be changed to the number of known courses received. A comment will be added to state the infant previously received the medication under a masked study protocol.

When the **type** is required for any medication question coded as “T” for Trial, record the type as ‘Other’ and add a comment detailing the medication used in the trial. If at another time during the hospitalization or at discharge the infant receives the known medication as part of clinical care, the type would be changed to the type of known medication received. A comment will be added to state the infant previously received the medication under a masked study protocol, and the drugs used in that masked study protocol should be detailed.

Unmasked Clinical Trials

For **unmasked** trials where known medications or treatment were received (i.e., head-to-head trial of two steroids), code 'Y' if the active medication or treatment was received.

When a **date of exposure/first dose** is required, record the date the infant receives the medication under an unmasked study protocol

When the **number of courses received** is required, record the total number of courses the infant completed under the unmasked study protocol.

When the **type** is required, record the type of medication the infant received under the unmasked study protocol.

NG02: GENERIC BASELINE FORM

3.2.2 Section A - MATERNAL INFORMATION

The following information is to be obtained primarily from the mother's chart or any other reliable source. If possible, collect information about the mother while she is still in the hospital. This is the biological (birth) mother at the time of this delivery. If this is a surrogate pregnancy, code the surrogate mother's information.

1. **Mother's age**
Record the age in completed years at the time of delivery
2. **Pregnancy history (include this pregnancy)**
 - a. **Gravidity**
The number of confirmed pregnancies, including this one
 - b. **Parity**
The number of pregnancies reaching 20 weeks and 0 days of gestation or beyond, regardless of the number of fetuses or outcomes (ACOG 2014), including this delivery. It is the number of pregnancies reaching viability, and not the number of fetuses delivered, that determines parity. Note that each infant of a multiple birth should be assigned the same parity, and parity is only increased with birth of the last fetus. For example, if twins were born and it was the first delivery, then the parity for each twin would be 1. The parity for any children resulting from the next birth (whether singleton or multiple) following these twins would be 2. Note also that parity increases regardless of whether the fetus is liveborn or stillborn, as long as the pregnancy has reached at least 20 weeks.
3. **Marital status**
Choose the appropriate marital status code.
Code as follows:
 - 1 = Married - If mother is married, currently married but separated (including legal separations), or common law marriage
 - 2 = Single - If mother is single, divorced or widowed
 - 6 = Unknown - If marital status is unknown
4. **Highest level of education achieved by the biological Mother**
Code the highest level of education achieved by the biological (birth) mother at the time

of this delivery. If this is a surrogate pregnancy, code the highest education level of the surrogate mother at the time of delivery. This information should be obtained from the mother's or infant's medical record or other reliable source.

Code as follows:

- 1 = 8th grade or less - if completed 8th grade or less
- 2 = 9th to 12th grade - if completed through 9th, 10th, 11th or 12th grade and high school diploma not received
- 3 = High School diploma - if a high school diploma or GED was received and one full year of college was not attended
- 4 = Trade or Technical School
- 5 = Partial college or Associate's degree - if at least one year of college was attended or an Associate's degree attained but a B.A. or B.S. was not received
- 6 = College degree - if a Bachelor's (B.S. or B.A.) was received but no higher degrees were obtained
- 7 = Graduate degree - for degrees completed higher than a Bachelor's degree
- 8 = Unknown

5. Mother's medical insurance

Code all types of medical insurance documented in the maternal medical record at the time of admission. This information may often be found on the admitting or face sheet in the hospital record.

Code as follows:

- 10 = Public Insurance - this may include Medicaid, Medicare, a state funded program, federally funded program, or insurance obtained through the Affordable Care Act
- 3 = Private - this is traditional insurance, managed care, etc. (include CHAMPUS, Tricare or any insurance that may be tied to work).
- 5 = Self-Pay/uninsured - if hospitalization is to be or has already been paid for by the mother or other responsible party
- 6 = Unknown
- 9 = Other

6. Mother's ethnic categories

Code the mother's ethnicity as follows:

- 1 = Hispanic or Latino - A person of Cuban, Mexican, Puerto Rican, South or Central American, or other Spanish culture or origin, regardless of race. The term, "Spanish origin," can be used in addition to "Hispanic or Latino".
- 2 = Not Hispanic or Latino - None above
- 3 = Unknown or Not Reported - A person not knowing or not reporting ethnicity

7. Mother's racial categories

Code the mother's race as follows:

- 1 = Black - A person having origins in any of the black racial groups of Africa. Terms such as "Haitian" or "Negro" can be used in addition to "Black or African American."
- 2 = White - A person having origins in any of the original peoples of Europe, the Middle East, or North Africa.
- 3 = American Indian or Alaskan Native - A person having origins in any of the original peoples of North, Central, or South America, and who maintains tribal affiliation or community attachment.
- 4 = Asian - A person having origins in any of the original peoples of the Far East, Southeast Asia, or the Indian subcontinent including, for example, Cambodia, China, India, Japan, Korea, Malaysia, Pakistan, the Philippine Islands, Thailand, and Vietnam. (Note: Individuals from the Philippine Islands have been recorded as Pacific Islanders in previous data collection strategies.)
- 5 = Native Hawaiian or Other Pacific Islander - A person having origins in any of the original peoples of Hawaii, Guam, Samoa, or other Pacific Islands.
- 6 = More Than One Race - A person having origins in multiple racial designations.
- 7 = Unknown or Not Reported - A person not knowing or not reporting race

8. Mother's height

Record the mother's height in inches or centimeters. If unknown select the code 'PM' to indicate permanently missing.

9. Mother's weight prior to pregnancy or during the first trimester

Record the mother's weight prior to pregnancy or during the first trimester in pounds or kilograms. If recording the first trimester weight, record the earliest weight available from the first trimester. If unknown select the code 'PM' to indicate permanently missing.

10. Mother's weight at delivery

Record the mother's weight at delivery in pounds or kilograms. If unknown select the code 'PM' to indicate permanently missing.

3.3.3 Section B - PREGNANCY COMPLICATIONS

The following information is to be obtained primarily from the mother's chart or any other reliable source. Items 4a and 5a may be coded as Yes, No or Unknown. Code 'Y' if the item is listed as a problem in the maternal or infant record. Otherwise code 'Unk' or 'N'.

1. Multiple birth?

Code 'Y' if there was a multiple birth.

If YES,

Complete only if this pregnancy is a multiple gestation. Do not include early (< 14 weeks) fetal reductions.

a. Number of fetuses

Include all fetuses, live or stillborn.

2. Is there evidence of prenatal health care in this pregnancy?

Use the following codes:

'1 = No' if no prenatal visit was noted.

'2 = Limited' if less than 3 visits or prenatal care started in third trimester.

'3 = Adequate' if greater than or equal to 3 visits and prenatal care started prior to third trimester.

3. Was fetal ultrasound dating obtained during pregnancy?

Code 'Y' if fetal ultrasound dating was obtained during pregnancy. Code 'N' if fetal ultrasound dating was not obtained. Code 'Unk' if it is unknown whether fetal ultrasound dating was obtained during pregnancy.

If YES, answer question 3.a. or 3.b. but not both:

a. Gestational age determined from first ultrasound:

Record the gestational age at the time of the first ultrasound as it was determined from first ultrasound in weeks and days. If the value for weeks is known and the value for days is unknown, record the data for days as missing (i.e., code "PM" or "**") rather than '0'.

b. If gestational age at ultrasound unknown, was ultrasound obtained during the first trimester?

Code 'Y' if ultrasound was obtained during the first trimester (<14 weeks' gestation). Code 'N' if there was no ultrasound obtained during the first trimester. Code 'Unk' if it is not known whether an ultrasound was obtained during the first trimester.

4. Diabetes prior to pregnancy?

Code 'Y' if diabetes mellitus was diagnosed prior to present pregnancy. Otherwise Code 'N'.

a. If YES, Type of diabetes?

Code:

'1 = Type 1' if Type 1 diabetes is documented. Type 1 diabetes is considered insulin deficient diabetes/ body does not produce insulin.

'2 = Type 2' if Type 2 diabetes is documented. Type 2 diabetes is considered insulin resistant diabetes/ body does not use insulin properly.

'3 = Unknown' if diabetes but type not documented.

b. If YES, Treatment(s) given

If multiple treatments used during the pregnancy, record the highest level using the following hierarchy:

1 = Insulin

2 = Oral hypoglycemic medication' if any oral hypoglycemic medication has been given.

See Appendix K for list of oral hypoglycemics

3 = Diet only

5 = None if no medication is given

6 = Unknown if type of medication given is not known.

5. Gestational diabetes (diagnosed during the pregnancy)?

Code 'Y' if gestational diabetes was diagnosed during the pregnancy.

a. If YES, Treatment(s) given

If multiple treatments used during the pregnancy, record the highest level using the following hierarchy:

Code:

1 = Insulin

2 = Oral hypoglycemic medication' if any oral hypoglycemic medication has been given.

See Appendix K for list of oral hypoglycemics

3 = Diet only

5 = None if no medication is given

6 = Unknown if type of medication given is not known.

6. Hypertension?

Code 'Y' if hypertension, chronic or pregnancy induced, is specifically recorded in the mother's chart. The standard definition of hypertension (maternal BP above 140 systolic or 90 diastolic) was recorded prior to or during the present pregnancy on at least 2 occasions.

If YES,

a. Hypertension existed prior to pregnancy?

Code 'Y' if patient had hypertension prior to this pregnancy documented in the chart and/or chronic hypertension.

7. Antepartum hemorrhage?

Code 'Y' if placenta previa, abruption or threatened abortion resulting in bleeding, which can be external (vaginal bleeding) or occult (retroplacental clot), is documented after 20 weeks of pregnancy. This does NOT include bloody show.

8. Was chorioamnionitis documented in the mother's medical record?

Code 'Y' if chorioamnionitis is specifically documented in the mother's medical record.

9. Was placental pathology performed?

If YES,

a. Was histologic chorioamnionitis documented?

See **Appendix Ia** for definitions.

b. Was acute funisitis documented?

Code 'Y' if acute funisitis is documented. Otherwise, code 'N'.

3.3.4 Section C - LABOR AND DELIVERY

1. Date and time of maternal hospital admission for this delivery:

a. Date:

From labor and delivery sheet, admission notes or other reliable source

b. Time:

From labor and delivery sheet, admission notes, or other reliable source. Use a 24 hour clock with midnight coded as 00:00. If time of admission is not specifically recorded in records, please use time of first maternal vitals.

2. Was there rupture of membranes (ROM) prior to delivery?

Code 'Y' if the time of rupture of membranes (ROM) was documented prior to the delivery. Note: If ROM at the same time as delivery, this question should be answered 'N'. For C-sections, if the date and time is not recorded, assume the time of ROM was at delivery and answer 'N'. Code 'Unk' if ROM is not documented.

If YES,

a. Date:

From labor and delivery sheet, admission notes or other reliable source

b. Time:

From labor and delivery sheet, admission notes, or other reliable source. Use a 24 hour clock with midnight coded as 00:00

c. If date and/or time unknown, was ROM estimated at >18 hours?

3. Were antenatal steroids given to accelerate fetal maturity?

Code 'Y' if corticosteroids (e.g. betamethasone, dexamethasone) were given during this pregnancy. Code 'N' if no steroids were given or 'Unk' if not documented.

a. If YES, type of antenatal steroids given:

Code the type of corticosteroid given, or both as documented in the maternal medical record. Code

- 1 = betamethasone
- 2 = dexamethasone
- 3 = both
- 4 = If unknown

If the mother is in a study and is receiving either the placebo or the drug, then answer 'T' to any questions concerning mother being on the drug.

b. Was a complete course of antenatal steroids given?

Code 'Y' if 2 doses of betamethasone or 4 doses of dexamethasone were given, as part of a single course, specifically to promote fetal maturation, with at least 24 hours from the first dose elapsed before delivery. If the time elapsed was less, this indicates that there was insufficient time for the drug to have an effect and would be considered incomplete. Information may be obtained from the maternal and/or infant chart. Count only doses that occurred prior to delivery. Code 'Unk' if unknown.

c. Was more than one course of antenatal steroids given?

Code 'Y' if any dose(s) beyond one complete course of antenatal steroids was given. Code 'N' if one or fewer courses of antenatal steroids were given. Code 'Unk' if the number of courses of antenatal steroids given is not known.

d. When was the last dose of antenatal steroids given?

i. Date:

Record the date on which last dose of antenatal steroids was given (day begins at 00:00, ends at 23:59).

ii. Time:

Use a 24-hour clock with midnight coded as 00:00.

4. Were maternal antibiotics given within 72 hours prior to birth?

Code 'Y' if any maternal antibiotics were used within 72 hours prior to birth. Do not include prophylactic interoperative antibiotics.

a. If YES, list antibiotics given

Refer to **Appendix B**

5. Was magnesium sulfate given during this admission prior to delivery?

Code 'Y' if mother received magnesium sulfate during the admission prior to this delivery. Code 'N' if the mother did not receive it. Code 'Unknown' for incomplete records or if documentation is unclear.

6. Was there documentation of electronic fetal heart rate monitoring within 12 hours prior to birth?

Code 'Y' if there is documentation of continuous or intermittent electronic fetal heart rate monitoring within 12 hours prior to birth. Do not include monitoring with stethoscope or sonograms. Code 'N' if there was no electronic fetal heart rate monitoring within 12 hours prior to birth. Code 'Unknown' for incomplete records or if documentation is unclear.

7. Final mode of delivery:

As documented on labor/delivery sheet. Code:

- 1 = Vaginal vertex
- 2 = Vaginal breech
- 3 = Cesarean section
- 5 = Unknown or not noted in chart.

3.3.5 Section D - NEONATAL INFORMATION

1. Date and time of birth:

a. Date:

Record the date on which child was born (day begins at 00:00, ends at 23:59).

b. Time:

Use a 24-hour clock with midnight coded as 00:00.

NOTE: The day of birth is considered Day of life 1 even though it is not a complete 24 hour period. Day of life 2 is the next calendar day.

2. Was the infant outborn?

Code 'N' if the infant was born within the walls of one of the designated Network hospitals. Code 'Y' if the infant was born outside of the Network hospital. This could include any other hospital, home delivery, taxi, etc. This will include infants who are enrolled in an NRN trial who were not born at the Network center hospital.

If YES, **date admitted to NICU**

a. **Date admitted to NICU:**

Day begins at 00:00, ends at 23:59.

3. **Was a prenatal diagnosis made that influenced the decision to withdraw or limit intensive care?**

Code 'Y' if there was documentation in the mother's or infant's chart of a prenatal diagnosis other than immaturity or prematurity that has led to limitations of care. This may include, but is not limited to, withdrawal of life support, DNAR/DNR status or not offering usual care such as respiratory support, nutrition, antibiotics, surgery or other diagnostic procedures.

a. If YES, diagnosis (code all that apply),

Use Birth Defects Codes (**Appendix H**) to code a prenatal diagnosis made. If 'Other' diagnosis, specify. 'Other' codes include 199, 299, 399, 499, 599, 699 and 799, according to system involvement.

i. If OTHER, specify.

4. **Did the infant die ≤12 hours after birth?**

This includes infants who die in the delivery room.

IF YES, COMPLETE FORM NG03E.

5. **Sex:**

Code the stated sex of the infant.

1 = Male

2 = Female

3 = Ambiguous, there must be confirmation of ambiguous genitalia by either a genetics consult

at time of birth or pathology report if infant expires.

6. **Gestational age in weeks and days:**

Record the best estimate of gestational age.:

a. **Method used to determine recorded gestational age**

Code the method used to determine recorded gestational age.

1 = Obstetric estimate

Obstetric estimate is based on last menstrual period, obstetrical parameters, and/or early (first trimester) prenatal ultrasound as recorded in the maternal chart.

2 = Neonatal estimate

Neonatologist's estimate is based on physical criteria, neurologic examination, or combined physical and gestational age exam (Ballard, Dubowitz or Finnstrom).

If the OB date based on last menstrual period (not early ultrasound) disagrees with the Neonatologist's estimate by more than 2 weeks, the Neonatologist's estimate should prevail.

7. **Apgar score - 1 minute:**
Use the official one minute Apgar score as assigned in the delivery room from delivery chart.
8. **Apgar score - 5 minute:**
Use the official five minute Apgar score as assigned in the delivery room from delivery chart..
9. **Apgar score – 10 minutes**
Use the official 10 minute Apgar score as assigned in the delivery room from delivery chart.
10. **Birth resuscitation/stabilization:**
Code 'Y' for all support provided to the infant at the time of birth.
 - a. **Oxygen?**
Supplemental O₂ (FiO₂ > .21) delivered to the infant via face mask, hood, CPAP, or ET tube.
 - b. **Positive pressure ventilation?**
Positive pressure ventilation with face mask or prongs and using an anesthesia bag or devices such as Neopuff.
 - c. **CPAP?**
Continuous positive airway pressure delivered by CPAP device or mask.
 - d. **Intubation?**
Insertion of a tube (even if transiently) into the trachea to allow positive pressure ventilation for breathing. If intubation was done for suctioning or to give surfactant and immediately removed, it should not be included here.
 - e. **Chest compression?**
External pressure over central chest to contract the heart.
 - f. **Epinephrine?**
Epinephrine delivered intravenously or intratracheally for resuscitation.
11. **Is there documentation of delayed cord clamping?**
Code 'Y' if infant had documented delayed cord clamping. Otherwise code 'N.'
 1. **If YES, timing of cord clamping delay**
Code the timing of cord clamping delay:
 - 1 = < 30 seconds
 - 2 = 30 – 60 seconds
 - 3 = > 1 min – 3 min
 - 4 = > 3 min – prior to stop of pulsation
 - 5 = Until stop of pulsation
 - 6 = Other, specify _____
12. **Is there documentation of cord milking?**
Code 'Y' if infant had documented cord milking. Otherwise code 'N.'.
13. **Birth weight:**
Record the birth weight in grams of the infant as recorded in one of the following places on the chart in order of preference (an individual center may employ a different ordering if it is deemed more reliable):

- 1) On the labor and delivery record
- 2) On the nursery admission record
- 3) On the admission physical examination form
- 4) The pathology report when an infant expires

14. Length:

Record the length in centimeters, as recorded on the admission physical or in the child's medical record within 72 hours of birth.

15. Head circumference:

Record the head circumference in centimeters as recorded or in the child's medical record within 72 hours of birth:

- 1) On the admission physical exam form
- 2) On the nursery admission record
- 3) The pathology report if the infant died

16. Was any thermal product used to improve temperature regulation?

Code 'Y' if source documentation is found on thermal wrap use in the patients chart.

Code "N" if there is no source documentation. At sites where thermal wrap use at delivery is "*standard of care*" but source documentation cannot be found, Coordinators should code this question with 'Unknown' to indicate unknown. Thermal wraps may include NeoWrap, saran wrap, any kind of polyethylene (plastic) wrap/bag, or exothermic mattress. Do not include plastic covers, warming bed or tents used to cover the infant in an open bed warmer while in the NICU.

17. Record infant's first temperature documented after admission to nursery?

Record the first temperature recorded following admission to the nursery. For centers who admit infants to a holding area or a delivery room stabilization area, record the first temperature taken in this location. Record the temperature in Centigrade or Fahrenheit.

b. Celsius

c. Fahrenheit

c. Date

d. Time

e. Source

Record the source of the temperature as:

- 1 = rectal
- 2 = axillary
- 3 = skin

Chapter 4

Survey of Morbidity and Mortality - Outcome Data Collection

4.1 Overview of Clinical Outcome Data Collection

Outcome data will be collected for each infant from time of birth until Status is reached. Detailed information is listed below.

4.2 Clinical Outcome - Forms NG03, NG03E and NG05

The outcome data form NG03 (to be completed from the baby's chart through day 120, discharge, transfer or death, whichever comes first) was designed to summarize each baby's clinical course. For infants who die in 12 hours or less, the early death form NG03E is completed in lieu of the NG03 and NG07, as there are many questions on the NG03 and NG07 which do not apply to infants who die early. In addition, because an infant's values for some items on NG03 could change after 120 days of hospitalization, form NG05 was created. The NG05 (late clinical outcome form) contains a subset of the items on NG03 and should be completed only for infants hospitalized greater than 120 days, after they have died, have been discharged or have been transferred.

It is very important that form NG03 or form NG03E be coded and entered in the DMS as soon as possible after the baby's discharge, transfer, death or as soon as possible after day 120. When applicable, paper documentation versus electronic should be obtained prior to the infant's chart leaving the NICU for Medical Records.

It should be noted that form NG03 or form NG03E should not be held back for coding or entry into the DMS by waiting for the autopsy report. If the autopsy result for cause of death is to be included and results are not yet available, enter the temporarily missing data code 'TM', and enter final autopsy cause of death in the DMS when it becomes available.

4.2.1 Discharge from the NICU to Home

This is the straightforward case: The first date the baby is discharged to home, the baby's clinical data are all obtainable from the medical record. If the baby is readmitted to the NICU from home, no further data will be collected.

4.2.2 Transfer to Another Location within the Center

When the baby is transferred from the NICU to a step down unit, another floor, or even another hospital within the same clinical center, the coordinator is still expected to keep track of the baby and complete form NG03 when the baby finally leaves the clinical center.

4.2.3 Transfer to Another Facility

The Network Coordinator should attempt to determine the final outcome (death or discharge) for each baby that is transferred out of the clinical center to another facility. If an infant is transferred back within 7 days (without being discharged to home), do not count this as a discharge but rather as a continuous admission.

Transfer to another NRN Center does not meet the NRN definition of transfer (status). The following procedures should be followed to document a transfer between NRN Centers and to allow for complete and accurate collection of GDB data for infants transferred from one NRN Center to another NRN Center during the hospitalization. Institutional and local IRB policy should be followed for the release and transfer of information between hospitals.

The procedures outlined below should be strongly considered when completing the hard copy paper forms for an infant that is transferred from one NRN Center to another. Completion of the electronic forms will depend on the individual case and the Centers involved.

The NRN Center of origin for the transferring infant will complete required GDB forms with the most current status of the infant for receiving NRN center.

NG02 will be the responsibility of the originating hospital, as this information is available from the maternal record and admission documents for the infant.

NG03E will not be included, as any infant born at an NRN center who dies within 12 hours of birth will not likely be a candidate for transfer.

For infants transferred ≤120 days of age, the following data forms should be provided by the transferring center to the receiving center:

NG03

Provide a Y response to any question where the infant has already 'met' the data point, even if it may occur again (i.e. sepsis or pneumothorax). Otherwise, leave question unanswered and this will be completed by the receiving center at discharge, death or 120 days per usual NRN practice.

Provide all responses, Y/N or dates for questions that are time limited if the infant has reached or surpassed the time point noted in the question. This would include questions that ask for data in the first 24 hours, <72 hours, first 28 days or at 36 weeks. If the infant has not reached the time point, leave blank.

Provide responses for questions even if the item may be ongoing (i.e. episode of sepsis, surgeries) or questions where the data may be updated after transfer (i.e. most severe cranial sonogram).

Provide all counts up to the time of transfer for questions that ask for number of days (i.e. breast milk) or number of courses (i.e. indomethacin/ibuprofen). Provide the end date of counts to allow the receiving center to continue the counting as needed.

NG07

Complete the highest mode of support for each of the time points met. For time points not yet met, leave blank and provide dates for each type of support the infant has received since the last time point met.

If an infant is transferred >120 days, the NG07 will have been completed by the transferring hospital.

For infants transferred >120 days of age, the following data forms should be provided by the originating center to the receiving center:

NG05

If the infant is >120 days of age and still hospitalized in the receiving NRN center, complete the data form starting with section B for questions that would be answered Y. If the infant has not had the problem (i.e. problem that has caused hospitalization >120 days or treatment, testing or ethics/palliative care) leave the questions blank.

4.2.4 *Death*

The cause of death is recorded using descriptions and codes listed in **Appendix C**.

4.3 **NG03: GENERIC CLINICAL OUTCOME FORM**

4.3.1 *Section A - STATUS*

1. **Status of infant at time of completion of form**

- 1 = if infant was discharged to home
- 2 = if infant is still in the hospital at 120 days
- 3 = if infant was transferred to another facility without returning to a Network hospital within 7 days. This includes transfers to hospitals outside the Network Center or chronic care facilities
- 5 = if the infant died

2. **Date of status**

Record date of status.

3. **Weight at status**

Weight in grams on day of status (preferably), or within 7 days.

4. **Length at status**

Length in centimeters on day of status (preferably), or within 7 days.

5. **Head circumference at status**

Head circumference in centimeters on day of status (preferably), or within 7 days.

4.3.2 *Section B - ETHICS/PALLIATIVE CARE*

1. **At any time after birth (prior to NG03 Status) was there documentation of discussion with parents to limit, withdraw or not escalate care?**

Code 'Y' if prior to 120 days, there was documentation of a discussion with parent to limit, withdraw or not escalate care. Otherwise, code 'N'.

2. **Were the following treatments withheld, limited or withdrawn at any time with the intent to limit care?**

Code 'Y' to all that apply, otherwise code 'N'.

a. **Intubation/ventilation**

b. **Nutrition/Hydration**

3. **Palliative care consult obtained?**

Code 'Y' if there was documentation that a palliative care consult was obtained. Otherwise code 'No'.

4. **Ethics consult obtained?**

Code 'Y' if there was documentation that an ethics consult was obtained. Otherwise code 'N.'

4.3.3 *Section C - TRANSFER*

This section refers to when the infant is sent to another hospital or chronic care facility without returning in 7 days. Complete this section if status (A1) of infant at time of completion of this form is "transferred."

1. **Date of transfer:**
Record date of transfer.
4. **Final outcome:**
 - 1 = if the baby died in hospital (informed via telephone contact or letter from the other hospital)
 - 2 = if the baby was discharged to home
 - 3 = if the baby remains in the hospital at one year of age

4.3.4 Section D - DISCHARGE ALIVE

Complete this section if the Status (A1) of the infant at the time of completion of this form is discharged home'

1. **Date of discharge to home**
Record date of discharge.
2. **Discharged home on continuous oxygen?**
Code 'Y' if discharged home on continuous oxygen. Code 'N' if otherwise'.
3. **Discharged home on any of the following medications?**
Code 'Y' if infant is discharged home on any medication(s) listed below. Code 'N' if otherwise. For questions 3.a.-3.i., code 'T' if infant discharged on a trial medication in a masked placebo controlled study. Refer to section 3.2.1 for details.
 - a. **Diuretics** - including but not limited to furosemide, spironolactone, chlorothiazide, or hydrochlorothiazide
 - b. **Bronchodilators** - including but not limited to albuterol
 - c. **Anticonvulsants** - including but not limited to phenobarbital, dilantin, valproic acid, depakane, carbamapexine
 - d. **Antireflux medications** - including but not limited to ranitidine, famotidine, metaclopramide.
 - e. **Antihypertensive medications** - including but not limited to hydralazine.
 - f. **Methylxanthines** - including but not limited to caffeine, theophylline, and aminophylline.
 - g. **Inhaled steroids** - including but not limited to budesonide (pulmicort) and fluticasone (flovent).
 - h. **Systemic steroids** – including but not limited to dexamethasone, hydrocortisone, and prednisone/prednisolone.
 - i. **Other, specify**
Specify other medication on which infant was discharged home. Do not include multivitamins or iron.
4. **Discharged home receiving any human milk?**
Code 'Y' if infant is discharged home receiving any human milk. Code 'N' if otherwise.
 - a. **If YES, type of milk**

Use the following codes:

- 1= Maternal Milk
- 2= Donor Milk
- 3= Both
- 4= Unknown

4.3.5 Section E - DEATH

Complete this section if the Status of the infant at time of completion of this form is '5=Death' or if the final outcome for a transferred infant is 'Died in Hospital.' Include additional information about death that becomes available after status (such as blood culture results, late entered discharge notes, etc.)

1. **Date of death:**
Record date of death.
2. **Autopsy performed?**
Code "Y" if autopsy performed. If an autopsy is performed, the cause of death should not be coded until the results of the autopsy are known. Code 'N' otherwise.
3. **Contributory cause of death:**
This should be the underlying, proximate disease which initiated the series of events leading to death. This underlying cause must be (1) etiologically specific and (2) antecedent to all other causes with respect to time and pathologic relationship. Without the underlying cause, death would not have occurred. The cause of death should be based on both clinical evidence and autopsy findings. In the absence of an autopsy, the clinical evidence will be used. Only one option is coded after consultation with the PI or alternate PI.

A list of the causes of death are in **Appendix C**.
4. **If cause of death is code 10 (Congenital malformation), code 31 (Sepsis, specify organism), or code 90 (Other), specify.**

Part 2: 36 WEEK INFORMATION

4.3.6 Section F. –36 Week Information

Complete this section for all infants.

1. **Status at 36 weeks**
 - 1 = infant discharged to home
 - 2 = infant in the hospital at 36 weeks gestational age
 - 3 = infant transferred to another facility without returning in 7 days
 - 5 = infant died**If "2," (In hospital)**
 - a. **Date of 36 week measurement**
Record the actual date that this measurement was taken.

NOTE: The measurements may not be taken on the same calendar day, but all should be within the window at 36 weeks (± 7 days). If all the measurements are not taken on the same day, then the date recorded should be that of the weight at 36 weeks ± 7 days.

- b. **Weight**
Record the weight in grams at 36 weeks (± 7 days) gestational age
- c. **Length**
Record the length in centimeters at 36 weeks (± 7 days) gestational age
- d. **Head circumference**
Record the head circumference in centimeters at 36 weeks (± 7 days) gestational age

Part 3: Organ Systems

4.3.7 Section G - PULMONARY

1. **Did the baby receive surfactant?**

This includes any surfactant preparation used at any location (delivery room, NICU or at referring hospital).

If YES, record the following:

a. **Date of first dose?**

Record the date the first dose was given (day begins at 00:00, ends at 23:59).

b. **Was first dose given in the delivery room?**

Code 'Y' if first dose was given in the delivery room. Code 'N' otherwise.

c. **Type of surfactant given:**

Record the type of surfactant given:

1 = Beractant (Survanta)

2 = Poractant alfa (Curosurf)

3 = Calfactant (Infasurf)

9 = Other

d. **Was the surfactant mixed with budesonide?**

Code 'Y' if the surfactant administered was mixed with budesonide. Code 'N' if otherwise. Code 'T' if the infant was enrolled in the BiB trial. If 'TI' refer to section 3.2.1 for detailed instructions.

e. **Method of administration:**

Record the method of administration:

1 = ETT (endotracheal tube) – includes INSURE (INTubate, SURFactant, REMove) given using an endotracheal tube

2 = LISA/MIST (less invasive surfactant administration or minimally invasive surfactant treatment) – defined by use of a specialized catheter used to inject surfactant below the vocal cords

3 = Aerosolized – may be given using a nebulizer via mask or prongs

4 = LMA (laryngeal mask airway)

9 = Other

2. **Pneumothorax?**

Code 'Y' if pneumothorax is documented in the infant's chart. Pneumothorax is a collection of air in the pleural space where a lucency is identified in the pleural space with displacement of the lung away from the chest wall. Do not include a pneumothorax resulting from a thoracotomy during surgery (e.g. PDA ligation).

3. **PIE?**

Code 'Y' if pulmonary interstitial emphysema (PIE) is documented in the infant's chart. PIE is a radiographic and pathologic diagnosis made when air ruptures from alveoli or small airways into the perivascular tissues of the lung. Include PIE documented in the patient notes. Otherwise code 'N'.

4. **Pulmonary hemorrhage?**

Code 'Y' if there was bright red blood per the ET tube associated with clinical deterioration.

5. **Steroids for BPD/CLD?**

Code 'Y' if the infant received any doses or courses of systemic steroids to prevent or treat bronchopulmonary dysplasia/chronic lung disease. Do not include inhaled steroids or doses of steroids given for extubation and/or stridor.

If initial treatment/medication is given under a masked or unmasked study protocol or if subsequent medication is received as part of clinical care post trial, refer to Section 3.2.1 for detailed instructions.

Code 'N' otherwise.

a. **If YES, Date of first dose:**

If 'Y' record the date the infant received the first dose of steroids or trial medication.

b. **Was more than one course given,?**

Code "Y" if more than one course was given. Code 'N' otherwise.

c. **Was total exposure > 14 days**

Code 'Y' if the total exposure was > 14 days. Code 'N' otherwise.

d. **Drugs prescribed**

Code the type of drugs prescribed.

2 = dexamethasone

6 = hydrocortisone

7 = prednisolone.

6. **Did infant receive inhaled nitric oxide?**

Code 'Y' if infant received inhaled nitric oxide.

If initial treatment/medication is given under a masked or unmasked study protocol, or if subsequent medication is received as part of clinical care post trial, refer to section 3.2.1 for detailed instructions.

a. **If YES or Trial, Date of first exposure**

If 'Y' or 'T', record the date of first exposure. If 'T' refer to section 3.2.1 for detailed instructions.

7. **Did the infant receive vitamin A?**

Code 'Y' if infant received vitamin A.

If the treatment/medication was given under a masked or unmasked study protocol, or if subsequent medication is received as part of clinical care post trial, refer to section 3.2.1 for detailed instructions.

4.3.8 *Section H - CARDIOVASCULAR*

1. **Patent ductus arteriosus (PDA)?**

Code 'Y' if clinical evidence of left to right PDA shunt documented by continuous murmur, hyperdynamic precordium, bounding pulses, wide pulse pressure, congestive heart failure, increased pulmonary vasculature or cardiomegaly by CXR, and/or increased oxygen requirement or ECHO evidence of PDA with documentation of left to right ductal shunting. Code 'N' if otherwise.

If YES, **Treatment**

a. **Indomethacin?**

Code 'Y' if infant received indomethacin. If infant treatment/medication was given under a masked or unmasked protocol, or if subsequent medication is received as part of clinical care post trial refer to section 3.2.1 for detailed instructions. Code 'N' otherwise.

a. **Ibuprofen?**

Code 'Y' if infant received ibuprofen. If infant treatment/medication was given under a masked or unmasked protocol or if subsequent medication is received as part of clinical care post trial refer to Section 3.2.1 for detailed instructions. Code 'N' if otherwise.

b. **Acetaminophen?**

Code 'Y' if infant received acetaminophen. If infant treatment/medication was given under a masked or unmasked protocol or if subsequent medication is received as part of clinical care post trial refer to Section 3.2.1 for detailed. Code 'N' if otherwise.

d. **Cardiac catheterization for PDA closure?**

Code 'Y' if the infant had a cardiac catheterization for PDA closure. Code 'N' if otherwise. Failed catheterization closures should also be included.

IF YES,

1. **Date of cardiac catheterization for PDA closure**

If 'Y' record the date the cardiac catheterization for PDA closure was performed.

2. **Was the cardiac catheterization for PDA closure successful?**

Code 'Y' if the cardiac catheterization for PDA closure was successful.
Code 'N' if otherwise.

3. Did the infant initiate any new vasoactive drugs within 24 hours after PDA closure?

Code 'Y' if the infant received any new vasoactive drugs within 24 hours after PDA closure. Code 'N' otherwise.

e. Surgery?

Code 'Y' if surgical ligation was required to close the PDA. Code 'N' otherwise.

1. Did the infant initiate any new vasoactive drugs within 24 hours after PDA closure?

Code 'Y' if the infant received any new vasoactive drugs within 24 hours after PDA closure. Code 'N' otherwise.

f. Other?

Code 'Y' if another treatment such as coils was used. Do not include fluid restriction.

If YES, specify treatment given. Code 'N' otherwise.

2. Was the infant treated for hypotension in the first 24 hrs of life?

Code 'Y' if the infant was treated for hypotension in first 24 hours of life. Code 'N' if otherwise.

If YES,,

a. Volume bolus?

Code 'Y' if a volume bolus was given. Code 'N' otherwise.

b. Initiation of vasopressors?

Code 'Y' if there was an initiation of vasopressors. Code 'N' otherwise.

c. Initiation of systemic steroid?

Code 'Y' if there was an initiation of systemic steroid. Code 'N' otherwise.

3. Was an echocardiogram done to monitor for pulmonary hypertension in the first week of life?

If YES,

a. Date

Record date of the echocardiogram done to monitor for pulmonary hypertension in the first week of life.

b. Treated?

Code 'Y' if the infant received treatment for pulmonary hypertension. Code 'N' otherwise.

4. Was an echocardiogram done to monitor for late or chronic pulmonary hypertension?

If YES,

a. Date

Record date of the echocardiogram done to monitor for late or chronic pulmonary hypertension.

b. Treated?

Code 'Y' if the infant received treatment for late or chronic pulmonary hypertension.
Code 'N' otherwise.

4.3.9 Section I - NEUROLOGY

1. Was indomethacin given within the first 24 hours of life?

If the infant was known to have received indomethacin within the first 24 hours of life, code as 'Y'. If infant was enrolled in a masked or unmasked protocol, or if subsequent medication is received as part of clinical care post trial, refer to Section 3.2.1 for detailed instructions. Code 'N' otherwise.

2. Were seizures confirmed by EEG?

Code 'Y' if there was documentation of EEG findings that were consistent with seizures.
Code 'N' otherwise.

3. Were any cranial sonograms done within 28 days of birth?

Code 'Y' if cranial sonograms were done within the first 28 days of life. Code 'N' otherwise.

IF **NO**, GO TO QUESTION I6

a. If YES, Are all studies without evidence of intracranial hemorrhage, periventricular leukomalacia or ventriculomegaly?

Intracranial hemorrhage refers to any bleed in the brain, including germinal matrix.

IF YES, GO TO QUESTION I6.

If No, continue with question I3b.

b. Date of sonogram with most severe findings:

Record the date of the cranial sonogram with the most severe findings within the first 28 days. Determination of the most severe findings should be based on the following rank order from least severe to most severe. Least severe is blood/echodensity in the germinal matrix/subependymal area followed by blood echodensity in the ventricle, ventricular dilatation and blood/echodensity in the parenchyma is the most severe. **Use the earliest scan of an infant in whom multiple scans have the same most severe findings.**

For discrete lesions, echodensity or increased echogenicity may be considered equivalent.

PLEASE NOTE: Document all the findings on the cranial imaging identified above. For all imaging studies (questions I3, I7 and I8) only record definite findings and do not record those that are interpreted as uncertain, probable or possible. **If it is unclear how to record the results of an image, discuss findings with the PI or his designee.** The documentation for items 3c-3g does not correspond to a grade of hemorrhage. Items 3c-3g should be considered independent of each other and therefore an infant may have more than one item recorded. For items d-g, indicate

side of involvement by coding 'Y' under right (R), left (L), or both. Questions about midline shifts are not right and left but yes and no answers.

- c. **Blood/echodensity in germinal matrix/subependymal area?**
Code 'Y' if blood/echodensity in the germinal matrix/subependymal area is documented. **When blood echodensity is seen in the ventricle but NOT in the germinal matrix, code 'N' for germinal matrix hemorrhage.**
- d. **Blood/echodensity in the ventricle?**
Code 'Y' if blood/echodensity in the ventricle is documented. This finding should be recorded independent of the size of the ventricle. Indicate side of involvement right, left or both.
- e. **Ventricular size enlarged with concurrent or prior blood in the ventricles?**
Code 'Y' if ventricular enlargement occurs in association with blood/echodensity in the ventricular system on any scan. Indicate side of involvement, right, left or both.
- f. **Ventricular size enlarged without concurrent or prior blood in the ventricles?**
Code 'Y' if ventricular enlargement occurs without blood/echodensity in the ventricular system on any scan. Indicate side of involvement, right, left or both.
- g. **Blood/echodensity in the parenchyma?**
Code 'Y' if blood/echodensity in the parenchyma is documented. Intraparenchymal echodensities may or may not be accompanied by blood/echodensity in the ventricle. Intraparenchymal echodensity differs from diffuse increased echogenicity. Indicate side of involvement, left, right or both.

NOTE: For sonographic reports that are limited to a grade of ICH (usually I-IV) without a description of the findings record as follows:

Grade I: record 'Y' for question 3c
 Grade II: record 'Y' for question 3d
 Grade III: record 'Y' for questions 3d and 3e
 Grade IV: record 'Y' for question 3g

For sonographic reports that are limited to isolated ventricular dilatation without other associated findings (**no blood/echodensity in ventricles**) record 'Y' for question 3f. If YES,

1. **Midline shift.**

Code 'Y' if midline shift was documented. Code 'N' otherwise.

h. **Blood/echodensity in the basal ganglia or the thalamus?**

Code 'Y' if blood/echodensity in the basal ganglia or the thalamus is documented. Code 'N' otherwise.

i. **Cerebellar hemorrhage?**

Code 'Y' if cerebellar hemorrhage was documented on the cranial ultrasound.
 Code 'N' otherwise.

For sonographic reports that contain both a grade of ICH and a description, prioritize the descriptive findings over the assigned grade if the two pieces of information are not consistent. **Confirm the latter prioritization with the PI.**

4. **Cystic Periventricular Leukomalacia (cPVL) within 28 days?**

Record as cPVL when this diagnosis is used. In the absence of a diagnosis of cystic PVL on sonographic reports, use cPVL when cysts (echolucencies) are described in the white matter around the ventricle. These cysts are most commonly dorsal and lateral to the external angle of the lateral ventricle. Echolucencies may be single or multiple, may be bilateral or unilateral, may vary in size, and may be diffuse or focal in distribution along the front to back axis of the head. **If PVL is reported and cysts or echolucencies within the brain parenchyma are not specified, review the scan with the PI or designee of the PI to verify the presence of this finding.** Indicate side of involvement by coding 'Y' under right (R), left (L), or both. Code 'N' otherwise.

5. **Porencephalic cyst within 28 days?**

Porencephalic cyst will be used to lump all cystic disease other than cystic PVL. In the absence of a diagnosis on sonographic reports, use porencephalic cyst when there is a single unitary cyst within the cerebral hemisphere that may or may not communicate with the lateral ventricle. These cysts may be congenital and can be present on sonograms shortly after birth, or may evolve over time at the site of a previous parenchymal blood/echodensity (3g from above). These cysts may also be termed post-hemorrhagic cysts, or if multiple, multi-cystic encephalomalacia. Do not include subependymal cysts or choroid plexus cysts as part of this category. The latter two are minor cysts that will not be recorded. Indicate side of involvement by coding 'Y' under right, left, or both. Code 'N' otherwise.

6. **Were any cranial imaging studies done after day 28?**

Code 'Y' if any cranial imaging was done after day 28. Code 'N' otherwise.

If No, Go to Section J.

7. **Was a sonogram performed after day 28?**

Code 'Y' if sonogram was performed after day 28. Code 'N' if otherwise.

If YES, record information for the sonogram closest to 36 weeks postmenstrual age.

a. **Date of image**

Record the date of the imaging study closest to 36 weeks PMA.

b. **Normal Study?**

Code 'Y' if the results of the cranial imaging closest to 36 weeks postmenstrual age was reported to be normal.

If Yes, go to Question I8.

If No, go to Question I7c-e.

If NO,

c. **Ventricular size enlarged?**

Code 'Y' if ventricular size was documented to be enlarged. Indicate side of involvement if enlarged by marking 'Y' under right, left or both.

d. **Cystic Periventricular Leukomalacia (cPVL)?**

Record as cPVL when this diagnosis is used. In the absence of a diagnosis of cystic PVL on sonographic reports, use cPVL when cysts (echolucencies) are described in the white matter around the ventricle. These cysts are most commonly dorsal and

lateral to the external angle of the lateral ventricle. Echolucencies may be single or multiple, may be bilateral or unilateral, may vary in size, and may be diffuse or focal in distribution along the front to back axis of the head. **If PVL is reported and cysts or echolucencies within the brain parenchyma are not specified, review the scan with the PI or designee of the PI to verify the presence of this finding.** Indicate side of involvement by marking 'Y' under right , left , or both.

e. **Porencephalic cyst?**

Porencephalic cyst will be used to lump all cystic disease other than cystic PVL. In the absence of a diagnosis on sonographic reports, use porencephalic cyst when there is a single unitary cyst within the cerebral hemisphere that may or may not communicate with the lateral ventricle. These cysts may be congenital and can be present on sonograms shortly after birth, or may evolve over time at the site of a previous parenchymal blood/echodensity (3g from above). These cysts may also be termed post-hemorrhagic cysts, or if more than one, multi-cystic encephalomalacia. Do not include subependymal cysts or choroid plexus cysts as part of this category. The latter two are minor cysts that will not be recorded. Indicate side of involvement by marking 'Y' under right , left , or both.

8. Was an MRI performed after day 28?

Code 'Y' if MRI was performed after day 28. Code 'N' if otherwise.

If YES , record information for the MRI closest to 36 weeks postmenstrual age.

a. **Date of image:**

Record the date of the imaging study closest to 36 weeks PCA.

b. **Normal Study?**

Code 'Y' if the results of the cranial imaging closest to 36 weeks postmenstrual age was reported to be normal.

If Yes, go to Section J.

If No, complete I8c-e.

c. **Ventricular size enlarged?**

Code 'Y' if ventricular size was documented to be enlarged. Indicate side of involvement if enlarged. Indicate side of involvement by marking 'Y' under right , left , or both.

d. **Cystic Periventricular Leukomalacia (cPVL)?**

Record as cPVL when this diagnosis is used. In the absence of this diagnosis, code cPVL when cysts are described in the white matter surrounding the ventricle. Cysts may be single or multiple, may be bilateral or unilateral, may vary in size, and may be diffuse or focal in distribution from the front to back (cephalad to caudad) axis of the head. Non-cystic PVL should not be recorded under this entry. Indicate side of involvement by marking 'Y' under right , left or both.

e. **Porencephalic/posthemorrhagic cyst/multicystic encephalomalacia?**

Porencephalic cyst will be used to lump all cystic disease other than cystic PVL. Use porencephalic cyst when there is a single unitary cyst within the cerebral hemisphere that may or may not communicate with the lateral ventricle. These cysts may be

congenital and be present on sonograms shortly after birth, or may evolve over time at the site of a previous parenchymal blood/echodensity (3g from above). These cysts may also be termed post-hemorrhagic cysts, or if multiple, multi-cystic encephalomalacia. Do not include subependymal cysts or choroid plexus cysts as part of this category. The latter two are minor cysts that will not be recorded. Indicate side of involvement by coding 'Y' under right , left , or both.

4.3.10 Section J - GASTROINTESTINAL

1. Total days of parenteral nutrition?

Record the number of days on which the baby received parenteral alimentation including amino acids or lipid solution at some point during the day. Enter '0' if baby never received parenteral nutrition of any type described above.

2. Did the baby receive enteral feeds?

Code 'Y' if the infant was fed enterally. Feeds consist of formula or breast milk. Colostrum priming is coded as enteral feeds if it reaches the stomach. Sterile water is not considered enteral feeding. Code 'N' otherwise.

If YES,

a. Type of first enteral feeding?

Code type of enteral feeds infant first received. If more than one type of feed is given record the type that made up the majority of the first feeding:

1 = Maternal milk

2 = Donor milk

3 = Formula

b. Date of first enteral feed

Record the date of the first enteral feed.

c. Did enteral feeds reach 120 ml/kg/day?

Code 'Y' if the enteral feeds reached 120 ml/kg/day. Calculate using the current weight. Code 'N' otherwise.

If YES,

1. Date first achieved?

Record the date first achieved.

d. Type of human milk infant received in the first 28 days?

Code all types of human milk received in the first 28 days:

1 = Maternal if the infant only received maternal milk in the first 28 days

2 = Donor if the infant only received donor milk in the first 28 days

4 = None if the infant did not receive maternal milk, donor milk or was not in a feeding trial in the first 28 days

d. Type of fortification of human milk received in the first 28 days?

Code the type of fortification of human milk received in the first 28 days:

1 = bovine-based fortifier

2 = human milk-based fortifier

3 = Both

4 = Neither

f. Did the infant receive probiotics in the first 28 days?

Code 'Y' if the infant received probiotics in the first 28 days.

If infant treatment/medication was given under a masked or unmasked study protocol, or if subsequent medication is received as part of clinical care post trial, refer to Section 3.2.1. Code 'N' otherwise.

See Appendix L for list of probiotics.

3. Proven necrotizing enterocolitis (NEC) diagnosis

Code the correct response using the Modified Bell's Staging Criteria for NEC in **Appendix G**. Also refer to **Appendix I**.

0 = Absent/Suspect when there is no necrotizing enterocolitis present or Bell's Stage IA or IB.

2 = Proven, no surgery, Stages IIA, IIB or IIIA

3 = Proven, surgery, Stage IIIB

4 = Proven, autopsy

Code type of surgery to include laparotomy or drain and other procedures in Section Q using Appendix E.

a. If proven NEC, record date of first episode.

Record date of first episode. If proven NEC is discovered during the autopsy then list the date of death as the date of first episode of proven NEC.

b. Did the infant restart feeding after first episode of NEC?

Code 'Y'; if the infant restarted feeding after the first episode of NEC. Code 'N' otherwise.

If YES,

i. Date feeding restarted after first episode of NEC

Record date feeding restarted after first episode of NEC.

c. Was creatinine measured?

Code 'Y' if at least one serum creatinine was measured within 10 days of the diagnosis. Code 'N' otherwise.

If YES,

1. Highest value

Record highest serum creatinine level obtained in mg/dL.

2. Date

Record date highest serum creatinine level was measured.

4. Spontaneous gastrointestinal perforation without proven NEC?

Code 'Y' if the infant has a spontaneous gastrointestinal perforation without necrotizing enterocolitis. Code 'N' otherwise..

a. If YES, Date of the first spontaneous GI perforation

Record the date on which the first spontaneous gastrointestinal perforation occurred.

5. Did the infant receive any type of enema as therapy for meconium ileus or obstruction?

Code 'Y' if infant received any type of enema as therapy for meconium ileus or obstruction. Code 'N' otherwise.

a. If YES, date of the first enema

Record date of the first enema.

6. Did the infant have GI surgery that resulted in short gut?

Code 'Y' if the infant had surgery involving the GI tract that resulted in the diagnosis of short gut which includes malabsorption, severe diarrhea, gastric hypersecretion, secondary bacterial overgrowth and failure to thrive. Record surgical procedure in Section Q. Code 'N' otherwise.

4.3.11 Section K - HEARING

1. Was a hearing screen performed prior to status?

Code 'Y' if a hearing screen was performed to evaluate the infant's hearing prior to status. Code 'N' otherwise.

If YES,

a. Was otoacoustic emissions (OAE) testing performed?

Code 'Y' if OAE testing was performed. Code 'N' otherwise.

If YES,

1. Was OAE failed?

Code 'Y' if infant failed the OAE. Otherwise, code 'N'.

If YES,

i. Unilateral or bilateral fail?

Code

1=Unilateral

2=Bilateral

b. Was automated auditory brainstem response (AABR) performed?

Code 'Y' if automated auditory brainstem response (AABR) was performed.
Code 'N' otherwise.

If YES,

1. Was AABR failed?

Code 'Y' if infant failed the AABR. Otherwise, code 'N'.

If YES,

i. Unilateral or bilateral fail?

Code
1=Unilateral
2=Bilateral

2. Was a diagnostic auditory brainstem response (ABR) performed prior to status?

Code 'Y' if there was documentation of an ABR performed prior to status. Otherwise, code 'N'.

If YES,

a. Was hearing loss documented?

Code 'Y' if hearing loss was documented. Code 'N' if no hearing loss was documented.

If YES,

1. Unilateral or bilateral hearing loss?

Code
1=Unilateral
2=Bilateral

4.3.12. Section L - OPHTHALMOLOGY

Appendix D contains a Retinopathy of Prematurity (ROP) Diagram.

1. Was an exam performed for Retinopathy of Prematurity (ROP)?

Review the medical record to determine if an examination was performed for ROP and flag all examinations found in order to answer the remainder of the questions. Code 'Y' if an ophthalmologist examined the infant's eyes for ROP. The examinations usually begin at 4 to 6 weeks and continue until the retinal vasculature is mature. If coded as 'N', no other questions in this section are required.

If YES,

a. Was ROP diagnosed in either eye?

Code 'Y' if ROP diagnosed prior to Status (any stage) in either eye in any of the examinations. Code 'N' otherwise.

If YES,

(1) Did ROP reach stage 3 or worse in either eye?

Code 'Y' if it reached stage 3 or worse in either eye. Code 'N' otherwise.

(2) Did plus disease develop in either eye?

Plus disease is noted by the ophthalmologist in addition to the stage and zone. It is recorded as present or not. Sometimes it is referred to as "posterior pole

vascular dilation and tortuosity". Usually if this is present on any examination, it is close to the worst ROP examination for the infant. Code 'Y' if plus disease was observed on the worst examination recorded above for either eye. When the posterior veins of the retina are enlarged and the arterioles tortuous, then the designation "plus" is added to the ROP stage number. For example, Stage 2 with plus disease is sometimes written 2+. Code 'N' otherwise.

(3) Did infant have stage 1 or stage 2 ROP diagnosed in Zone 1?

Code 'Y' if infant had stage 1 or stage 2 ROP diagnosed in Zone 1. Code 'N' otherwise.

b. Intervention therapies:

(1) Was retinal ablation performed in either eye (laser and/or cryotherapy)?

If 'Y' code type and date of surgery in Section K using Appendix E. Code 'N' otherwise.

(2) Was scleral buckle or vitrectomy performed in either eye?

If 'Y' code type and date of surgery in Section K using Appendix E. Code 'N' otherwise.

(3) Avastin or other anti-VEGF drug

Code "Y" if the infant received Avastin or other anti-VEGF medication.

If infant treatment/medication was given under a masked or unmasked study protocol, or if subsequent medication is received as part of clinical care post trial, refer to Section 3.2.1. Code 'N' otherwise.

c. At the time of reaching status, indicate the most appropriate description as described below:

1 = Determined, favorable in both eyes

EACH eye met one of the following criteria

- Mature vessels (fully vascularized)
- Immature vessels in zone III for two consecutive examinations
- Acute ROP of stage 1 or 2 in zone III for two consecutive examinations
- ROP in zone II or zone III but determined to be clearly regressing

2 = Determined, severe ROP in either eye

Severe: EITHER eye met one of the following criteria

- Received surgery for ROP
- Retinal detachment from ROP
- Avastin injection or Anti-VEGF drug

3 = Undetermined ROP status in either eye (and neither had "severe ROP")

Undetermined: EITHER eye met one of the following criteria, code reason under (1):

- 1 = Immature vessels in zone I or zone II
- 2 = Immature vessels reaching zone III for only 1 examination
- 3 = Stage 1 or 2 ROP in zone III for only 1 examination
- 4 = Stage 3 ROP in zone III
- 5 = ROP in zone I or zone II

6 = Plus disease

4.3.13 Section M - HEMATOLOGY

1. **Blood type of infant**

1 = A

2 = B

3 = AB

4 = O

5 = Unknown

a. Rh (Rhesus) factor?

Code

1 = Positive

2 = Negative

3 = Unknown

2. **Was the infant transfused with pRBC?**

Code 'Y' (if infant received packed red blood cells at any time prior to status. Code 'N' otherwise.

If YES,

a. Date of first pRBC transfusion

Record the date of the first pRBC transfusion.

b. Lowest hemoglobin OR hematocrit prior to first transfusion

Record either the lowest hemoglobin (g/dL) or hematocrit (%) prior to first transfusion.

Only one of these values needs to be entered. Use the one that is used most often for any clinical judgments at your center.

c. Total number of pRBC transfusions

Record the number of transfusions

3. **Was the infant transfused with platelets?**

If Yes, (complete M3a-b); If "No" (skip to M4)

a. Date of first platelet transfusion:

Record the date of the first platelet transfusion

b. Lowest platelet count before transfusion

Record the lowest platelet count

4. **Did the infant receive erythropoietin or another erythropoiesis stimulating agent?**

Code 'Yes' if infant received erythropoietin or another erythropoiesis stimulating agent.

i.e. Erythropoietin alfa (Procrit, Epogen). Darbepoietin alfa (Aranesp).

If initial treatment/medication is given under a masked or unmasked study protocol or if subsequent medication is received as part of clinical care post trial, refer to section 3.2.1 for detailed instructions. Code 'n' otherwise.

5. **Highest total bilirubin in first 14 days (mg/dL)**

4.13.14 Section N – RENAL

1. Clinical Diagnosis of Acute Kidney Injury - AKI?

Code 'Y' if there is a clinical diagnosis of acute kidney injury (AKI) or renal failure at any time during the NICU stay. Code 'N' otherwise.

If YES,

a. Date of diagnosis

Record date clinical diagnosis of AKI or renal failure was made.

b. Inpatient consult with or referral to outpatient nephrology?

Code 'Y' if the clinical team either requested an inpatient pediatric nephrology consultation or made a referral for an outpatient evaluation by pediatric nephrology. Code 'N' otherwise.

2. Blood Pressure at status?

Record blood pressure (systolic and diastolic) measured within ± 1 week of status status in mmHg. If there is more than one blood pressure reported, the coordinator should discuss with the PI to determine which one to record. The clinical team may be consulted if needed. For example, if there is a high BP that is then repeated within 24 hours that is normal, if there is a note that the infant was crying during the high measurement then the site should record the repeat measurement. If the BP is consistently elevated, then the elevated value should be recorded if they are similar to one another.

3. Was there a diagnosis of hypertension?

Code 'Y' if there was a diagnosis of hypertension. Code 'N' otherwise.

If YES,

a. Were antihypertensives used?

Code 'Y' if the infant was treated for hypertension with any of the following: amlodipine, captopril, enalapril, enalaprilat, hydralazine, isradipine, labetalol, Lisinopril, nicardipine, propranolol. Code 'No' otherwise.

4. Were any serum creatinines obtained in the first 6 weeks?

Code 'Y' if there were any serum creatinines obtained in the first 6 weeks. Code 'N' otherwise.

If YES,

a. Week

Record days 1-4 in the first row and days 5-7 in the second row. In the next rows record the week number for week's 2-6.

b. Maximum creatinine date

Record the date the maximum serum creatinine was obtained for each timepoint.

1. Maximum creatinine value

Record the maximum serum creatinine value in mg/dL for each timepoint.

c. Minimum creatinine date

Record the date the minimum serum creatinine was obtained for each timepoint.

1. Minimum creatinine value

Record the minimum serum creatinine value in mg/dL for each timepoint.

Part 4: Diagnoses and Treatments

4.3.15 Section O - INFECTION

Organism codes for this section are listed in **Appendix A**. If the organism is not included in the list, code as '010' (to be assigned), and notify the DCC that a new code should be assigned. The DCC will issue new lists periodically.

1. Early onset septicemia/bacteremia (≤ 72 hrs)?

Code 'Y' if there was a positive blood culture drawn within the first 72 hours. Code 'N' otherwise.

a. If YES, organism code(s):

Record the code(s) of organism(s) identified. If there were more than two organisms identified, list the organisms felt to be most important.

2. Did the infant receive antibiotics for ≥ 5 days, starting within the first 72 hours?

Code 'Y' if the infant was treated with antibiotics for five or more days beginning by 72 hours. Include cases where the infant died before an intended therapy of five or more days was completed. Code 'N' otherwise.

a. If YES, reason:

Code reason antibiotics were given:

- 1 = Sepsis
- 2 = Meningitis
- 3 = UTI
- 4 = Pneumonia
- 5 = Other

3. Number of episodes of late onset blood culture negative clinical infection (> 72 hours to status) treated with antibiotics for ≥ 5 days?

Record the number of blood culture negative episodes, occurring after 72 hours, treated with antibiotics for five or more days. Include cases where the infant died before an intended therapy of five or more days was completed.

4. Late onset blood culture positive septicemia/bacteremia (> 72 hrs)?

Code 'Y' if there was a positive blood culture, obtained in the presence of compatible

clinical signs of septicemia, occurring after 72 hours, treated with antibiotics for ≥ 5 days. Include blood culture positive episodes in which the infant dies before an intended therapy of five or more days is completed. A new organism cultured at any time is considered an additional episode. If the same bacterial organism is cultured after 10 days of appropriate antibiotic therapy, this is considered a new episode. There are only 6 entries on the form, however, the DMS will accept up to 10 entries. Code 'N' otherwise.

If YES,

For each episode (a), record the start and stop date (b1 and b2), organism code(s) (c1-c3), and whether there was Central Line Associated Blood Stream Infection (CLABSI) (d) (record 'Y' or 'N') from positive blood culture(s) for which the infant was treated, or where there was intent to treat for ≥ 5 days. The stop date is the last day of antibiotics. A **Central Line Associated Blood Stream Infection (CLABSI)** is a laboratory-confirmed bloodstream infection (BSI) in a patient who had a central line within the 48 hour period before the development of the BSI and that is not related to an infection at another site. If more than 3 organisms are identified, list the 3 organisms felt to be the most important. If the same organism is isolated under the circumstances described in a. (above), the code is repeated.

d. If late onset sepsis, was creatinine measured?

Code 'Y' if at least one serum creatinine was measured within the 10 day period after initial diagnosis. Code 'N' otherwise.

1. If YES, highest value

Record the highest serum creatinine level obtained.

2. If YES, Date

Record the date the highest serum creatinine level was measured.

5. Was the infant diagnosed with UTI?

Code 'Y' if infant was diagnosed with a urinary tract infection (UTI) that was treated with antibiotics for ≥ 5 days. Code 'N' otherwise.

If YES,

Organism code(s):

For each episode number (a) record the date (b) and code(s) of organism(s) (1, 2, 3) associated with the UTI. **Use Appendix A to obtain code(s) for identified organism(s).** If there were more than three organisms identified, list the three organisms felt to be most important.

6. Did the infant have an LP as part of a sepsis evaluation (not for hydrocephalus)?

Code 'Y' if the infant had a lumbar puncture performed or attempted as part of a septic work up. Exclude an LP performed for a ventricular tap for hydrocephalus. There are 5 fields on the form, however, the DMS will accept up to 10 fields. Code 'N' otherwise.

If YES, **a. Record date(s) lumbar puncture** performed. Record up to 5 LPs only if they were successfully performed with CSF sent for analysis.

7. Meningitis?

Code 'Y' if there was a positive cerebrospinal fluid (CSF) culture and treatment (or intent to treat) with antibiotics or antifungals for 7 or more days. Code 'N' otherwise.

If YES,

Organism code(s):

For each episode number (a) record the date, (b) code(s) of organism(s) (1, 2, 3) identified, and (c) if the episode was confirmed by PCR. If there were more than three organisms identified, list the three organisms felt to be most important.

Use Appendix A to obtain code(s) for identified organism(s).

8. Was the infant diagnosed with any of the following other proven infections?

Code 'Y' if the infant was diagnosed with any of the following other proven infections listed below. Code 'N' otherwise.

If YES:

For each diagnosis record:

a. Diagnosis number

b. Date of Diagnosis,

c. Diagnosis code

Code:

4 = Syphilis

5 = Toxoplasmosis

6 = Human immunodeficiency virus (HIV)

7 = SARS-CoV-2 (COVID-19)

8 = Herpes simplex virus (HSV)

9 = Cytomegalovirus (CMV)

10 = Influenza

11 = Respiratory syncytial virus (RSV)

12 = Other, specify

If Other, specify.

d. Treated

Code 'Y' if the infant was treated for the infection. Code 'N' otherwise

4.3.16 Section P – SYNDROMES AND/OR MALFORMATIONS

1. Syndromes and/or major malformations?

Code 'Y' if any syndromes and/or major malformations were diagnosed, including Down syndrome, chromosomal abnormalities, and other syndromes with multi-organ involvement. Code 'N' otherwise.

a. If YES, code

Record the code(s) of the syndromes and/or major malformations diagnosed. The syndrome/major malformation codes are listed in **Appendix H: Birth Defect Codes**.

b. **If a syndrome is coded as 'Other', specify**

Document the specific name of the syndrome. 'Other' codes include 199, 299, 399, 499, 599, 699 and 799, according to system involvement.

4.13.17 Section Q – SURGERIES AND CATHETERIZATION PROCEDURES

1. **Did the infant have surgery or catheterization procedure?**

Code 'Y' for all surgical procedures and/or catheterization procedures that occurred prior to status, including surgery for ROP and fetal surgeries. Code "N" otherwise.

- a. If YES, record the date (i), all surgical codes (ii) for procedures performed on that date, and if there was surgical site infection (iii) for each surgery

i. **Date (mm/dd/yyyy)**

Record date of surgery.

ii. **Surgery code(s)**

See **Appendix E** for major surgery codes. There are 6 entries on the form, however, the DMS will accept up to 10 entries. Failed catheterization closures should also be included.

iii. **Surgical Site Infection?**

Code 'Y' if there was a surgical site infection. Code 'N' if there was not a surgical site infection. Code 'Unk' if this information is not known.

b. **If a surgery is coded as 'Other' (codes ending in 99), specify**

Record the specific name of the surgery performed.

4.4 NG03E: GENERIC EARLY DEATH FORM

4.4.1 NG03E Form Questions – Section A – Early Death

Complete this form if the infant died at ≤ 12 hours of age. This includes infants who died in the delivery room.

1. **Medical therapy initiated**

Code 'Y' to all medical therapies that were initiated anytime after birth. Code 'N' otherwise.

a. **Antibiotics?**

b. **Surfactant replacement therapy?**

If YES, record the following:

1. **Type of surfactant given:**

Record the type of surfactant given:

- 1 = Beractant (Survanta)
- 2 = Poractant alfa (Curosurf)
- 3 = Calfactant (Infasurf)
- 9 = Other

2. Was the surfactant mixed with budesonide?

Code 'Y' if the surfactant administered was mixed with budesonide. Code 'N' if otherwise. Code 'T' if the infant was enrolled in the BiB trial. If 'TI' refer to section 3.2.1 for detailed instructions.

3. Method of administration:

Record the method of administration:

- 1 = ETT (endotracheal tube) – includes INSURE (INtubate, SURFactant, REmove) given using an endotracheal tube
- 2 = LISA/MIST (less invasive surfactant administration or minimally invasive surfactant treatment) – defined by use of a specialized catheter used to inject surfactant below the vocal cords
- 3 = Aerosolized – may be given using a nebulizer via mask or prongs
- 4 = LMA (laryngeal mask airway)
- 9 = Other

c. Pressor support?

This includes dopamine, dobutamine, epinephrine and isuprel.

d. Volume support?

This includes albumin 5% or 20%, blood transfusion, fresh frozen plasma, plasmanate and Ringer's lactate.

e. Intubation and Ventilation?

Code 'Y' if infant was intubated and received mechanical ventilation with an endotracheal tube in place.

f. IV fluids?

2. Autopsy performed?

Code 'Y' if an autopsy was performed. If an autopsy is performed the cause of death should not be coded until the results of the autopsy are known. The NG03E should not be held back for coding or entry into the EDC waiting for the autopsy report. If the autopsy result for cause of death is to be included and results are not yet available, enter the temporarily missing data code 'TM', and enter final autopsy cause of death in the EDC when it becomes available.

3. At any time after birth (prior to NG03E Status), was there documentation of discussion with parents to limit, withdraw or not escalate care?

Code 'Y' if there is documentation of discussion with parents to limit, withdraw or not escalate care. Otherwise code 'N.'

4. **Were the following treatments withheld, limited or withdrawn at any time with the intent to limit care?**

a. Intubation/ventilation

Code 'Y' if intubation or ventilation was withheld, limited or withdrawn at any time with the intent to limit care. Code 'N' otherwise.

b. Nutrition/hydration

Code 'Y' if nutrition or hydration was withheld, limited or withdrawn at any time with the intent to limit care. Code 'N' otherwise.

5. **Palliative care consult obtained?**

Code 'Y' if palliative care consult was obtained. Code 'N' otherwise.

6. **Ethics consult obtained?**

Code 'Y' if an ethics consult was obtained. Code 'N' otherwise.

7. **No intention to resuscitate (comfort care only)**

Code 'Y' if there was no intention to resuscitate and only comfort care was provided. Code 'N' otherwise.

8. **Contributory cause of death:**

This should be the underlying, proximate disease which initiated the series of events leading to the cause of death. This underlying cause must be (1) etiologically specific and (2) antecedent to all other causes with respect to time and pathologic relationship. Without the underlying cause, death would not have occurred. The cause of death should be based on both clinical evidence and autopsy findings. In the absence of an autopsy, the clinical evidence will be used. Only one option is coded after consultation with the PI or alternate PI.

A list of the causes of death are in **Appendix C**.

9. **If contributory cause of death is code 10 (Congenital malformation) or code 90 (Other), specify:**

NG05: LATE CLINICAL OUTCOME FORM

4.5 Coding Instructions for Form NG05

The NG05 is to be completed for infants who are hospitalized greater than 120 days and reach status (death, discharge, transfer or one year postnatal age). The NG05 collects diagnosis and treatments that occur after day of life 120.

4.5.1 Section A - STATUS

1. **Status of infant at time of completion of form**

1 = if infant was discharged to home

3 = if infant was transferred to another facility (including hospital, chronic care facility or hospice facility, etc.) without returning in 7 days

5 = if the infant died

6 = if the infant remains in hospital after one year postnatal age

2. **Date of status**
Record date of status.
3. **Weight at status**
Weight in grams at status (preferably), or within 7 days.
4. **Length at status**
Length in centimeters at status (preferably), or within 7 days.
5. **Head circumference at status**
Head circumference in centimeters at status (preferably), or within 7 days.

4.5.2 *Section B - EXTENDED STAY INFORMATION*

1. **What problem (s) caused hospitalization greater than 120 days:**
Answer 'Y' to all that apply. Answer 'N' otherwise.

- a. **Pulmonary?**
- b. **Cardiac?**
- c. **Neurologic?**
- d. **Gastrointestinal?**
- e. **Multiple Malformations?**
- f. **Social?**
- g. **Ophthalmologic?**
- h. **Sepsis/Infection?**
- i. **Renal?**
- j. **Apnea?**
- k. **Feeding?**
- l. **Other? (If Yes, specify)**

2. **Did either eye receive therapy for ROP after 120 days?**

Code 'Y' if either eye received therapy for ROP after 120 days. Code 'N' otherwise.

- a. If YES or Trial **List all therapies done for either eye (Use codes below):**

- 1 = laser
- 2 = cryotherapy
- 3 = scleral buckle
- 4 = vitrectomy
- 5 = avastin or anti-VEGF
- 6 = other (specify) for either eye

If initial treatment/medication is given under a masked or unmasked study protocol or if subsequent treatment/medication is received as part of clinical care post trial, refer to Section 3.2.1 for detailed instructions.

3. **Was a hearing screen performed after 120 days?**

Code 'Y' if a hearing screen was performed after 120 days to evaluate the infant's hearing while in the hospital. Code 'N' otherwise.

If YES,

a. Was otoacoustic emissions (OAE) testing performed?

Code 'Y' if OAE testing was performed. Code 'N' otherwise.

If YES,

1. Was OAE failed?

Code 'Y' if infant failed the OAE. Code 'N' otherwise.

If YES,

i. Unilateral or bilateral fail?

Code

1=Unilateral

2=Bilateral

b. Was automated auditory brainstem response (AABR) performed?

Code 'Y' if automated auditory brainstem response (AABR) was performed.

Code 'N' otherwise.

If YES,

1. Was AABR failed?

Code 'Y' if infant failed the AABR. Code 'N' otherwise.

If YES,

i. Unilateral or bilateral fail?

Code

1=Unilateral

2=Bilateral

4. Was a diagnostic auditory brainstem response (ABR) performed prior to discharge?

Code 'Y' if there was documentation of an ABR performed prior to discharge. Code 'N' otherwise.

If YES,

a. Was hearing loss documented?

Code 'Y' if hearing loss was documented. Code 'N' if no hearing loss was documented. Code 'Unk' if report not available to make determination.

If YES,

b. Unilateral or bilateral hearing loss?

Code

1=Unilateral

2=Bilateral

4.5.3 Section C - ETHICS/PALLIATIVE CARE

1. Was there documentation of discussion with parents to limit, withdraw or not escalate care after 120 days?

Code 'Y' if after 120 days, there was documentation of a discussion with parent to limit, withdraw or not escalate care. Code 'N' otherwise.

2. **Were the following treatments withheld, limited or withdrawn after 120 days with the intent to limit care?**

Code 'Y' to all that apply. Code 'N' otherwise.

a. **Intubation/ventilation**

b. **Nutrition/hydration**

3. **Palliative care consult was obtained?**

Code 'Y' if there was documentation that a palliative care consult was obtained. Code 'N' if otherwise.

4. **Ethics consult was obtained?**

Code 'Y' if there was documentation that an ethics consult was obtained. Code 'N' if otherwise.

4.5.4 Section D - TRANSFER

This section refers to when the infant is sent to another hospital or chronic care facility without returning in 7 days. Complete this section if status of infant is "transferred."

1. **Date of transfer:**

Record date of transfer

2. **Final outcome:**

Code the final outcome:

1 = if the baby died in hospital (informed via telephone contact or letter from the other hospital)

2 = if the baby was discharged to home

3 = if the baby remains in the hospital at one year of age

4.5.5 Section E - DISCHARGE ALIVE

Complete this section if the Status of the infant is 1 =discharged home'

1. **Date of discharge to home**

Record date of discharge.

2. **Discharged home on continuous oxygen?**

Code 'Y' if discharged home on continuous oxygen. Code 'N' otherwise.

3. **Discharged home on any of the following medications?**

Code 'Y' if infant is discharged home on any medication(s) listed below. Otherwise code 'N.' For questions 3.a.-3.g., code 'Trial' if infant discharged on a trial medication in a masked placebo controlled study. Refer to section 3.2.1 for details.

a. **Diuretics-** including but not limited to furosemide, spironolactone, chlorothiazide, or hydrochlorothiazide

b. **Bronchodilators-** including but not limited to albuterol

c. **Anticonvulsants** including but not limited to phenobarbital, dilantin, valproic acid, depakane, carbamapazine

- d. **Antireflux medications**- including but not limited to ranitidine, famotidine, metaclopramide.
- e. **Antihypertensive medications** including but not limited to hydrazaline.
- f. **Methylxanthines** - including but not limited to caffeine, theophylline, and aminophylline.
- g. **Inhaled steroids** - including but not limited to budesonide (pulmicort) and fluticasone (flovent).
- h. **Systemic steroids** – including but not limited to dexamethasone, hydrocortisone, and prednisone/prednisolone.
- i. **Other, specify**

Specify other medication on which infant was discharged home. Do not include multivitamins or iron.

4. Discharged home receiving any human milk?

Code 'Yes' (complete D4a) if infant is discharged home receiving any human milk. Otherwise answer "no or Unknown."

a. If YES, type of milk

Code the type of milk:

- 1= Maternal Milk
- 2= Donor Milk
- 3= Both
- 4= Unknown

4.5.6 Section F - DEATH

Complete this section if the Status of the infant is '5=Death'

1. **Date of death:**
Record date of death.
2. **Autopsy performed?**
Code 'Y' if an autopsy was performed. If an autopsy is performed, the cause of death should not be coded until the results of the autopsy are known. Code 'N' otherwise.
3. **Contributory cause of death:**
This should be the underlying, proximate disease which initiated the series of events leading to death. This underlying cause must be (1) etiologically specific and (2) antecedent to all other causes with respect to time and pathologic relationship. Without the underlying cause, death would not have occurred. The cause of death should be based on both clinical evidence and autopsy findings. In the absence of an autopsy, the clinical evidence will be used. Only one option is coded after consultation with the PI or alternate PI.

A list of the causes of death are in **Appendix C**.
4. **If cause of death is code 10 (Congenital malformation), code 31 (Sepsis, specify organism), or code 90 (Other), specify.** If the cause of death is COVID-19 use code 90 (Other) and specify.

Chapter 5

THE RESPIRATORY SUPPORT DATA

5.1 The Respiratory Support data on the NG07

5.1.1 Respiratory Support Form Overview

The NG07 form is completed for all GDB infants who have survived for greater than 12 hours and have reached status (death, discharge, transfer or 120 days). Respiratory support the infant received during the hospitalization is documented at the time points noted, including 40 weeks PMA. For any infant who survived >12 hours but death occurred <24 hours, data is completed in the Status column only. If Status (death, discharge or transfer) is reached before any time point, some column(s) will be left blank. The Status column will always be completed.

The NG07 also collects information to define the outcome of bronchopulmonary dysplasia (BPD) for infants in the GDB. Infants enrolled in a trial requiring the Physiologic Definition of BPD will be evaluated for their eligibility for the Oxygen Reduction Challenge and the results of this evaluation, if performed, will be documented. The Jensen BPD criteria (see Appendix M) will be used to standardize the definition and grade of BPD for infants in the GDB and for any protocol which has a primary or secondary outcome of BPD except for trials that specifically use the Physiologic Definition of BPD.

5.1.2 Section A - RESPIRATORY SUPPORT

Respiratory Support data is collected as follows:

Questions A1. through A5.:

1. Highest FiO₂ on day

2. Highest mode of support on day, code

- 1 = HFV
- 2 = CV
- 3 = Nasal ventilation
- 4 = CPAP
- 5 = NC
- 6 = Hood
- 7 = No Support
- 8 = Temporarily out of unit

3. If mode '5', record Flow Rate

4. Weight (grams)

5. Date of last HFV use (evaluated up through 40 0/7 weeks)

6. Date of last SIMV/CV use (evaluated up through 40 0/7 weeks)

7. Date of last NIPPV/nasal ventilation use (evaluated up through 40 0/7 weeks)

8. Date of last CPAP use (evaluated up through 40 0/7 weeks)

9. Date of last respiratory support (evaluated up through 40 0/7 weeks)

The respiratory support box collects FiO₂ and mode of support **at** 24 hours of life (snapshot **at** 24 hours of life) and at each time point on days 3, 7, 14, 28, 36 and 40 weeks' PMA ,and Status

during hospitalization. If Status is reached before any time point(s), no data will be entered for the time point(s) that would occur after the date of Status.

Respiratory support is collected using a hierarchy, counting only the highest type of respiratory support (HFV, CV, Nasal ventilation, CPAP, Nasal Cannula, Hood) for any given day. **The hierarchy for respiratory support is HFV as highest, next CV, nasal ventilation, CPAP, nasal cannula, hood.** If an infant is on multiple modes of support on any day, only the highest level would count for that day.

Example: If the infant was receiving conventional ventilation (CV) and high frequency (HFV) on the same calendar day, count only the HFV for that day.

Modes of support are HFV (including oscillator, jet), CV (including IMV, SIMV and/or assist control via an endotracheal tube), nasal ventilation (via nasal prongs or cannula), CPAP (nasal prongs, cannula or mask), and supplemental oxygen (delivered by any method including ventilator, CPAP, nasal cannula, hood or incubator).

The day of birth is considered day of life 1 even though it is not a complete 24 hour period. Day of life 2 is the next calendar day.

The total number of days of all support cannot exceed the day of life.

Example: On day 7, the infant may have received 4 days of CV and 2 days nasal ventilation and 1 day CPAP for a total number of days of support to equal 7 days. The infant could receive less than 7 days if there was no respiratory support on any of the first 7 days.

Questions A.1. and A.2.:

The highest FiO₂ and highest mode of support are recorded for each time point.

Question A.1. Record the highest FiO₂ delivered (by HFV, CV, nasal ventilation, CPAP, nasal cannula or hood) at 24 hours of life, and any time during the day on days 3, 7, 14, 28, 36 and 40 weeks' PMA, and Status. When determining the highest FiO₂ for each day, disregard any temporary increases in FiO₂ for desaturation episodes, apnea, bradycardia or procedures, when the infant returns to the previous FiO₂ in a reasonable amount of time (<2 hours). DO NOT include supplemental oxygen given only with feedings if the infant is not receiving oxygen at any other time during that day.

Question A.2. Record the highest mode of support the infant received on each day using the codes listed on the form. If the infant is on more than one mode of support, use the hierarchy (HFV as highest, CV next, nasal ventilation, next then CPAP, nasal cannula, hood) to record the mode of support for that day. Additionally:

1. Respiratory support delivered by nasal mask or prongs that is only flow is coded as CPAP.
2. Respiratory support delivered by nasal mask or prongs that delivers flow and intermittent breaths is Nasal ventilation.

Question A.3. Record the flow rate if nasal cannula (code=5) is recorded as the highest level of support on any day. The flow rate recorded is the flow delivered with the highest FiO₂ on that day.

Example: If any time during a day, the infant was on FiO₂ 1.00 with flow rate 0.03 lpm and also during that day was switched to an FiO₂ .27 with a flow rate of 0.5 lpm, record FiO₂ 1.00, flow 0.03 lpm.

Any GDB infant that is enrolled in a trial requiring the Physiologic Definition of BPD should be evaluated for Oxygen Reduction Challenge eligibility **if on day of 36 weeks (A.1. 36 weeks) the highest mode of support (A.2.) is answered 5 = nasal cannula with or without supplemental oxygen or 6 = hood (to include oxygen delivered into an incubator).**

Supplemental oxygen requirements should be determined with the infant at rest. Disregard any temporary increases in oxygen requirement (for desaturation episodes, apnea, bradycardia or procedures, when the infant returns to baseline oxygen in a reasonable amount of time (<2 hours)). Do not include supplemental oxygen given only with feeds.

If the Section 'A - 36 weeks' question 2 is NOT answered with mode = 5 or 6, **the infant should NOT be evaluated for Oxygen Reduction Challenge eligibility** and the form is complete.

Question A.4. Record the infant's weight at each time point. The following windows may be used for each time point:

- Day 3: (weight collected between day 2 and day 4)
- Day 7: (weight collected between day 6 and day 8)
- Day 14: (weight collected between day 12 and day 16)
- Day 28: (weight collected between day 26 and day 30)
- 36 weeks: (weight collected between 36-0/7 to 37-0/7 weeks)
- 40 weeks: (weight collected between 40-0/7 to 41-0/7 weeks)

Question A.5. through A.9. Record the date the last mode of respiratory support was used (evaluated up through 40 0/7 weeks). Please leave blank if infant never received mode of respiratory support. Please check box and leave blank if infant on mode of respiratory support (or higher, if noted) at 40 0/7 weeks PMA.

5.1.2 Section B – OXYGEN REDUCTION CHALLENGE ELIGIBILITY

Oxygen Reduction Challenge eligibility should only be conducted on infants enrolled in a trial requiring the Physiologic Definition of BPD and if Section 'A - 36 weeks' question 2 is answered with mode = 5 or 6.

1. Is infant enrolled in a trial requiring the Oxygen Reduction Challenge?

If **NO**, the infant is not enrolled in a trial requiring the Oxygen Reduction Challenge, then the infant is not eligible and the form is complete.

If **YES**, proceed to question B1a.

To determine eligibility for the Oxygen Reduction Challenge:

An **Effective FiO_2** will be calculated for infants receiving supplemental oxygen by nasal cannula using the infant's weight, flow (lpm) and oxygen concentration. Conversion tables in **Appendix J** are included to determine the **Effective FiO_2** .

The actual FiO_2 concentration will be used for infants receiving oxygen by hood (including oxygen delivered into an incubator).

If eligibility criteria are met, the infant's chart will be reviewed for oxygen saturation readings in the prior 24 hours. The 'majority' of saturations is defined as >90% of saturation readings during the prior 24 hour period.

Any infant deemed eligible will have an Oxygen Reduction Challenge performed.
The protocol for the Physiologic Definition of Bronchopulmonary Dysplasia is included as
Appendix J.

a. Weight on day of 36 weeks

Record the infant's weight in grams on day of 36 weeks' post menstrual age.

b. Is the infant eligible for an Oxygen Reduction Challenge?

Infants eligible for an Oxygen Reduction Challenge must meet one of the following:

- Effective oxygen <27% AND majority of saturations ≥90%
- Effective oxygen 27%-30% AND majority of saturations ≥96%
- Room air by nasal cannula.

c. Was the Oxygen Reduction Challenge performed?

If YES,

1. Date of challenge

2. Did the infant pass challenge?**

See Physiologic Definition of BPD protocol (Appendix J**)

3. Reason not performed (use codes)

1 = Increased FiO₂

Infant's supplemental oxygen requirement has increased and is no longer eligible

2 = Increased respiratory support-

Infant has been placed on CPAP or ventilator and is no longer eligible

3 = Instability

To include, but not limited to surgery, sepsis where the infant may not tolerate the challenge

4 = Parent/Physician refusal

5 = Weaned to room air on/before day of evaluation/challenge

No longer eligible because no longer requiring supplemental oxygen and is not receiving any respiratory support.

9 = Other- Explain

Chapter 6

Revisions/Additions to the GDB

6.1 Overview

It is recognized that changes (additions, deletions, revisions) to the GDB will be necessary periodically. Such changes may be: 1) permanent changes to the core GDB; or 2) time-limited data collection for studies which relate to areas of special interest. All proposed changes to the GDB must undergo thorough, prospective and formal review to determine whether the proposed changes are appropriate and acceptable. Historical changes are recorded in a separate document.

The core GDB is a limited data set which represents important/essential information (which is expected to change over time) for all extremely premature infants. The core GDB is not intended to include detailed data on areas of special interest in the extremely premature population. Such data, in sufficient detail to provide meaningful information, can be collected as an addendum to the core GDB after appropriate approval.

It is the understanding of the GDB Subcommittee that areas of special interest generally lend themselves to data collection for a finite period of time, i.e. special interest data collection forms are not considered part of the core GDB. The process for proposing changes to the GDB, either permanent or temporary, is described in Section 7.2

6.2 Process of Proposing Revisions/Additions to the GDB

All proposed changes to the GDB must be submitted as a formal proposal to the Protocol Review Subcommittee (PR Subcommittee).

The Chair of the PR Subcommittee will determine whether the proposal is acceptable (contains all information required for review as identified by policies established by the Protocol Review Subcommittee). The Chair will also determine whether the proposal requires full review by the PR Subcommittee or whether it may be reviewed by the GDB Subcommittee (without prior full review by the PR Subcommittee).

If a proposal is reviewed and approved by either the PR or the GDB Subcommittee, the proposal will be reviewed by the Steering Committee for final approval prior to implementation.

Appendix A Organism Codes List

Code	Genus	Species
<u>Bacteria</u>		
310	Achromobacter	sp. [inc. Achromobacter xylosoxidans and others]
320	Acinetobacter	sp. [antiratus, baumannii calcoaceticus, haemolyticus, johnsonii, junii, lwoffii, radioresistens]
330	Aeromonas	sp.
340	Alcaligenes	sp. [Alcaligenes xylosoxidans and others]
140	Bacillus	sp.
410	Bacteroides	sp.
160	Bifidobacterium	sp. [Bifidum, lactis, infantis, thermophilum, and others]
350	Burkholderia	sp. [Burkholderia capeciae and others]
360	Campylobacter	sp. [Campylobacter fetus, C. jejuni and others]
620	Chlamydia	trachomatis
370	Chryseobacterium	sp.
226	Citrobacter	sp. [Citrobacter diversus, C. freundii, C. koseri and others]
420	Clostridia	sp.
500	Corynebacterium	sp.
240	Enterobacter	sp. [Enterobacter aerogenes, E. cloacae, and others]
380	Enterococcus	sp. [Enterococcus faecalis (a.k.a. Streptococcus faecalis and Streptococcus Group D), E faecium, and other Enterococcus species]
200	Escherichia	coli
390	Flavobacterium	sp.
590	Haemophilus	sp. [Haemophilus influenzae, H. vaginalis and others]
325	Herellea	vaginicola
230	Klebsiella	sp. [Klebsiella oxytoca, K. pneumoniae and others]
170	Lactobacillus	sp. [Acidophilus, casei, and others]
150	Listeria	sp. [Listeria monocytogenes]
105	Micrococcus	sp.
430	Moraxella	sp. [Moraxella catarrhalis (a.k.a. Branhamella catarrhalis) and others]
650	Morganella	morganii
640	Mycobacterium	sp. [tuberculosis]
570	Neisseria	sp. [Neisseria meningitides, N. gonorrhoeae and others]
440	Pasteurella	sp.
460	Peptostreptococcus	sp.
480	Prevotella	sp.

470	Propionibacterium	sp.
260	Proteus	sp. [Proteus mirabilis, P. vulgaris and others]
300	Pseudomonas	sp.
301	Pseudomonas	aeruginosa
210	Salmonella	sp.
250	Serratia	sp. [Serratia liquefaciens, S. marcescens odorifera, ficaria, plymuthica and others]
101*	Staphylococcus	aureus [coagulase positive]
104	Staphylococcus	coagulase negative (including S. epidermis, saprophyticus, haemolyticus, hominis, lugdunensis, simulans, cohnii, warnei, saccharolyticus)
510	Stenotrophomonas	maltophilia
111	Streptococcus	viridans
112	Streptococcus	Group A
113	Streptococcus	Group B

*Note- call lab if unclear whether staphylococcus is coagulase negative, coagulase positive, aureus or epi.

133	Streptococcus	pneumoniae
561	Treponema	pallidum [syphilis]
610	Vibrio	
600	Yersinia	
699	Other; Code to be assigned	

Fungi

834	Aureobasidium	sp.
810	Candida	sp.
811	Candida	albicans
813	Candida	krusei
816	Candida	parapsilosis
817	Candida	tropicalis
819	Candida	glabrata
820	Candida	guilliermondi
821	Candida	lusitaniae
881	Malassezia	fur
872	Saccharomyces	sp. [yeast]
899	Other; Code to be assigned	

Resistant Organisms

750	Methicillin resistant staphylococcus aureus (MRSA)
760	Vancomycin resistant enterococci (VRE)

Viruses

901	Cytomegalovirus
903	Enteroviruses [includes coxsackie, echoviruses]
920	Hepatitis A
923	Hepatitis B

924	Hepatitis C	
905	Herpes Simplex	
907	HIV	
910	Rubella	
960	Varicella	zoster
<u>Other Organisms</u>		
911	Toxoplasma	gondii [toxoplasmosis]
010	Other; Code to be assigned	

Appendix B Drug Therapeutic Agent List

Penicillins

- 01 = Ampicillin
- 02 = Carbenicillin
- 03 = Oxacillin
- 04 = Penicillin G
- 05 = Piperacillin
- 06 = Ticarcillin
- 07 = Mezlocillin
- 08 = Methicillin
- 09 = Nafcillin
- 10 = Amoxicillin
- 11 = Amoxicillin/clavulanate (Augmentin)

Cephalosporins

- 19 = Cephalexin (Keflex)
- 20 = Cephalothin
- 21 = Cefazolin (Kefzol)
- 22 = Cefotaxime (Claforan)
- 23 = Cefoxitin
- 24 = Moxalactam
- 25 = Ceftazidime (Fortaz)
- 26 = Ceftriaxone (Rocephin)
- 27 = Ceftizoxime
- 28 = Cefuroxime
- 29 = Cefotetan
- 30 = Cefixime

Aminoglycosides

- 31 = Amikacin
- 32 = Gentamicin
- 33 = Kanamycin
- 34 = Tobramycin

Other Antibiotics

- 44 = Vancomycin
- 46 = Bactrim
- 47 = Chloramphenicol (Chloromycetin)
- 49 = Clindamycin
- 50 = Erythromycin

Other Antibiotics (continued)

- 62 = Imipenem
- 63 = Metronidazole
- 64 = Aztreonam
- 67 = Ampicillin + sulbactam (unasyn)
- 68 = Azithromycin
- 70 = Cefepime
- 75 = Meropenem
- 76 = Piperacillin + tazobactam (zosyn)
- 77 = Rifampin
- 78 = Ticarcillin + clavulanate
- 80 = Ciprofloxacin
- 81 = Clarithromycin (Biaxin)
- 82 = Doxycycline
- 83 = Trimethopin/sulfa (TMP/SMX)
- 91 = Linezolid

Antifungals

- 43 = Flucytosine (5FC)
- 48 = Nystatin
- 61 = Fluconazole
- 69 = Caspofungin
- 92 = Itraconazole
- 93 = Posaconazole
- 94 = Voriconazole
- 41 = Amphotericin B
- 66 = Amphotericin B Liposome (Ambisome)
- 95 = Amphotericin B lipid complex (Abelcet)
- 97 = Anidulafungin
- 98 = Micafungin

Antivirals

- 45 = Vidarabine
- 71 = Acyclovir (Zovirax)
- 72 = Ganciclovir
- 73 = Nevarapine
- 74 = Zidovudine (AZT)
- 96 = Valacyclovir hydrochloride (Valtrex)

Other Drugs Not Listed

- 99 = Other

Appendix C

Causes of Death

Malformation

10 - Congenital malformation

Code '10' for major congenital malformations that are incompatible with life, or incompatible with life without drastic surgical or other measures to maintain life such as a chromosomal defect, inborn error of metabolism, neural tube defect, congenital heart disease, or renal abnormality. If an infant has respiratory distress syndrome or intracranial hemorrhage but is allowed to die because his/her main problem is the congenital malformation, then malformation should be coded as the cause of death.

Pulmonary

20 - RDS

Code '20', RDS, for severe respiratory insufficiency in the presence of RDS during the first 28 days of life, i.e., increasing oxygen and ventilatory pressure requirements, pneumothorax, pneumopericardium, etc.

21 - RDS with severe intracranial hemorrhage

Code '21' for infants with severe respiratory insufficiency in the presence of RDS during the first 28 days of life with severe (grade III-IV) intracranial hemorrhage.

22 - RDS with infection

Code '22' for infants with severe respiratory distress in the presence of RDS during the first 28 days who have a fulminant pulmonary or other infection. For example, if an infant has group B streptococcal sepsis with a clinical presentation of respiratory distress, or respiratory distress syndrome with an intercurrent infection, then RDS with infection should be coded as the cause of death.

23 - RDS with massive pulmonary hemorrhage

Code '23' for infants with respiratory distress syndrome diagnosed during the first 28 days of life and massive pulmonary hemorrhage.

25 - BPD

Code '25', for infants with chronic lung disease requiring respiratory support (nasal cannula, CPAP, or positive-pressure ventilation, either noninvasive or invasive) at 36 weeks' postmenstrual age.

26 - BPD with infection

Code '26' for infants with chronic lung disease requiring respiratory support (nasal cannula, CPAP, or positive-pressure ventilation, either noninvasive or invasive) at 36 weeks' postmenstrual age, who develop a severe intercurrent infection.

27 - BPD with severe CNS insult

Code '27' for infants with chronic lung disease requiring respiratory support (nasal cannula, CPAP, or positive-pressure ventilation, either noninvasive or invasive) at 36

weeks' postmenstrual age and extubated prior to death because of severe brain atrophy, hydrocephalus, etc.

28 – Pulmonary Hypertension (PPHN)

Code '28' for infants with pulmonary hypertension (PPHN) diagnosed clinically or by echocardiography.

29 – BPD with Pulmonary Hypertension (PPHN)

Code '29' for infants with chronic lung disease requiring respiratory support (nasal cannula, CPAP, or positive-pressure ventilation, either noninvasive or invasive) at 36 weeks' postmenstrual age AND with pulmonary hypertension (PPHN) diagnosed clinically or by echocardiography.

Infection**30 - Suspect sepsis/infection**

Code '30' for infants with clinical presentation of septicemia or localized infection without positive cultures during life or on autopsy. The autopsy may reveal polymorph infiltrate of organs and other indications of infection, however cultures are negative.

31 - Proven sepsis/infection

Code '31' for septicemia or localized infection with positive blood or organ cultures. For example, septicemia, meningitis, pulmonary abscess.

GI**40 - NEC**

Code '40' for proven NEC, stage IIA or higher by Bell's criteria as shown in **Appendix G**.

41 - NEC with sepsis

Code '41' for infants with proven NEC (stage IIA or higher by Bell's criteria in **Appendix G**) together with positive blood or peritoneal fluid cultures.

42 - Spontaneous perforation

Code '42' for infants with acute gastrointestinal perforation diagnosed by x-ray without classic radiographic findings of necrotizing enterocolitis or findings of necrotizing enterocolitis at surgery or autopsy, including pathologic specimens.

43 - Short Bowel Syndrome

Code '43' for infants with small bowel resection characterized by malabsorption, diarrhea and failure to thrive.

44 – Liver failure

Code '44' for liver failure if the clinician documents cause of death as liver failure; may include elevated liver enzymes with a presentation of coagulopathy.

CNS insult**50 - Severe intracranial hemorrhage**

Code '50' for infants with severe intracranial hemorrhage (grade III-IV) with a clinical presentation of central nervous system decompensation including seizures, apneas, etc., in the absence of severe respiratory distress syndrome requiring high ventilator settings.

51 - Severe intracranial hemorrhage with infection with culture proven or suspected

Code '51' for infants with severe intracranial hemorrhage (grade III or IV) i.e., documented blood/echodensity/echogenicity in the ventricle with ventriculomegaly and/or blood/echodensity/echogenicity in the parenchyma.

Renal

70 – Renal failure

Code '70' for infants with renal failure noted in the chart as cause of death, characterized by Stage 3 acute kidney injury as described:

Serum Creatinine	Urine Output
SCr rise ≥ 3 X reference SCr * <i>or</i>	<0.3 ml/kg/hour for > 24 hours
SCr ≥ 2.5 mg/dl** <i>or</i>	<i>or</i>
Receipt of dialysis	anuria for > 12 hours

Other

60 – Immaturity without active neonatal treatment

Code '60' should be used for infants receiving no active treatment (i.e. comfort care only) due to extreme prematurity. Infants provided comfort care because of a severe congenital anomaly or other specific cause aside from extreme prematurity should have the specific cause indicated under pre-specified causes listed above. This code should also be used for infants who receive no further active treatment following unsuccessful attempts to resuscitate at birth.

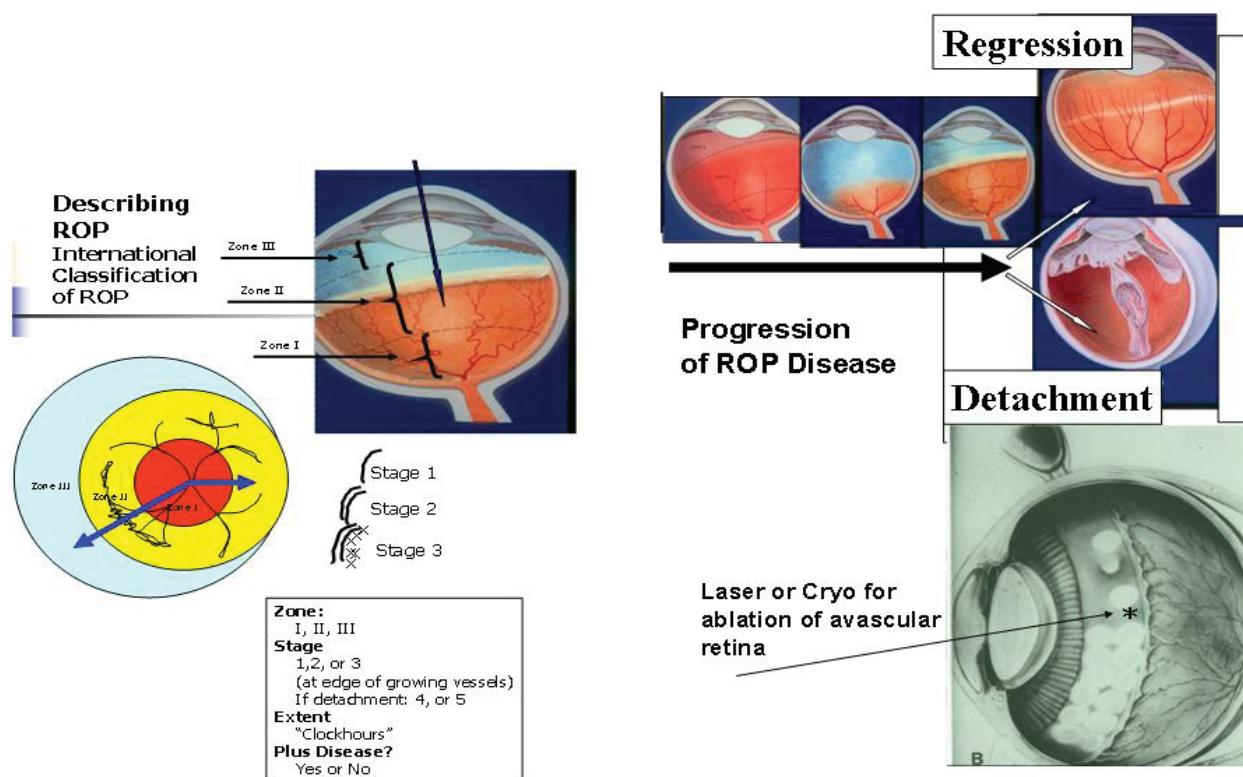
90 - Other

Code '90' for infants with other causes of death such as severe asphyxia with multi-system failure, severe metabolic disease, and severe trauma. Cardiorespiratory arrest is not to be used.

99 - Unknown

Code '99' only if the cause of death has been investigated but could not be established.

Appendix D Retinopathy of Prematurity Diagram



Appendix E Surgical Procedures

GI – 200 Series	
201	Laparotomy
202	Bowel resection (end to end anastomosis) or repair of perforation
203	Jejunostomy/ileostomy/colostomy—primary surgery or revision
204	Ostomy takedown/reanastomosis
205	Peritoneal drain (used for any reason including NEC or IP)
206	Gastrostomy/G Tube
207	Appendectomy
208	Fundoplication or repair of hiatal hernia
209	T-E fistula/esophageal atresia repair
210	Gastroschisis/omphalocele repair
211	Diaphragmatic hernia repair/hernia of Morgagni
212	Inguinal hernia repair
213	Umbilical hernia repair
214	Open Liver biopsy
215	Strictureplasty
216	Ladd's procedure/malrotation
217	Abdominal wound debridement and closure of abdominal wound/abdomen
218	Pyloromyotomy
219	Anorectoplasty
299	GI other
Pulmonary - 300 Series	
301	Tracheostomy
302	Anterior cricoid split
303	Resection of cystic adenomatoid malformation
304	Bronchoscopy
305	Open lung biopsy
306	Lobectomy
307	Pulmonary sequestration—extralobar
399	Pulmonary other
GU – 400 Series	
401	Repair of extrophy of the bladder
402	Urinary diversion/stent
499	GU other
Head & Neck - 500 Series	
501	Correction of choanal atresia
502	Repair of cleft lip/palate
503	Oral surgical procedure
599	Head & Neck other

CNS – 600 Series	
601	Shunt for hydrocephalus (post hemorrhagic)
602	Shunt for hydrocephalus (not post hemorrhagic)
603	Ventricular reservoir
604	Exteriorization of shunt or shunt revision
605	Shunt removal
606	Repair of meningomyelocele
699	CNS other
Cardiac – 700 Series	
701	Repair of CHD
702	Cardiac shunt procedure
799	Cardiac other
Fetal Surgeries – 800 Series	
801	Congenital diaphragmatic hernia
802	Twin-to-twin ablation
803	Congenital myelomeningocele
899	Fetal Surgeries other
Other - 900 Series	
901	Central line placement (requiring anesthesia; not including umbilical or percutaneous central lines)
902	Retinal Laser Ablation
903	Scleral Buckle for ROP
904	Vitrectomy for ROP
905	Excision of teratoma
906	Ophthalmology other—not ROP related
999	Other surgeries not listed

Appendix F Data Forms

The following pages contain the data forms for the Survey of Morbidity and Mortality Among Preterm Infants (GDB)

NGO1	SCREENING LOG
NGO2	GENERIC BASELINE FORM
NGO3	GENERIC CLINICAL OUTCOME FORM
NGO3E	GENERIC EARLY DEATH FORM
NGO5	GENERIC LATE CLINICAL OUTCOME FORM
NGO7	GENERIC RESPIRATORY SUPPORT FORM

Appendix G Modified Bell's Staging Criteria for NEC

Stage	Systemic	Intestinal Signs	Radiologic Signs
IA - Suspected NEC	Temperature instability, apnea, bradycardia, lethargy	Elevated pre-gavage residuals, mild abdominal distension, emesis, guaiac positive stool	Normal or intestinal dilation, mild ileus
IB - Suspected NEC	Same as above	Bright red blood from rectum	Same as above
IIA - Definite NEC Moderately ill	Same as above	Same as above, plus absent bowel sounds, ± abdominal tenderness	Intestinal dilation, ileus, pneumatosis intestinalis
IIB - Definite NEC	Same as above, plus mild metabolic acidosis, mild thrombocytopenia	Same as above, plus absent bowel sounds, definite abdominal tenderness, ± abdominal cellulitis or right lower quadrant mass	Same as IIA, plus portal vein gas, ± ascites
IIIA - Advanced NEC Severely ill Bowel intact	Same as IIB, plus hypotension, bradycardia, severe apnea, combined respiratory and metabolic acidosis, disseminated intravascular coagulation, neutropenia	Same as above, plus signs of generalized peritonitis, marked tenderness and distention of abdomen	Same as IIB, plus definite ascites
IIIB - Advanced NEC Severely ill Bowel perforated	Same as IIIA	Same as IIIA	Same as IIB, plus pneumoperitoneum

From: Ped. Clinics of North America, February 1986, Walsh, M., Kliegman, R., Necrotizing enterocolitis: Treatment based staging criteria.

Appendix H Birth Defects Codes

CODE	TYPE OF DEFECT
Central Nervous System Defects - 100 Series	
101	Anencephaly
102	Meningomyelocele
103	Hydranencephaly
104	Congenital Hydrocephalus
105	Holoprosencephaly
199	Other Central Nervous System Defects
Congenital Heart Defects - 200 Series	
201	Truncus Arteriosus
202	Transposition of the Great Vessels
203	Tetralogy of Fallot
204	Single Ventricle
205	Double Outlet Right Ventricle
206	Complete Atrio-Ventricular Canal
207	Pulmonary Atresia
208	Tricuspid Atresia
209	Hypoplastic Left Heart Syndrome
210	Interrupted Aortic Arch
211	Total Anomalous Pulmonary Venous Return
299	Other Congenital Heart Defects
Gastrointestinal Defects - 300 Series	
301	Cleft Palate
302	Tracheo-Esophageal Fistula
303	Esophageal Atresia
304	Duodenal Atresia
305	Jejunal Atresia
306	Ileal Atresia
307	Atresia of large bowel or rectum
308	Imperforate anus
309	Omphalocele
310	Gastroschisis
399	Other Gastrointestinal Defects
Genitourinary Defects - 400 Series	
401	Bilateral Renal Agenesis
402	Bilateral polycystic, multicystic, or dysplastic kidneys
403	Obstructive Uropathy with Congenital Hydronephrosis
404	Exstrophy of the Urinary Bladder
499	Other Genitourinary Defects

Chromosomal Abnormalities – 500 Series

501	Trisomy 13
502	Trisomy 18
503	Trisomy 21
599	Other Chromosomal Abnormality

Other Birth Defects – 600 Series

601	Skeletal Dysplasia
602	Congenital Diaphragmatic Hernia
605	Inborn Error of Metabolism (include G6PD)
699	Other Serious and/or Life-Threatening Birth Defect

Pulmonary - 700 Series

701	Cystic Adenomatoid Malformation (CAM)
799	Other Pulmonary

The following conditions should **NOT** be coded as a Major Birth Defect.

- Extreme Prematurity
- Intrauterine Growth Restriction
- Small Size for Gestational Age
- Fetal Alcohol Syndrome
- Hypothyroidism
- Intrauterine Infection
- Cleft Lip without Cleft Palate
- Club Feet
- Congenital Dislocation of the Hips
- Limb Abnormalities
- Syndactyly
- Hypospadias
- Patent Ductus Arteriosus
- Pulmonary Hypoplasia (use code 401 for bilateral renal agenesis)

Appendix I Clarification of Infection/NEC

BLOOD CULTURE	NEC	ANTIBIOTICS	FINDINGS
Positive	Absent/suspect	≥ 5 days	Late onset sepsis
Positive	Proven	≥ 5 days	NEC and late onset sepsis
Negative	Absent/suspect	≥ 5 days	No LOS, no NEC
Negative	Proven	≥ 5 days	NEC no LOS

Appendix IA

Chorioamnionitis – Pathologic Findings

If “chorioamnionitis” is noted on the placental pathology report or if **any** of the following findings (as defined by the Stillbirth Collaborative Research Network [SCRN] Pathology Protocol) are documented on the placental pathology report, Code YES to histologic chorioamnionitis.

ACUTE CHORIOAMNIONITIS

- | | |
|-----------|---|
| Grade I | Acute subchorionitis – early acute chorionitis or both. Patchy-diffuse accumulations of neutrophils in the subchorionic plate fibrin and/or membranous chorionic trophoblast layer, (a few scattered neutrophils in the lower ½ of chorionic plate and/or the membranous chorionic connective tissue allowed) |
| Grade II | More than 10 PMNs per high power field scattered in the chorionic plate or membranous chorionic connective tissue and/or the amnion. |
| Grade III | Numerous PMNs present without necrosis of the amnion. |
| Grade IV | Necrotizing chorioamnionitis. Degenerating neutrophils (karyorrhexis), thickened eosinophilic amniotic basement membrane and at least focal amnionic epithelial degeneration/sloughing. |

SUBACUTE CHORIOAMNIONITIS

More than occasional mononuclear cells (usually macrophages) in the placental membranes or chorionic plate. Neutrophils may be rare or abundant, but a coexistent acute chorioamnionitis should be present in at least one section. (Mononuclear cells are most frequent near the amnion).

CHRONIC CHORIONITIS

Mononuclear cell infiltrates in chorion without evidence of acute chorioamnionitis.

FUNISITIS

Inflammation of umbilical cord.

VASCULITIS

Inflammation of vessels.

From SCRN Pathology Protocol, Manual of Operations

Appendix J

Physiologic Definition of BPD

Purpose

The Physiologic Definition of BPD protocol will be utilized for the infants in the PDA Trial or the BiB Trial. This protocol will allow the definition of BPD to become standardized across network sites.

Objective

To conduct a physiologic monitored reduction of oxygen in eligible infants at 36 +1 weeks corrected age who are receiving oxygen, to establish the definition of bronchopulmonary dysplasia. Infants are screened at exactly 36 weeks of age and if eligible and enrolled in either the PDA Trial or the BiB Trial, receive the challenge as close as possible to 36 weeks but no later than 37 weeks PMA.

Eligibility Criteria

Inclusion Criteria

1. Infants who are part of the Generic Data Base and alive at 36+1 weeks corrected age and enrolled in the PDA Trial or the BiB Trial
2. Supplemental oxygen as follows:
 - A. Infants receiving oxygen by hood at rest^Δ:
 1. Oxygen by hood <27% with majority* of saturations ≥ 90% in prior 24 hours.
 2. Oxygen by hood 27-30% with majority* of saturations ≥ 96% in prior 24 hours
 - B. Infants receiving oxygen by nasal cannula at rest^Δ:
 1. Oxygen by nasal cannula <27% *EFFECTIVE*** oxygen and majority* of saturations ≥ 90% in prior 24 hours.
 2. Oxygen by nasal cannula 27-30% *EFFECTIVE*** oxygen and majority* saturations ≥ 96% in prior 24 hours.
 - C. Infants receiving room air by nasal cannula at ANY liter per minute (lpm) flow.

Exclusion Criteria

1. Need for mechanical ventilation or continuous positive airway pressure (CPAP).
2. Oxygen by hood >30%.
3. Oxygen by nasal cannula >30% *EFFECTIVE*** oxygen.

Please Note:

*Majority defined as ≥90% of saturation readings during the 24 hour time period.

***EFFECTIVE* oxygen applies to infants receiving oxygen via nasal cannula only. *EFFECTIVE* oxygen is determined from *EFFECTIVE FiO₂ CONVERSION TABLES FOR INFANTS ON NASAL CANNULA* which are included.

^ΔSupplemental oxygen requirement is determined at rest. Disregard any temporary increases in O₂ requirement (for desaturation episodes, apnea, bradycardia or procedures when infant returns to baseline in a reasonable amount of time (< 2 hours). Do not include supplemental oxygen given only with feeds.

Methods

Preparation for Oxygen Reduction Challenge

If needed at your institution, permission from parent(s) and/or attending of record will be secured prior to the evaluation.

All infants should be studied in the supine position. Feedings and medications should be given 30 minutes before the evaluation. Bolus feedings should not be given during the evaluation. Infants on continuous feeding may continue feeds during the evaluation.

A pulse oximeter will be placed (if not already in use), the infant positioned and allowed to rest for 5 minutes before baseline readings are started.

Worksheets are provided to document the Oxygen Reduction Challenge for all eligible infants. Results of the challenge will be recorded on the GBD Form NG07.

Oxygen Reduction Phase:

Infants are weaned as follows:

For infants receiving oxygen by hood:

Decrease FiO_2 by 2% every 5 minutes.

For infants receiving oxygen by nasal cannula $\geq 22\%$:

Wean flow in increments of 0.5 lpm every 5 minutes until a flow of 0.5 lpm is reached.

Continue weaning by 0.1 lpm every 5 minutes until a flow of 0.1 lpm is reached.

Continue with Method A or Method B:

Method A - using oxygen blender: with flow at 0.1 lpm, wean FiO_2 by 20% every 5 minutes to room air

Method B - using low flow with 100% oxygen concentration: continue to wean flow every 5 minutes to 0.06 lpm, then 0.03 lpm

For both methods turn off flow, gently remove cannula from nares.

For infants receiving room air by nasal cannula:

If flow > 2 lpm: Wean flow in increments of 2.0 lpm every 5 minutes until a flow of 2 lpm is reached. Continue weaning as below to 0.5 lpm.

If flow 0.5-2.0 lpm: Wean flow in increments of 0.5 lpm every 5 minutes until a flow of 0.5 lpm is reached. Then turn off flow.

If flow is 0.001-0.49 lpm: turn off flow, gently remove cannula from nares.

ALL INFANTS SHOULD:

CONTINUE TO THE NEXT REDUCTION STEP IF SATURATION $\geq 90\%$ DURING THE 5 MINUTE MONITORING PHASE.

STOP REDUCTION IF SATURATION $<90\%$ FOR 5 CONTINUOUS MINUTES OR $<80\%$ FOR 15 SECONDS with good signal fidelity.

Monitoring the oxygen reduction phase:

Document each oxygen reduction step sequentially starting with #1. Use worksheets to record the following.

At each '0' timepoint record:

- a. Starting time
- b. Saturation

- c. FiO_2
- d. If NC, lpm flow

For each minute during the reduction step, record:

- a. Time
- b. Saturation

Continue with reduction steps until the infant is in room air.

Room Air Phase

The infant will be monitored in room air for 30 minutes. If saturations remain $\geq 90\%$ the infant will be considered to have passed the oxygen reduction challenge. The infant should then be placed back in his/her baseline oxygen.

The infant may qualify for a RAPID PASS if all saturations $\geq 96\%$ in room air for 15 consecutive minutes.

If the infant has saturations $<90\%$ for 5 continuous minutes or $<80\%$ for 15 seconds, the infant should be immediately placed back in his/her baseline oxygen. The infant will be considered to have NOT passed the challenge.

Outcome of Challenge

The date and results of the oxygen reduction challenge will be recorded on the NG07 question B.3.

CERTIFICATION

It is important that all personnel performing the oxygen reduction challenge are familiar with patient population and procedures. All personnel are required to read the protocol and review the worksheets used to document the oxygen reduction challenge. It is suggested that a certification exam be employed for personnel. An example of a certification exam follows.

Certification Exam:

Each individual who desires to become certified to perform the Physiologic Evaluation should complete the certification exam to assure familiarity with the methods and evaluation procedures.

CERTIFICATION EXAM

(Duplicate as many as needed for your site before completion)

Name _____ Center _____

1. Consult the Table in the Methods section and indicate whether the following infants should undergo physiologic evaluation for BPD.

A. An infant on a mechanical ventilator	Y	N
B. Infant at 33 weeks gestation	Y	N
C. Infant in room air	Y	N
D. 2.0 kg infant in 2 liters nasal cannula with 100% oxygen, saturation 90 -96%	Y	N
E. 1.8 kg infant in 1.0 liters nasal cannula, 60% oxygen, saturation 90 - 96%	Y	N
F. 1.9 kg infant in 0.25 liters nasal cannula, 70% oxygen, saturation >96%	Y	N
G. 2.1 kg infant in 0.1 liters nasal cannula, and 30% oxygen conc., saturation >96%	Y	N
H. 1.7 kg infant in 0.5 liters nasal cannula, 40% oxygen, saturation >96%	Y	N

2. Determine the final classification (BPD/no BPD) of the following scenarios:

A. Infant has all saturations $\geq 96\%$ for 15 minutes in room air	BPD	No BPD
B. Infant has saturation $< 90\%$ for 6 minutes	BPD	No BPD
C. Infant has saturation $\geq 90\%$ for 30 minutes	BPD	No BPD
D. Infant has most saturation $\geq 90\%$, but has one minute period of saturation of 85% that resolves on its own, and returns to $\geq 90\%$	BPD	No BPD

3. During the physiologic evaluation, the infant has an apnea. What should the researcher do:

- Observe the infant without intervention.
- Institute the usual routines used in that institution for an apneic infant that generally include increasing levels of support after a brief period of observation to allow the infant to arouse on his/her own.

EFFECTIVE FIO₂ CONVERSION TABLES FOR INFANTS ON NASAL CANNULA.

Tables 1 and 2 allow the determination of the *Effective FiO₂* for infants receiving supplemental oxygen via nasal cannula.

Table 1 identifies a 'factor' by using the infant's current liter flow per minute and the infant's weight. If exact values are not included in the table, round up or down to find the value closest to your patient. If the value is exactly half way in between the two values, then round up.

TABLE 1. FACTOR AS A FUNCTION OF FLOW AND WEIGHT

FLOW (LPM)	FLOW (LPM)	WEIGHT (KG)								
		0.7	1.0	1.25	1.5	2	2.5	3	3.5	4
0.01		1	1	1	1	1	0	0	0	0
0.03	1/32	4	3	2	2	2	1	1	1	1
0.06	1/16	9	6	5	4	3	2	2	2	2
0.125	1/8	18	12	10	8	6	4	4	4	4
0.15		21	15	12	10	8	6	5	4	4
0.25	1/4	36	25	20	17	13	10	8	7	6
0.5	1/2	71	50	40	33	25	20	17	14	13
0.75	3/4	100	75	60	50	38	30	25	21	19
1.0	1.0	100	100	80	67	50	40	33	29	25
1.25		100	100	100	83	63	50	42	36	31
1.5		100	100	100	100	75	60	50	43	38
2.0		100	100	100	100	100	80	67	57	50
3.0		100	100	100	100	100	100	100	86	75
4.0		100	100	100	100	100	100	100	100	100
5.0		100	100	100	100	100	100	100	100	100
6.0		100	100	100	100	100	100	100	100	100

Table 2 determines the *Effective FiO₂* by using the 'factor' and the current oxygen concentration. If exact values are not included in the table, round up or down to find the value closest to your patient. If the value is exactly half way in between the two values, then round up. (See Next page for Table 2)

**TABLE 2. EFFECTIVE FiO_2 (x100) AS A FUNCTION
OF FACTOR AND CONCENTRATION**

FACTOR	CONCENTRATION (%)						
	21	22	25	30	40	50	100
0	21	21	21	21	21	21	21
1	21	21	21	21	21	21	22
2	21	21	21	21	21	22	23
3	21	21	21	21	22	22	23
4	21	21	21	21	22	22	24
5	21	21	21	21	22	22	25
6	21	21	21	22	22	23	26
7	21	21	21	22	22	23	27
8	21	21	21	22	23	23	27
9	21	21	21	22	23	24	28
10	21	21	21	22	23	24	29
11	21	21	21	22	23	24	30
12	21	21	21	22	23	24	30
13	21	21	22	22	23	25	31
14	21	21	22	22	24	25	32
15	21	21	22	22	23	25	33
17	21	21	22	23	24	26	34
18	21	21	22	23	24	26	35
19	21	21	22	23	25	27	36
20	21	21	22	23	25	27	37
21	21	21	22	23	25	27	38
22	21	21	22	23	25	27	38
23	21	21	22	23	25	28	39
25	21	21	22	23	25	28	41
27	21	21	22	23	25	29	42
28	21	21	22	24	26	29	43
29	21	21	22	24	27	29	44
30	21	21	22	24	27	30	45
31	21	21	22	24	27	31	47
33	21	21	22	24	27	31	47
36	21	21	22	24	28	31	49
38	21	21	23	24	28	32	51
40	21	21	23	25	29	33	53
42	21	21	23	25	29	33	54
43	21	21	23	25	29	33	55
44	21	21	23	25	29	34	56
50	21	21	23	25	30	35	60
55	21	22	23	26	31	37	64
57	21	22	23	26	32	38	66
60	21	22	23	26	32	38	68
63	21	22	24	27	33	39	71
67	21	22	24	27	34	40	74
71	21	22	24	27	34	42	77
75	21	22	24	28	35	43	80
80	21	22	24	28	36	44	84
83	21	22	24	28	37	45	87
86	21	22	24	29	37	46	89
100	21	22	25	30	40	50	100

Example:

What is the effective FiO_2 in a 2.0 kg infant on 100% cannula at a flow of 0.15 lpm?

ANSWER: Use 2.0 kg and 0.15 lpm in Table 1 to get a factor of 8. Then use Table 2, and the factor of 8 and 100% oxygen to yield an effective FiO_2 of 27%. Thus the effective oxygen concentration is less than 30% and the infant is eligible for the physiologic evaluation.

PHYSIOLOGIC DEFINITION OF BRONCHOPULMONARY DYSPLASIA WORKSHEET

Date of Oxygen Reduction Challenge ____/____/____

Network No ____-____-____
BN

Page 1 of ____

	0 min	1 min	2 min	3 min	4 min	5 min
Baseline (time)	:	:	:	:	:	:
Saturation						
FiO ₂						
If NC, flow						
Reduction step # (time)	:	:	:	:	:	:
Saturation						
FiO ₂						
If NC, flow						
Reduction step # (time)	:	:	:	:	:	:
Saturation						
FiO ₂						
If NC, flow						
Reduction step # (time)	:	:	:	:	:	:
Saturation						
FiO ₂						
If NC, flow						
Reduction step # (time)	:	:	:	:	:	:
Saturation						
FiO ₂						
If NC, flow						
Reduction step # (time)	:	:	:	:	:	:
Saturation						
FiO ₂						
If NC, flow						

GUIDELINES FOR OXYGEN REDUCTION	WEANING BY HOOD	WEANING BY NASAL CANNULA (NC)			
		<u>Oxygen by nasal cannula</u> ≥22%-	<i>Method A</i>	<i>Method B</i>	Room Air by nasal cannula
- Continue to next reduction step if saturations ≥90% - Stop reduction if saturation <90% x 5 continuous minutes <u>OR</u> <80% for 15 seconds	Decrease FiO ₂ by 2% every 5 minutes if saturation ≥90%	♦ Wean flow by 0.5 lpm every 5 min until flow of 0.5 lpm reached. ♦ Continue wean by 0.1 lpm until flow of 0.1 lpm reached. ♦ Continue with either: ♦ Method A or Method B	Using oxygen blender w/ flow at 0.1 lpm: ♦♦ Wean FiO ₂ by 20% every 5 min to room air. ♦♦ Gently remove cannula ♦♦ Monitor in RA 30 min	Using low flow w/ 100% O ₂ concentration and flow at 0.1 lpm: ♦♦ Continue to wean flow every 5 min to 0.06 lpm then to 0.03 lpm ♦♦ Gently remove cannula ♦♦ Monitor in RA 30 min	♦♦ Wean flow by 2.0 lpm until flow of 2.0 lpm is reached ♦♦ Wean flow by 0.5 lpm until flow of 0.5 lpm is reached. ♦♦ Gently remove cannula ♦♦ Monitor in RA 30 min

Use as many copies of this form to complete the reduction.

Network No _____

Page ____ of ____

	0 min	1 min	2 min	3 min	4 min	5 min
Reduction step # _____ (time) _____	_____	_____	_____	_____	_____	_____
Saturation	_____	_____	_____	_____	_____	_____
FiO ₂	_____	_____	_____	_____	_____	_____
If NC, flow	_____	_____	_____	_____	_____	_____
Reduction step # _____ (time) _____	_____	_____	_____	_____	_____	_____
Saturation	_____	_____	_____	_____	_____	_____
FiO ₂	_____	_____	_____	_____	_____	_____
If NC, flow	_____	_____	_____	_____	_____	_____
Reduction step # _____ (time) _____	_____	_____	_____	_____	_____	_____
Saturation	_____	_____	_____	_____	_____	_____
FiO ₂	_____	_____	_____	_____	_____	_____
If NC, flow	_____	_____	_____	_____	_____	_____
Reduction step # _____ (time) _____	_____	_____	_____	_____	_____	_____
Saturation	_____	_____	_____	_____	_____	_____
FiO ₂	_____	_____	_____	_____	_____	_____
If NC, flow	_____	_____	_____	_____	_____	_____
Reduction step # _____ (time) _____	_____	_____	_____	_____	_____	_____
Saturation	_____	_____	_____	_____	_____	_____
FiO ₂	_____	_____	_____	_____	_____	_____
If NC, flow	_____	_____	_____	_____	_____	_____
Reduction step # _____ (time) _____	_____	_____	_____	_____	_____	_____
Saturation	_____	_____	_____	_____	_____	_____
FiO ₂	_____	_____	_____	_____	_____	_____
If NC, flow	_____	_____	_____	_____	_____	_____

GUIDELINES FOR OXYGEN REDUCTION	WEANING BY HOOD	WEANING BY NASAL CANNULA (NC)			
		Oxygen by nasal cannula ≥22%-	Method A	Method B	Room Air by nasal cannula
- Continue to next reduction step if saturations ≥90% - Stop reduction if saturation <90% x 5 continuous minutes <u>OR</u> <80% for 15 seconds	Decrease FiO ₂ by 2% every 5 minutes if saturation ≥90%	- Wean flow by 0.5 lpm every 5 min until flow of 0.5 lpm reached. - Continue wean by 0.1 lpm until flow of 0.1 lpm reached. - Continue with either: - Method A or Method B	Using oxygen blender w/ flow at 0.1 lpm: ♦♦ Wean FiO ₂ by 20% every 5 min to room air. ♦♦ Gently remove cannula ♦♦ Monitor in RA 30 min	Using low flow w/ 100% O ₂ concentration and flow at 0.1 lpm: ♦♦ Continue to wean flow every 5 min to 0.06 lpm then to 0.03 lpm ♦♦ Gently remove cannula ♦♦ Monitor in RA 30 min	♦♦ Wean flow by 2.0 lpm until flow of 2.0 lpm is reached ♦♦ Wean flow by 0.5 lpm until flow of 0.5 lpm is reached. ♦♦ Gently remove cannula ♦♦ Monitor in RA 30 min

ROOM AIR Monitoring

Network No _____ BN

Page ____ of ____

	0 min	1 min	2 min	3 min	4 min	5 min
Time	:	:	:	:	:	:
saturation						
Time	6 min	7 min	8 min	9 min	10 min	11 min
saturation	:	:	:	:	:	:
Time	12 min	13 min	14 min	15 min	16 min	17 min
saturation						
Time	18 min	19 min	20 min	21 min	22 min	23 min
saturation						
Time	24 min	25 min	26 min	27 min	28 min	29 min
saturation						
Time	30 min					
saturation						

FAIL CRITERIA:

Saturation <90% for 5 continuous minutes OR <80% for 15 seconds

RAPID PASS CRITERIA:

Saturation ≥96% for 15 consecutive minutes

Appendix K Oral Hypoglycemic Medications

Biguanides

Metformin (Glucophage)
Metformin liquid (Riomet)
Metformin extended release (Glucophage XR, Fortamet, Glumetza)

Sulfonylureas

Glimepiride (Amaryl)
Glyburide (Diabeta, Micronase)
Micronized glyburide (Glynase)

Meglitinides

Repaglinide (Prandin)

D-Phenylalanine Derivatives

Nateglinide (Starlix)

Thiazolidinediones Pioglitazone (TZDs)

Pioglitazone (Actos)

DPP-4 Inhibitors

Sitagliptin (Januvia)
Saxagliptin (Onglyza)
Linagliptin (Tradjenta)

Alpha-glucosidase Inhibitors

Acarbose (Presose)
Miglitol (Glyset)

Bile Acid Sequestrants

Colesevelam (Welchol)

Combination Pills

Pioglitazone & metformin (Actoplus Met)
Glyburide & metformin (Glucovance)
Glipizide & metformin (Metaglip)
Sitagliptin & metformin (Janumet)
Saxagliptin & metformin (Kombiglyze)
Repaglinide & metformin (Prandimet)
Pioglitazone & glimepiride (Duetact)

Appendix L Probiotics

Product*	Strain(s)
ABC Dophilus	Bifidobacterium infantis Streptococcus thermophilus Bifidobacterium bifidum
Align	Bifidobacterium infantis
Bios Life Probiotic	Bifidobacterium breve Bifidobacterium lactis Lactobacillus acidophilus Lactobacillus rhamnosus
Culturelle for Kids	Lactobacillus rhamnosus
Culturelle Natural Health and Wellness	Lactobacillus rhamnosus
Discoflor 60	Lactobacillus rhamnosus
UltimateFlora	Bifidobacteria Lactobacilli

*Contact the Data Coordinating Center if a particular probiotic is not listed.

Appendix M

Jensen BPD Criteria

BPD classification based on respiratory support at 36 weeks' postmenstrual age:

BPD grade 0 (no BPD): room air, no respiratory support

BPD grade 1: nasal cannula flow ≤ 2 lpm (regardless of FiO_2)

BPD grade 2: nasal cannula flow > 2 lpm, CPAP, or noninvasive ventilation (regardless of FiO_2)

BPD grade 3: invasive (intubated) respiratory support (regardless of FiO_2 ; includes support via tracheostomy)

Jensen EA et al. *Am J Respir Crit Care Med* 2019;200:751-9